

Fortinet

FCP_FMG_AD-7.4 Exam

Fortinet Certified Professional

Questions & Answers

(Full Version)

Thank you for Purchasing FCP_FMG_AD-7.4 Exam

➤ TOTAL QUESTIONS : 133

Question: 1

Which of the following statements are true regarding schedule backup of FortiManager? (Choose two)

- A. Backs up all devices and the FortiGuard database
- B. Supports FTP, SCP, and SFTP
- C. Can be configured from the CLI and GUI
- D. Does not back up firmware images saved on FortiManager

Answer: B, D

Explanation:

At any time, you can backup the FortiManager configuration in the System Information widget on the GUI

Backs up:

- All devices
- Global database
- Flash configuration

Does not back up logs, FortiGuard objects, and saved firmware images

Scheduled backups from the GUI and CLI

Supports FTP, SCP and SFTP

The screenshot displays the FortiManager GUI. At the top, the 'Dashboard' tab is selected. Below it, the 'System Information' widget shows details for the FortiManager instance, including Host Name (FortiManager), Serial Number (FMG-VM0A16001583), Platform Type (FMG-VM64-KVM), HA Status (Standalone), System Time (Tue Sep 05 11:44:35 2023 PDT), Firmware Version (v7.4.1-build2308 230831 (GA)), System Configuration (Last Backup: N/A), Current Administrator (admin / 1 in total), Up Time (14 hours 23 minutes 37 seconds), Administrative Domain (disabled), and FortiAnalyzer Features (disabled). A red box highlights the 'Backup' icon in the System Configuration row. Below the System Information widget, the 'Backup System' configuration window is open. It has three tabs: 'Backup Now', 'Schedule Backup', and 'Backup History'. The 'Schedule Backup' tab is active, showing options to 'Enable Schedule Backup' (checked), 'Backup Configuration File to' (server name / IP:port), 'Directory' (empty), and 'Protocol' (SFTP). A red arrow points from the 'Backup' icon in the System Information widget to the 'Schedule Backup' tab. Another red arrow points from the 'Backup System' window to the 'Backup System' configuration window below it. The 'Backup System' configuration window shows 'Encryption' (checked), 'Password' (Maximum password length: 63), and 'Confirm Password' (Maximum password length: 63). The 'OK' and 'Cancel' buttons are at the bottom.

Question: 2

An administrator has enabled Service Access on FortiManager. What is the purpose of Service Access on the FortiManager interface?

- A. Allows FortiManager to run real-time debugs on the managed devices
- B. Allows FortiManager to download IPS packages
- C. Allows FortiManager to automatically configure a default route
- D. Allows FortiManager to respond to request for FortiGuard services from FortiGate devices

Answer: D

Explanation:

Initial Configuration

Initial configuration involves configuring:

- Network interfaces
- Administrative access
- Primary and secondary DNS server IP addresses (FortiGuard DNS servers are configured by default)
- Default GW

Services access allows you to enable the FortiManager response to the requests from managed devices for FortiGuard services on each interface. This includes FortiGate updates and web filtering. By default, all services to managed devices are enabled on port1 and disabled on other ports.

The screenshot displays the FortiManager configuration interface with several key sections and annotations:

- System Settings > Network**: The main configuration window for the network interface.
- Edit Network Interface**: A sub-window for configuring the 'port1' interface.
- DNS**: Shows the Primary DNS Server (208.91.112.52) and Secondary DNS Server (208.91.112.53).
- Routing Table**: A table showing the default gateway configuration.
- Annotations**: Blue callouts point to specific configuration elements:
 - Configure management address**: Points to the IP Address/Netmask field (10.0.1.241/255.255.255.0).
 - Enable administrative protocols**: Points to the Administrative Access checkboxes (HTTPS, HTTP, PING, SSH).
 - Enable response to FortiGuard queries and update poll from clients**: Points to the Service Access checkboxes (FortiGate Updates, Web Filtering).
 - Configure DNS**: Points to the DNS server IP addresses.
 - Default gateway**: Points to the default gateway entry in the Routing Table (ID 1, Type IPv4, IP/Netmask 0.0.0.0/0.0.0.0, Gateway 10.200.1.254, Interface port2).

Question: 3

What does a policy package status of Modified indicate?

- A. The policy package was never imported after a device was registered on FortiManager
- B. FortiManager is unable to determine the policy package status

- C. Policy configuration has been changed on a managed device and changes have not yet been imported into FortiManager
- D. Policy package configuration has been changed on FortiManager and changes have not yet been installed on the managed device

Answer: D

Explanation:

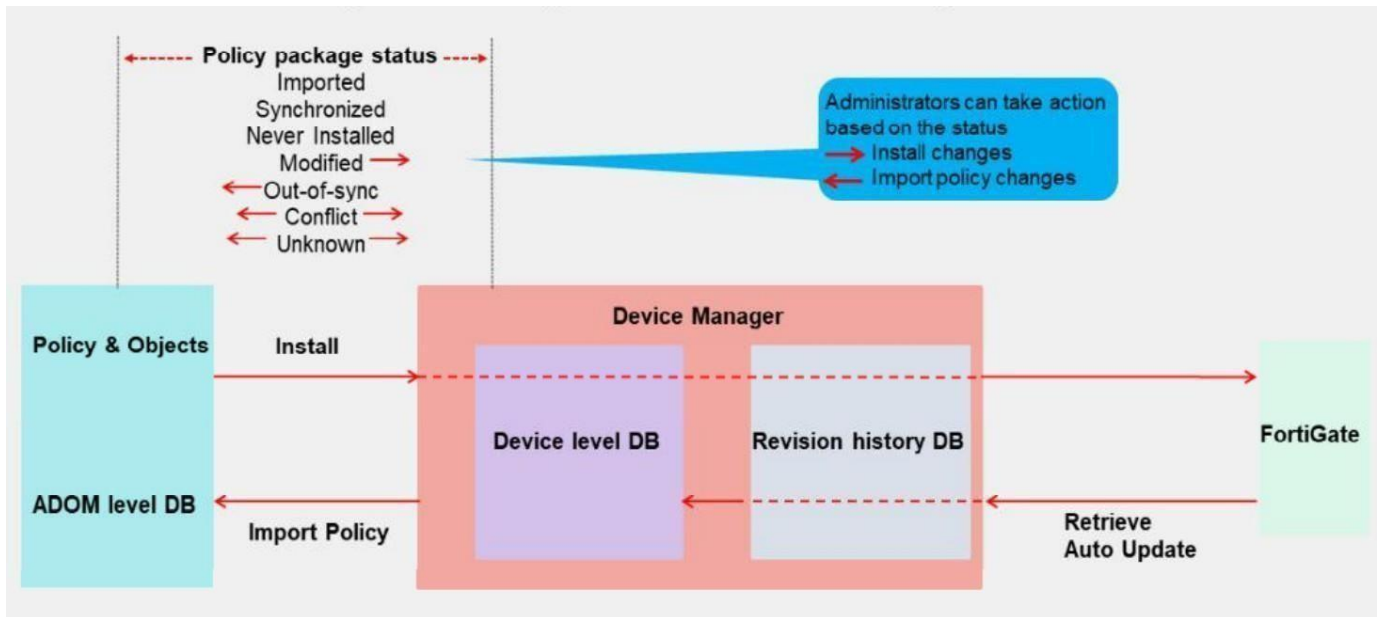
Status of Policy Package on FortiManager

After every retrieve, auto-update, and installation operation, FortiManager stores the FortiGate configuration in the revision history.

The illustration on this slide shows the status of the policy package:

- Imported: Indicates that a policy package was successfully imported for managed device.
- Synchronized: Indicates that a policies and objects are synchronized between FortiManager and the managed device.
- Never Installed: Policy package was never created, hence it was never imported for a managed device.
- Modified: Policy package configuration is changed on FortiManager and changes have not yet pushed to the managed device
- Out-of-sync: The latest policy package does not match the policies and objects configuration on the latest revision history because of configuration changes made locally on FortiGate or a previous partial installation failure. You should perform a retrieve, and then import policies from FortiManager.
- Conflict: If you make policy configuration changes locally on FortiGate and don't import the changes into the policy packages, and you also made the changes on FortiManager, the status enters conflict state. Depending on the configuration changes, you can either import a policy package or install the changes from FortiManager.
- Unknown: FortiManager is unable to determine the policy package status.

You can resolve most policy status issues by importing a policy package or installing policy package.



Question: 4

What configuration setting for FortiGate is part of a device-level database on FortiManager?

- A. Firewall policies
- B. VIP and IP Pools
- C. Security profiles
- D. Routing

Answer: D

Explanation:

Device-Level Settings

Device-level settings can be viewed and configured for individual managed devices.

You can configure device-level settings by selecting the desired device under **Device & Groups > Managed FortiGate**. The available settings can be managed by selecting them from toolbar near the top, which includes tabs to configure network, VPN, and system settings, among other. Most of the settings that you find here have a one-to-one correlation with the options that you would see if you logged in locally using the FortiGate GUI or CLI.

As an example, this slide shows how you can create a new static route on the device named ISFW. Keep in mind that changes made here are not applied to the device automatically, and they must be installed to take effect.

Not all available options are shown by default in the toolbar. To include more options, or to hide ones you don't need, you can click Feature Visibility to select them as shown in the slide. Note that the available options vary according to the device supported features, and firmware version.

The screenshot shows the FortiManager interface. On the left, the 'Device Manager' menu is open, showing 'Managed FortiGate (1)' and 'ISFW'. A red box highlights the 'Static Route' option under 'Create New'. A blue callout points to the 'Feature Visibility' tab, stating: 'Select which features you want available on the top bar'. The 'Feature Visibility' tab is open, showing a list of features that can be enabled or disabled. The 'Network' section is expanded, showing options like 'Interfaces', 'DNS', 'Static Routes', 'RIP', 'BGP', and 'Multicast'. The 'VPN' section is also expanded, showing options like 'IPsec Phase 1', 'IPsec Aggregate', 'Concentrator', 'IPsec Phase 2', and 'Manual Key'.

Question: 5

An administrator configure a new firewall policy on FortiManager and has not yet pushed the changes to the managed FortiGate.

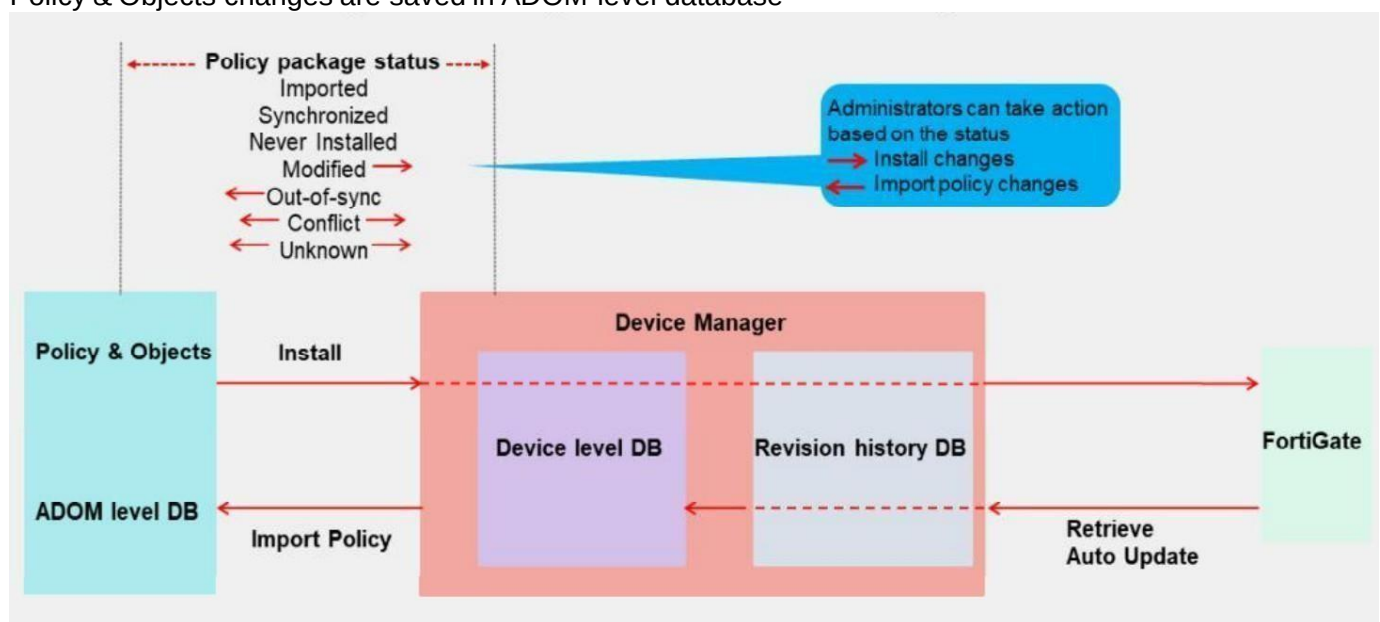
In which database will the configuration be saved?

- A. Configuration-level database
- B. Revision history database
- C. ADOM-level database
- D. Device-level database

Answer: C

Explanation:

Policy & Objects changes are saved in ADOM-level database



Question: 6

Which types of scripts can you create in FortiManager? (Choose two)

- A. Tcl scripts
- B. Bash scripts
- C. Linux Shell scripts
- D. CLI scripts

Answer: A, D

Explanation:

Scripts

Scripts allow you to make configuration changes to many managed devices

Can be used to provision FortiGate devices

Can be used to automate configuration changes

Help to simplify bulk configuration changes and maintain consistency across multiple devices

There are two types of scripts:

Command Line Interface (CLI)

- A sequence of FortiGate CLI commands, as you would type on the FortiGate CLI
- CLI scripts use the FGFM tunnel
- FGFM tunnel is authenticated using the FortiManager and FortiGate serial numbers
- CLI scripts are enabled by default.

Tool Command Language (Tcl)

- A dynamic scripting language that provides more functionality to your scripts
- Including global variables and decision structures
- Tcl scripts are not run through the FGFM tunnel like CLI scripts. They use SSH and require SSH authentication to work.

Tcl scripts can be enabled on the FortiManager CLI:

```
config system admin settings
set show_tcl_script enable
end
```

Question: 7

In addition to the default ADOMs, an administrator has created a new ADOM named Training for FortiGate devices. The administrator sent a device registration to FortiManager from a remote FortiGate. Which one of the following statements is true?

- A. The FortiGate will be added automatically to the default ADOM named FortiGate
- B. The FortiManager administrator must add the unregistered device manually to the unregistered device manually to the Training ADOM using the Add Device wizard
- C. By default, the unregistered FortiGate will appear in the root ADOM

D. The FortiGate will be automatically added to the Training ADOM

Answer: D

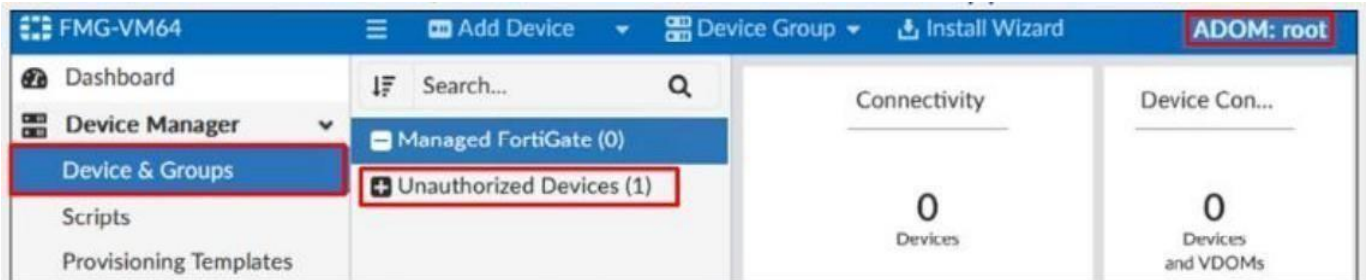
Explanation:

Unauthorized Devices

Unauthorized devices requesting registration appear under Device Manager > Device & Groups >

Unauthorized Devices

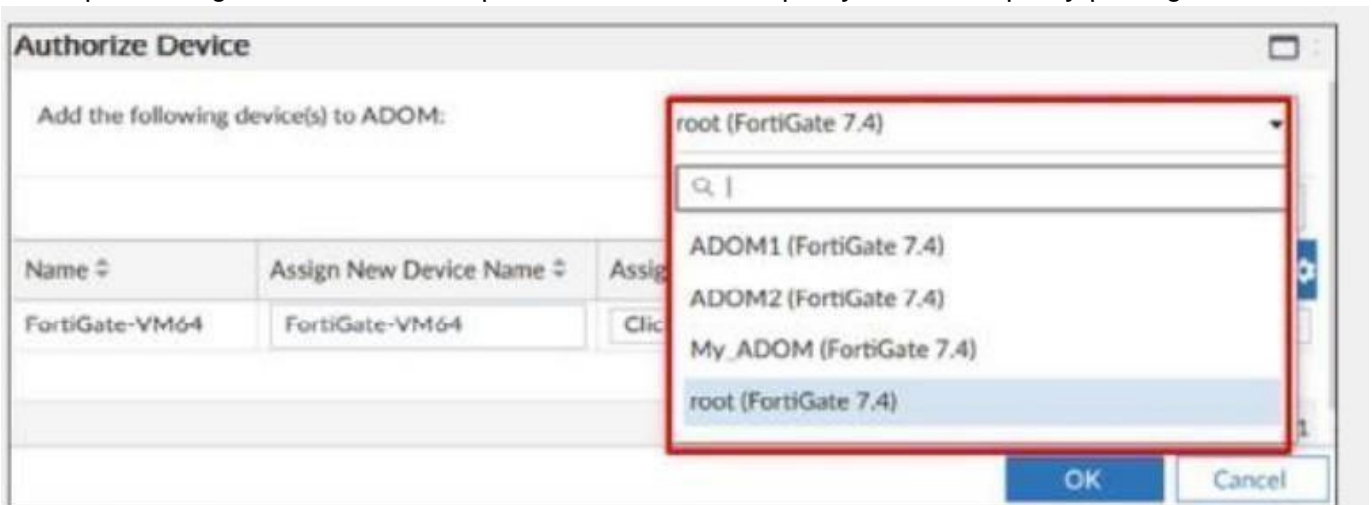
If ADOMs are enabled, unauthorized FortiGate devices appear in the root ADOM



Can enable automatic registration - the default is disabled

```
config system admin setting
set allow_register {enable | disable}
Can add FortiGate to a different ADOM
```

Finally, authorizing a device does not create a policy packages automatically. For that reason, you must run Import Configuration wizard to import the device firewall policy into a new policy package.



Question: 8

Which statement about a FortiManager HA cluster is true?

- A. FortiManager in HA operations are not interrupted during upgrade
- B. The HA heartbeat packets use TCP port 5199
- C. Upgrading the primary FortiManager upgrades all secondary devices at the same time
- D. The HA heartbeat packets use TCP port 5121

Answer: B, C

Explanation:

The HA heartbeat packets use TCP port 5199. FortiManager HA supports geographic redundancy so the

primary unit and secondary units can be in different locations attached to different networks as long as communication is possible between them.

When performing a firmware upgrade on the cluster, always schedule a maintenance window because upgrading the firmware on the primary FortiManager also upgrades the firmware on all the secondary devices, and reboots all the devices in the cluster. Administrators may not be able to connect to the GUI until the upgrade synchronization process is completed. During the upgrade, SSH or telnet connections to the CLI may also be slow. You can still use the console to connect to the CLI of the primary device.

Question: 9

An administrator is replacing a device on FortiManager by running the following command:

execute device replace sn

What device name and serial number must the administrator use?

- A. Device name of the original device and serial number of the replacement device
- B. Device name of the replacement device and serial number of the original device
- C. Device name and serial number of the original device
- D. Device name and serial number of the replacement device

Answer: A

Explanation:

Steps to replace a managed device are:

1. Note original FortiGate device name

`diagnose dvm device list`

If the replacement device is already listed as unregistered, then you will need to delete it from the unregistered device list in the root ADOM.

2. Update the serial number of the replacement FortiGate

`execute device replace sn`

`devname` = Original FortiGate device name

`serial_number` = Serial number of replacement FortiGate

After the replace command is executed, FortiManager updates the serial number in its database.

3. Verify that FortiManager updated the serial number in its database

`diagnose dvm device list`

The new device serial number must be associated with the faulty device in FortiManager

4. Send a request from the replacement device to register it with FortiManager

5. If connectivity is down after you update the serial number, you might need to reclaim the management tunnel. The device name is optional. If you run the command without the device name, FortiManager tries to reclaim tunnels from all managed device.

`execute fgfm reclaim-dev-tunnel`

Optionally, you can change the device password that you used when you added the device by running the `execute device replace pw` command.

Question: 10

Refer to the exhibit



An administrator logs into the FortiManager GUI and sees the panes shown in the exhibit. Which two reasons can explain why the FortiAnalyzer feature panes do not appear? (Choose two)

- A. FortiAnalyzer features are not enabled on FortiManager
- B. The administrator profile does not have full access privileges like the Super_User profile
- C. The administrator IP address is not a part of the trusted hosts configured on FortiManager interfaces
- D. The administrator logged in using the unsecure protocol HTTP, so the view is restricted

Answer: A, B

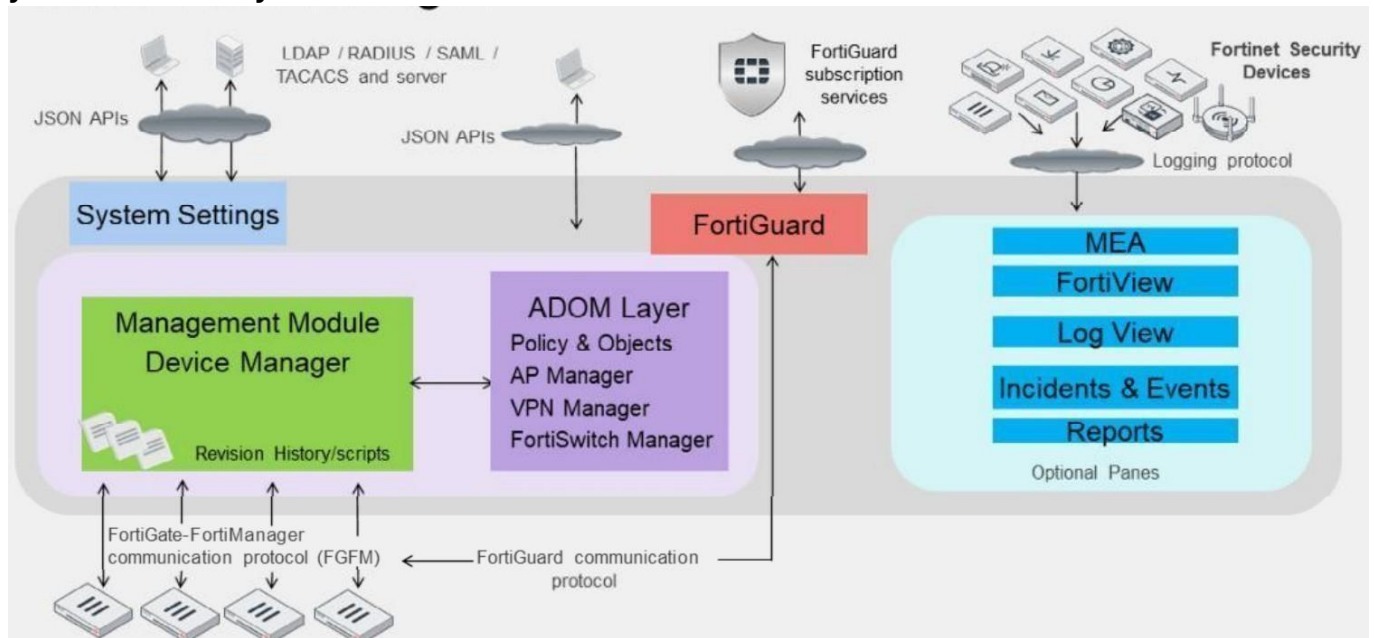
Explanation:

Inside FortiManager

Inside FortiManager there are several management layers that are represented as panes in the GUI. The default panes include:

- Device Manager: To add and authorize devices, create device configuration changes and install device and policy packages.
- Policy & Objects: To centralize the management of firewall policies, objects, and security profiles among others.
- VPN Manager: To view and configure IPsec VPN and SSL-VPN settings that you can install to one or more devices.
- AP Manager: To manage FortiAP access points that are controlled by FortiGate devices.
- FortiSwitch Manager: To manage FortiSwitch devices that are controlled by FortiGate devices.
- Extender Manager: To manage connected FortiExtenders
- Fabric View: To view Security Fabric Ratings of configurations for FortiGate Security Fabric groups, as well as create fabric connectors.
- FortiGuard: To provide FortiGuard services for your FortiManager system and its managed devices and FortiClient agents.
- System Settings: To manage system options for your FortiManager device.

Other panes can be added depending on the requirements of your network, for example, when you add FortiAnalyzer features.



Multiple Administrators and Security

- Most used access control methods are:

- ADOMs
- Administrative profiles
- Trusted hosts

Select one or more ADOMs

Select administrative profile type

Trusted hosts restrict login access to specific IP addresses or subnets

The screenshot shows the 'Create New Administrator' form in the FortiManager GUI. The form includes fields for User Name (Training), Avatar, Description, Admin Type (LOCAL), New Password, Confirm Password, and FortiToken Cloud options (Disable, FortiToken Mobile, Email, SMS). The Administrative Domain section shows a search for 'My_ADOM' with 1 entry selected. The Admin Profile section shows 'Single' and 'Per-ADOM' options, with 'Restricted_User' selected. The Specify Device Group to Access section shows 'All Packages' and 'Specify' options. The JSON API Access section shows 'None'. The Theme Mode section shows 'Use Global Theme' and 'Use Own Theme' options. The Trusted Hosts section shows two fields for Trusted IPv4 Host 1 (0.0.0.0/0.0.0.0) and Trusted IPv4 Host 2 (255.255.255.255/255.255.255.255). The form has OK and Cancel buttons.

ADOMs: Administrators have access to only those ADOMs to which they are assigned.

Administrative profiles: The level of access is determined by the administrative profile selected. When the administrators are restricted to specific ADOMs, you can assign different profiles for each ADOM, or the same profile for all ADOMs.

Trusted hosts: This option restricts administrators to log in only from specific IP addresses or subnets. The trusted host you define apply to both the GUI and the CLI when accessed through SSH. This feature is available for IPv4 and IPv6 addresses.

Question: 11

An administrator has added all the devices in a Security Fabric group to FortiManager.

How does the administrator identify the root FortiGate?

- A. By a question mark(?) at the end of the device name
- B. By a dollar symbol (\$) at the end of the device name
- C. By an Asterisk (*) at the end of the device name
- D. By an at symbol (@) at the end of the device name

Answer: C

Explanation:

The * beside External indicates that it is the root FortiGate in the Security Fabric.

Question: 12

An administrator has assigned a global policy package to custom ADOM1. Then the administrator creates a new policy package, Fortinet, in the custom ADOM1. Which statement about the global policy package assignment the newly-created policy package Fortinet is true?

- A. When a new policy package is created, you can select the option to assing the global policies to the new package
- B. When a new policy package is created, it automatically assigns the global policies to the new package
- C. When a new policy package is created, you need to reapply the global policy package to the ADOM
- D. When a new policy package is created, you need to assign the global policy package from the global ADOM

Answer: C

Explanation:

Viewing Assigned Global Policy

After you assign the global database ADOM objects, they appear on the Policy & Objects pane that ADOM. All global objects start with "g" and are edited or deleted in the global database ADOM only. A word of caution. Never create address objects or security profiles starting with letter "g" on an ADOM database or directly on the managed device because this may create conflicts with global database ADOM objects and cause configuration failure issues.

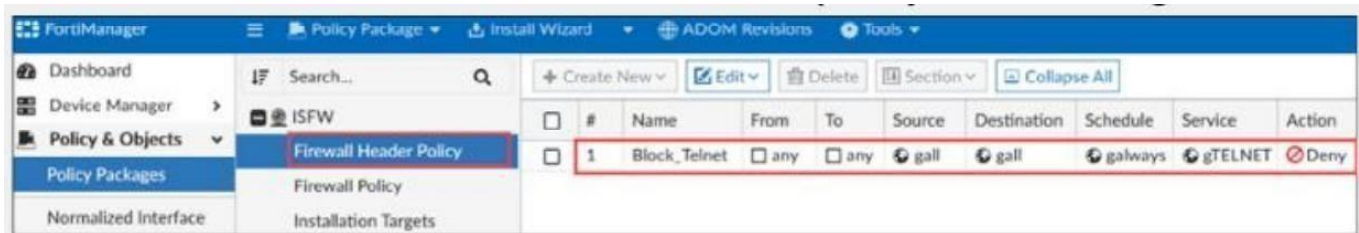
You can assing only one global policy package ti an individual ADOM policy package. Assigning an additional global policy package to the same individual ADOM policy package removes previously assigned policies. Also, you cannot edit and move the header and footer policies between the rules in an individual ADOM policy package.

You must install policy packages on the managed devices for the new rules to work. A header policy is installed at the top of the list of the firewall rules on the target device.

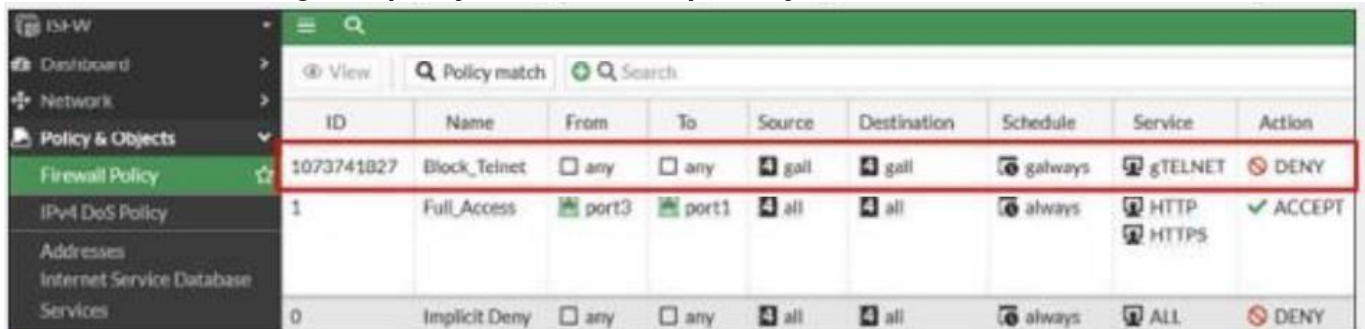
Header policy created in global database ADOM

#	Name	From	To	Source	Destination	Schedule	Service	Action
1	Block_Telnet	any	any	gall	gall		galways gTELNET	Deny

Local-FortiGate now includes the heder policy created in global database ADOM



After installation, the global policy shows at the top locally on FortiGate



Question: 13

What does the diagnose dvm check-integrity command do? (Choose two)

- A. Verifies and corrects duplicate VDOM entries
- B. Verifies and corrects database schemas in all objects
- C. Internally upgrades existing ADOMs to the same ADOM version in order to clean up and correct the ADOM syntax.
- D. Verifies and corrects unregistered, registered, and deleted device states

Answer: A, D

Explanation:

This slide list several commands that you can use to verify and maintain database integrity.

Command	Information
# diagnose dvm check-integrity	Verify and correct parts of the device manager databases, including: - Inconsistent device-to-group and group-to-ADOM memberships - Unregistered, registered, and deleted device states - Device lock statuses - Duplicate VDOM entries
# diagnose cdb upgrade check objcfg-integrity	Object config database integrity—perform a check to see if upgrade and repair is necessary
# diagnose cdb upgrade check reference-integrity	Perform a check to see if upgrade and repair is necessary for reference table.
# diagnose cdb check update-devinfo	Update device info by directly changing the database
# diagnose cdb check adom-integrity	Upgrade existing ADOMs to the same ADOM version in order to clean up and correct the ADOM syntax
# diagnose cdb check policy-packages	Verify and check dynamic mappings, and remove invalid dynamic mappings

Not all integrity problems are detected or corrected! After executing a database integrity command that performs corrections to the database, you should re-run the command to verify the proper implementation of those corrections.

Question: 14

As a result of enabling FortiAnalyzer features on FortiManager, which of the following statements is true?

- A. FortiManager will reboot
- B. FortiManager will enable ADOMs automatically to collect logs from non-FortiGate devices
- C. FortiManager can be used only as a logging device
- D. FortiManager will send the logging configuration to the managed devices so the managed devices will start sending logs to FortiManager

Answer: A

Explanation:

FortiManager with FortiAnalyzer Features

Enable FortiAnalyzer features manually to make FortiManager a logging and reporting device

More CPU, memory and disk space are required for the additional functions

Log rate and daily log quota restrictions are in place

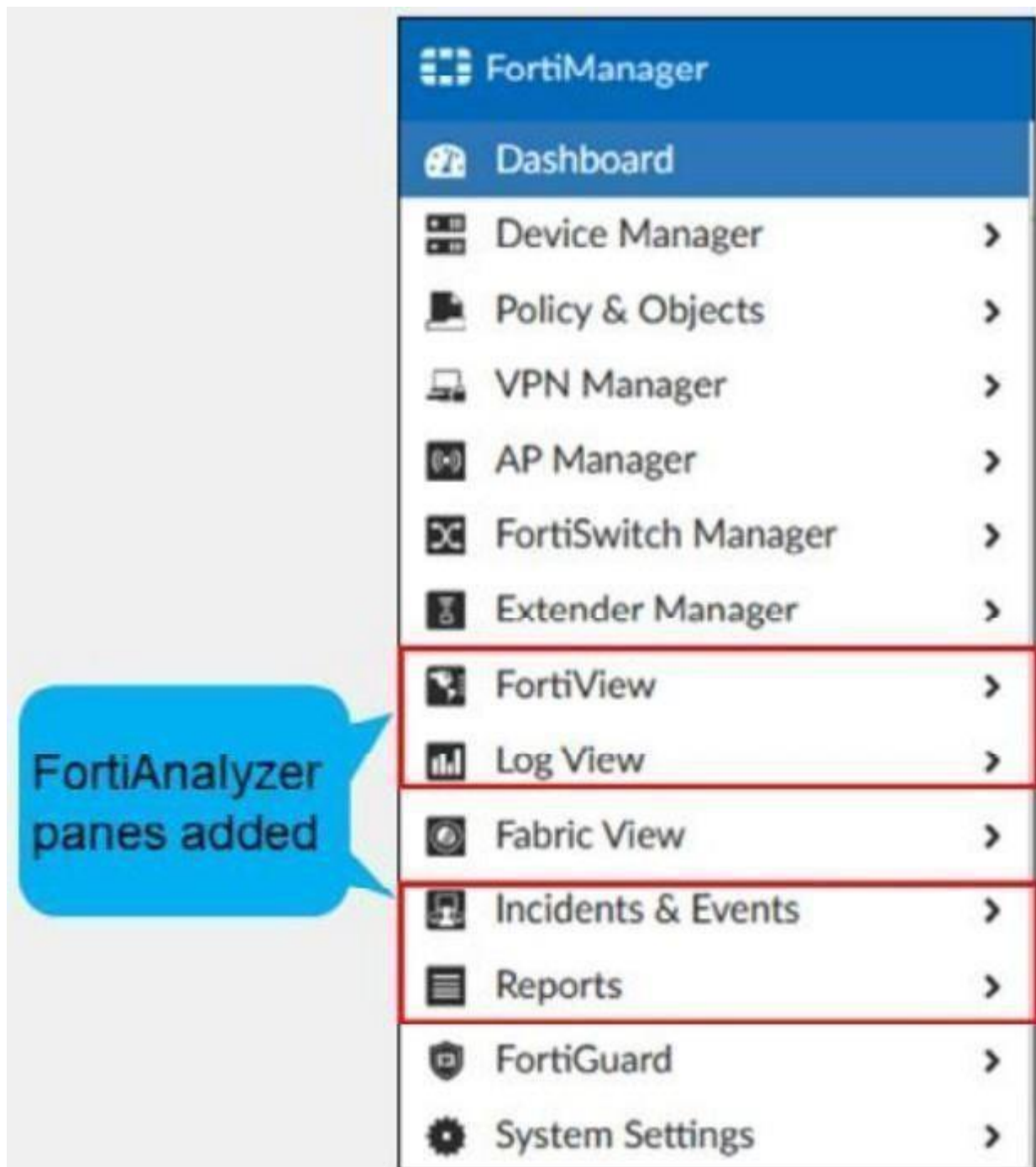
FortiManager supports up to 150 logs/sec only

FortiAnalyzer-related licensing is not stackable

FortiManager stores and manages logs

Solution feasible for low volume of logs only

For example, a testing environment



If FortiAnalyzer features are enabled, you cannot add FortiAnalyzer to FortiManager, and you cannot enable FortiManager high availability (HA)

Question: 15

An administrator would like to review, approve, or reject all the firewall policy changes made by junior administrators.

How should the Workspace mode be configured on FortiManager?

- A. Set to workflow and use the ADOM locking feature
- B. Set to disable and use the policy locking feature
- C. Set to normal and use the policy locking feature

D. Set read/write and use the policy locking feature

Answer: A

Explanation:

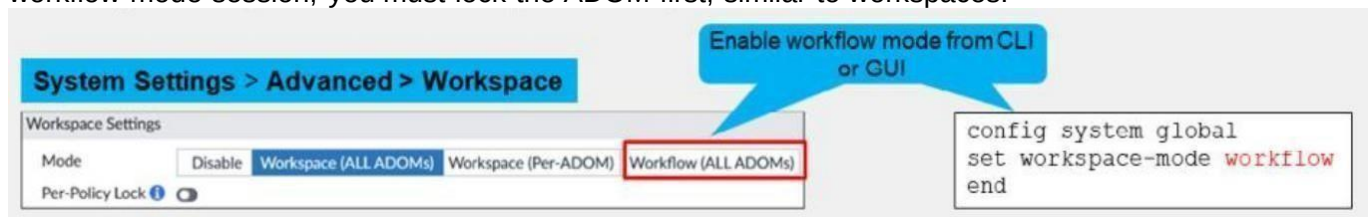
Workflow Mode

Instead of workspaces, you can use workflow mode. As with the other workspace modes, enabling workflow terminates all management sessions. You must notify other administrators to save their work to prevent any data loss. You can enable workflow mode on the CLI or GUI.

You can use workflow mode to control the creation, configuration, and installation of several settings. However this feature is mostly used to control changes to firewall policies and objects.

Approval is required before changes can be installed on a device. All the modifications made in a workflow mode session must be discarded or submitted for approval at the end of the session. Sessions that are rejected can be repaired and resubmitted for approval as new sessions. All sessions must be approved in the same order in which they were created to prevent any conflicts.

In workflow mode, panes related to FortiGate configuration are read-only at first. To create a new workflow mode session, you must lock the ADOM first, similar to workspaces.



Question: 16

What is the purpose of ADOM revisions?

- A. To save the current state of the whole ADOM
- B. To save the current state of all policy and objects for an ADOM
- C. To create System Checkpoints for the FortiManager configuration
- D. To revert individual policy packages and device-level settings for a managed FortiGate by reverting to a specific ADOM revision

Answer: B

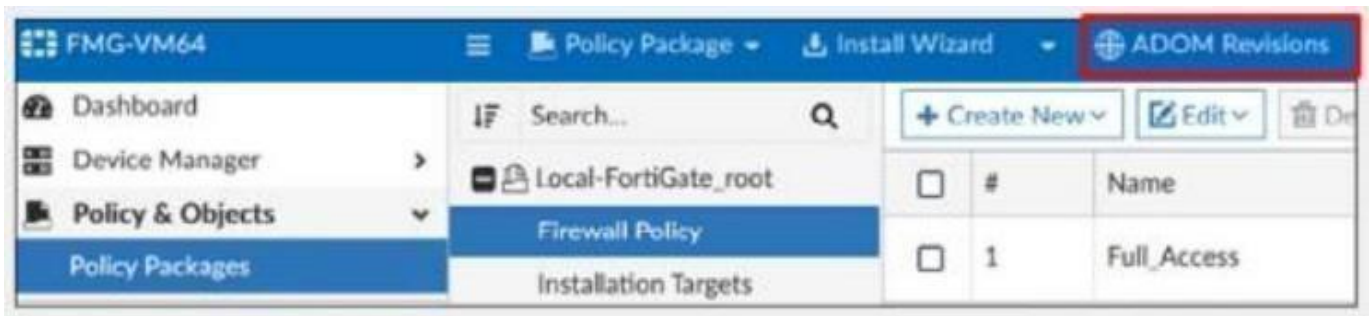
Explanation:

ADOM revisions allow you to save locally in FortiManager snapshots of the policy packages, objects, and VPN console settings contained in an ADOM.

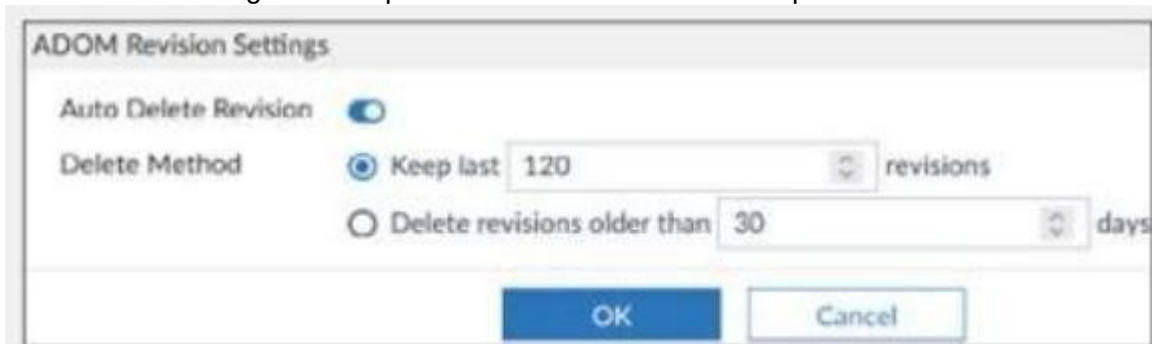
When you create multiple ADOM revisions, you can view differences between them, or revert to a specific revision. As word of caution, if you choose to revert to a specific ADOM revision, you will revert all the policy packages and objects based on that revision.

FortiManager can delete ADOM revisions automatically based on specified parameters as shown on this slide. Optionally, you can lock individual revisions to prevent them from being automatically deleted.

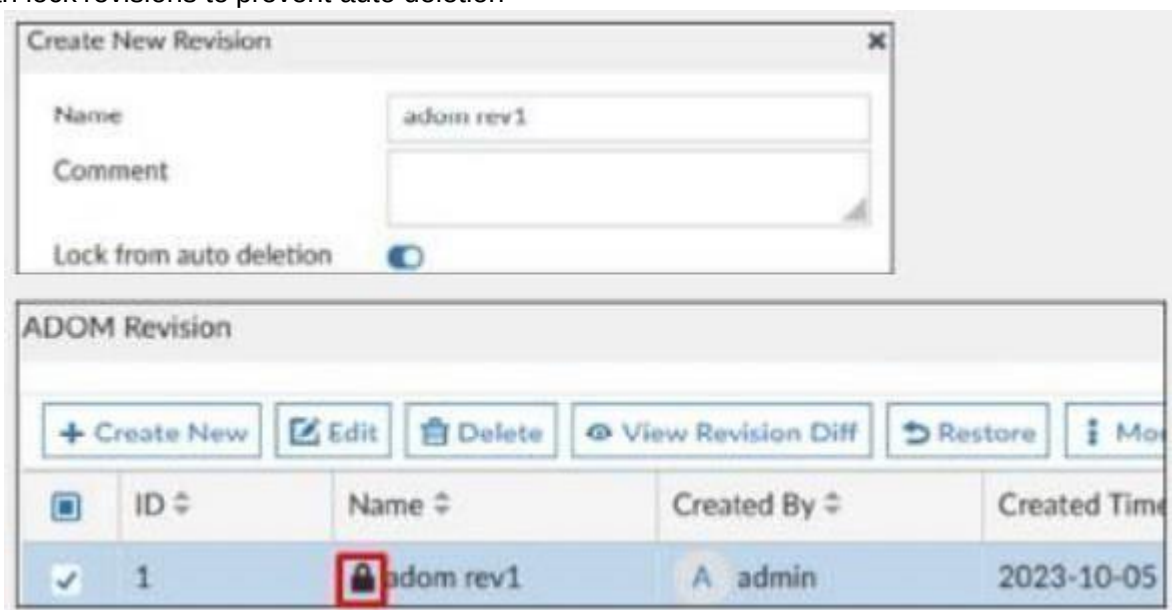
An ADOM Revision creates a snapshot of all policy and object configurations for the ADOM



The ADOM Revision Settings window provides access to autodeletion parameters



You can lock revisions to prevent auto deletion



Warning: Keep in mind that ADOM revisions can significantly increase the size of configuration backup.

Question: 17

Which of the following conditions trigger FortiManager to create a new revision history? (Choose two)

- A. When FortiManager installs device-level changes to a managed device
- B. When FortiManager is auto-updated with the configuration changes made directly on a managed device
- C. When changes to device-level database is made on FortiManager

D. When configuration revision is reverted to previous revision in the revision history

Answer: A,B

Explanation:

FortiManager creates a configuration revision when:

- FortiGate is added to FortiManager
- Device changes are installed from FortiManager
- FortiGate configurations is retrieved from FortiManager
- A local change on FortiGate causes an automatic update
- Scripts run directly on remote devices also cause automatic updates and create a revision history

Question: 18

View the following exhibit, which shows the Download Import Report

Start to import config from devices(Remote-FortiGate) vdom (root)to adom (MyADOM),

Package(Remote-FortiGate)

"firewall address", SUCCESS,"(name=REMOTE_SUBNET,oid=580, new object)"

"firewall policy",SUCCESS,"(name=1, oid=990,new object)"

"firewall policy",FAIL,"(name=ID:2(#2), oid=991, reason=interface(interface binding

Contradiction.detail:any<-port6)binding fail)"

Why it is falling to import firewall policy ID2?

- A. The address object used in policy ID2 already exist in ADOM database with any as interface association and conflicts with address object interface association locally on FortiGate
- B. Policy ID2 is configured from interface any to port6 FortiManager rejects to import this policy because any interfaces does not exist on FortiManager
- C. Policy ID2 does not have ADOM Interface mapping configured on FortiManager
- D. Policy ID2 for this managed FortiGate already exists on FortiManager in policy package named Remote-FortiGate.

Answer: A

Explanation:

When you add a FortiGate device using the Add Device wizard, or import policies using the Import Configuration wizard, always make sure that the policies and objects are successfully imported.

The screenshot shows the 'Import Device - Remote-FortiGate - Importing (5/5)' window. The 'Policy Import Summary' tab is active, showing a 'Download Import Report' button and a list of encountered import errors. Two errors are listed: 'firewall address' and 'firewall policy', both failing due to interface binding issues. The 'firewall address' error message is: "firewall address", FAIL, "(name=Test_PC, oid=2243, reason=interface((firewall address:Test_PC) any<-port6) binding fail)". The 'firewall policy' error message is: "firewall policy", FAIL, "(name=2, oid=3097, reason=interface(interface binding contradiction. detail: (firewall address:Test_PC) any<-port6) binding fail)".

Below the summary, the 'Firewall Objects' tab is selected, showing a table of objects. The 'Test_PC' object is highlighted, showing its address as 'IP/Netmask: 10.0.1.10/255.255.255.255' and interface as 'any'. A blue callout points to this object, stating 'Address object in ADOM database'.

On the right, the 'Test_PC' object details are shown. The 'Interface' field is set to 'port6', which is highlighted with a red box. A blue callout points to this field, stating 'Address object in device'.

In this example, the FortiManager ADOM database has a firewall address object named Test_PC that is associated with the interface any. However, Remote-FortiGate also has a firewall address object named Test_PC, but that is associated with the interface port6. This firewall address object is referenced in a firewall policy on the Remote-FortiGate.

When a policy package was added or imported to Remote-FortiGate, the operation failed to import the firewall address object Test_PC, as well as the associated firewall policy. The partial output of the import report shown in the image provides the reason for the failed import.

Question: 19

Which of the following items does an FGFM keepalive message include? (Choose two)

- A. FortiGate IPS version
- B. FortiGate license information
- C. FortiGate configuration checksum
- D. FortiGate uptime

Answer: C, D

Explanation:

The FortiGate-FortiManager (FGFM) protocol is used for communication between a FortiGate device and FortiManager. The **keepalive messages** are essential for maintaining communication and monitoring the health of the FortiGate devices connected to FortiManager. These messages provide important status information about the device.

Here are the items included in an FGFM keepalive message:

A. FortiGate IPS version

This is **false**. The IPS (Intrusion Prevention System) version is not included in the keepalive message. While IPS information can be part of other system syncs or monitoring processes, it is not part of the FGFM keepalive message.

B. FortiGate license information

This is **false**. The license information is not typically sent in the keepalive message. Licensing is checked and managed separately through other system operations and licensing checks.

C. FortiGate configuration checksum

This is **true**. The configuration checksum is a critical part of the keepalive message, as it ensures that

the configuration on the FortiGate matches the one managed by FortiManager. Any discrepancy would alert FortiManager to potential out-of-sync configurations.

D. FortiGate uptime

This is **true**. The keepalive message includes the FortiGate's uptime, which allows FortiManager to track the health and stability of the connected FortiGate device.

Question: 20

An administrator would like to create an SD-WAN using central management. What steps does the administrator need to perform to create an SD-WAN using central management?

- A. You must specify a gateway address when you create a default static route
- B. Enable SD-WAN central management in the ADOM, add member interfaces, create a static route and SDWAN firewall policies
- C. First create an SD-WAN firewall policy, add member interfaces to the SD-WAN template and create a static route
- D. Remove all the interface references such as routes or policies

Answer: C

Explanation:

Enabling central SD-WAN management

Central SD-WAN management can be enabled per ADOM. When enabled, the SD-WAN tab shows the following items on the left pane:

- Assigned Devices
- SD-WAN Templates
- Interface Members
- Health-Check Servers
- Monitor

When you create a SD-WAN configuration using central management, it's necessary to remove all the interface references such as routes or policies that will be a part of SD-WAN member interfaces

Option C is wrong because indicates "Remove all the interface references such as routes or policies" in general.

Question: 21

Which port is used between FortiManager and FortiGate for IPv4 remote configuration management of FortiGate devices?

- A. TCP 514
- B. RDP
- C. SSH
- D. TCP 541

Answer: D

Explanation:

FortiManager uses many TCP and UDP ports to perform tasks. The table below shows the ports FortiManager uses to manage FortiGate devices

Entries with an asterisk are used only when FortiManager acts as a local FortiGuard server

Functionality	Protocol and Port(s)
Incoming ports	
Webfilter queries, antivirus, and IPS updates*	UDP 53, UDP/TCP 8888, TCP 80
Antispam*	UDP/TCP 8889
Registration for license validation and UTM updates (antivirus, IPS)*	TCP 8890, TCP 443
FortiGuard and FortiClient Web Filter and Email Filter*	TCP 8900, TCP 443
Outgoing ports	
Antivirus and IPS push updates*	UDP 9443
Incoming and outgoing ports	
FGFM management	TCP 541 (IPv4), TCP 542 (IPv6)

Question: 22

Refer to the following exhibit:

```
config system global
set workspace-mode normal
end
```

Which of the following statements are true based on this configuration? (Choose two)

- A. Unlocking an ADOM will install configuration automatically on managed devices
- B. The same administrator can lock more than one ADOM at the same time
- C. Unlocking an ADOM will submit configuration changes automatically to approval administrator
- D. Ungraceful closed sessions will keep the ADOM in a locked state until the administrator session times out

Answer: B, D

Explanation:

Policy Lock

Allows administrators to lock a single policy package instead of whole ADOM

- Works in conjunction with workspace-mode normal or workspace per-adom
- Locks only a policy package instead of locking the whole ADOM
- You can edit locked policy package in a private workspace
- Multiple administrators can lock and work on separate policy packages at the same time

Question: 23

What is the main purpose of backup ADOM mode?

- A. To maintain the backup configuration of managed devices
- B. To facilitate the upgrade of managed devices
- C. To lock changes made directly on managed devices
- D. To install policy package changes offline

Answer: A

Explanation:

ADOM Operation Modes

When you create or configure ADOMs, you can choose between two operation modes: normal or backup.

Normal Mode

- Default mode
- ADOM is read/write
- All management panes available
- Can make configuration changes to ADOMs and managed devices

Backup Mode

- ADOM is read-only
- Not all management panes available
- Must make changes directly on managed devices

Only scripts can be used to make changes from FortiManager to the managed devices

In backup operation mode, you can import firewall address and service objects into FortiManager, and FortiManager stores the objects in the Device Manager database. You can view the objects on the Policy & Objects pane, but they are not stored in the central database. This lets you maintain a repository of objects used by all devices in the backup ADOM that is separate from the central database.

Question: 24

View the following exhibit.

Start to import config from device(Local-FortiGate) vdom(root) to adom(My_ADOM), package(Local-Fortigate_root)

"firewall service category",SKIPPED,"(name=General,oid=697, DUPLICATE)"

"firewall address", SUCCESS,"(name=LOCAL_SUBNET,oid=684,new object)"

"firewall service custom",SUCCESS,"(name=ALL,oid=863,update previous object)"

"firewall policy",SUCCESS,"(name=1,oid-1090, new object)"

Which one of the following statements is true regarding the object named ALL?

- A. FortiManager installed the object ALL with the updated value
- B. FortiManager created the object ALL as a unique entity in its database, which can be only used by this managed FortiGate
- C. FortiManager updated the object ALL using FortiManager's value in its database
- D. FortiManager updated the object ALL using FortiGate's value in its database

Answer: D

Explanation:

After the import configuration is complete, the wizard provides the Policy Import Summary and the Download Import Report. You can download the import report, which is only available on the Import Device page. You can view the report using any text editor.

As a best practice, you should download the report.

The import report provides information about FortiGate, the ADOM name on FortiManager, and the policy package name. The report also provides additional information, such as the objects that have been added as new objects. Existing objects that have the same values locally on FortiGate and FortiManager are referred to as DUPLICATE. If the value of an existing object is changed, FortiManager updates that in its database and shows update previous object in the import report.



- You can save import report in .txt format

```
Start to import config from device(Local-FortiGate) vdom(root) to adom(root),
package(Local-FortiGate)
"application list",SUCCESS,"(name=wifi-default, oid=2459, update previous object)"
...
"authentication setting",SUCCESS,"(name=, oid=3393, new object)"
...
"firewall address",SKIPPED,"(name=all, oid=2264, DUPLICATE)"
"firewall address",SKIPPED,"(name=none, oid=2265, DUPLICATE)"
"firewall address",SUCCESS,"(name=REMOTE_ETH1, oid=2266, new object)"
"firewall address",SUCCESS,"(name=REMOTE_SUBNET, oid=2267, new object)"
```

Question: 25

Which FGFM processes run on both FortiGate and FortiManager?

- A. In FortiGate: process fgfmsd
- B. In FortiGate: process fgfmd
- C. In FortiManager: process fgfmd
- D. In FortiManager: process fgfmsd

Answer: B, D

Explanation:

The management protocol FGFM runs on both FortiGate (fgfmd) and FortiManager (fgfmsd). FortiManager and FortiGate create a secure tunnel using port TCP 541. Being TCP-based, the connection works with port-based NAT, which allows both FortiGate and FortiManager to be behind a NAT device. On the FortiGate side, you must enable the FMG-Access setting for each interface facing FortiManager.

Question: 26

What will be the result of reverting to a previous revision in the revision history?

- A. It will modify the device-level database
- B. It will install configuration changes to managed device automatically
- C. It will tag the device settings status as Auto-Update
- D. It will generate a new version ID and remove all other revision history versions

Answer: A

Explanation:

Configuration Revision - Revert

The green checkmark in the revision history indicates which revision history configuration corresponds to the device manager database configuration. It is usually the top entry, which is synchronized with the FortiGate configuration.

A revert operation changes the device database configuration to a previous configuration state. You must install these reverted changes on FortiGate, which then creates a new revision entry. This new revision is a copy of the reverted one and in sync with the FortiGate configuration.

You can revert to any previous revision by right-clicking that entry and then clicking Revert. The selected entry automatically updates the Installation column to Revision Revert. FortiManager also updates the Comments column with the number of the revision it is reverted and indicating that new revision ID will be generated on an install.

Performing a revert operation followed by installation only reverts device-level changes and does not revert policy packages. To achieve full synchronization, you must run the Import Configuration tool on FortiManager to sync the policy package.

The left screenshot shows the 'Configuration Revision History' table with the following data:

ID	Date & Time	Name
5	2023-10-03 14:49:55	
4	2023-10-03 14:46:30	
3	2023-10-03 14:43:53	Local-FortiGate
2	2023-09-28 14:43:53	
1	2023-09-26 14:43:53	

The right screenshot shows the detailed view of revision 3:

ID	Date & Time	Name	Created by	Installation	Comments
5	2023-10-03 14:49:55		admin	Retrieved	Retrieve
4	2023-10-03 14:46:30		admin	Retrieved	Retrieve
3	2023-10-03 14:43:53	Local-FortiGate	admin	Revision Revert	Reverted from ID 5 by admin. A new r...

Question: 27

Refer to the exhibit

Create New CLI Script

Script Name: [View Sample Script]

Comments: 0/255

Type:

Run script on:

Script details:

```
config router prefix-list
edit public
config rule
edit 1
set prefix 0.0.0.0/0
set action permit
next
edit 2
set prefix 8.8.8.8/32
set action deny
end
```

Advanced Device Filters >

Which two statements are true if the script is executed using the Device Database option? (Choose two)

- A. The script history will show successful installation of the script on the remote FortiGate
- B. You must install these changes using the Install Wizard to a managed device
- C. The Device Setting Status will be tagged as Modified
- D. The successful execution of a script on the Device Database will create a new revision history

Answer: B, C

Explanation:

Configuring Scripts

Scripts can be run on:

Device Database

- By default option.
- This allows you to check what configuration changes are sent to the managed device.
- Once scripts are run on the device database, you can then install the changes on a managed device using the installation wizard.

Policy Package or ADOM Database

- This option is used if a script contains changes related to ADOM-level objects and policies
- Then install the changes using the installation wizard

Remote FortiGate Directly (via CLI)

- A script can be executed directly on the device, and you don't need to install the changes using the installation wizard.
- As the changes are directly installed on the managed device, no option is provided to check the configuration changes through FortiManager before executing them.

Select advanced filters to restrict the script to run on managed devices, only if the device matches the set criteria.

For example you can limit the script to run only on a subnet of the managed devices, or on devices with a specific platform.

The screenshot displays the FortiManager 'Edit Script' window. The left sidebar shows the navigation menu with 'Scripts' selected. The main area shows the 'Script Details' tab with the following configuration:

- Type:** CLI Script
- Run script on:** Device Database (highlighted with a red box and a blue callout: "By default, scripts run on the device database")
- Script details:** A sample CLI script (highlighted with a red box and a blue callout: "Sample CLI script")

```
1 config system interface
2   edit "port1"
3     set vdom "root"
4     set ip 10.200.1.1 255.255.255.0
5     set allowaccess ping https ssh http https
6     set type physical
7   next
8   edit "port2"
9     set vdom "root"
10    set ip 10.200.2.1 255.255.255.0
11    set allowaccess ping https ssh http
12    set type physical
13  next
```
- Advanced Device Filters:** A red box highlights the 'Advanced Device Filters' link, with a blue callout: "Select advanced filters to restrict the script to be executed on devices matching the set criteria".

At the bottom, there are 'OK' and 'Cancel' buttons.

Question: 28

Which two statements regarding device management on FortiManager are true? (Choose two.)

- A. FortiGate devices in HA cluster devices are counted as a single device.
- B. FortiGate in transparent mode configurations are not counted toward the device count on FortiManager.
- C. FortiGate devices in an HA cluster that has five VDOMs are counted as five separate devices.
- D. The maximum number of managed devices for each ADOM is 500.

Answer: A, C

Explanation:

Device Count on FortiManager

- Maximum number of managed devices/VMs is determined by model and license
 - Check product data sheet model limits
- Each FortiGate HA cluster count as one device. This is because the bulk of the configuration relates to the firewall policies and objects, and devices in the cluster are running the same configuration.
- Each VDOM counts as one device. This is because each VDOM is logically a separate firewall and has its own configuration.

Question: 29

Which two items are included in the FortiManager backup? (Choose two)

- A. Logs
- B. FortiGuard database
- C. All devices
- D. Global database

Answer: C, D

Explanation:

At any time, you can backup the FortiManager configuration in the System Information widget on the GUI.

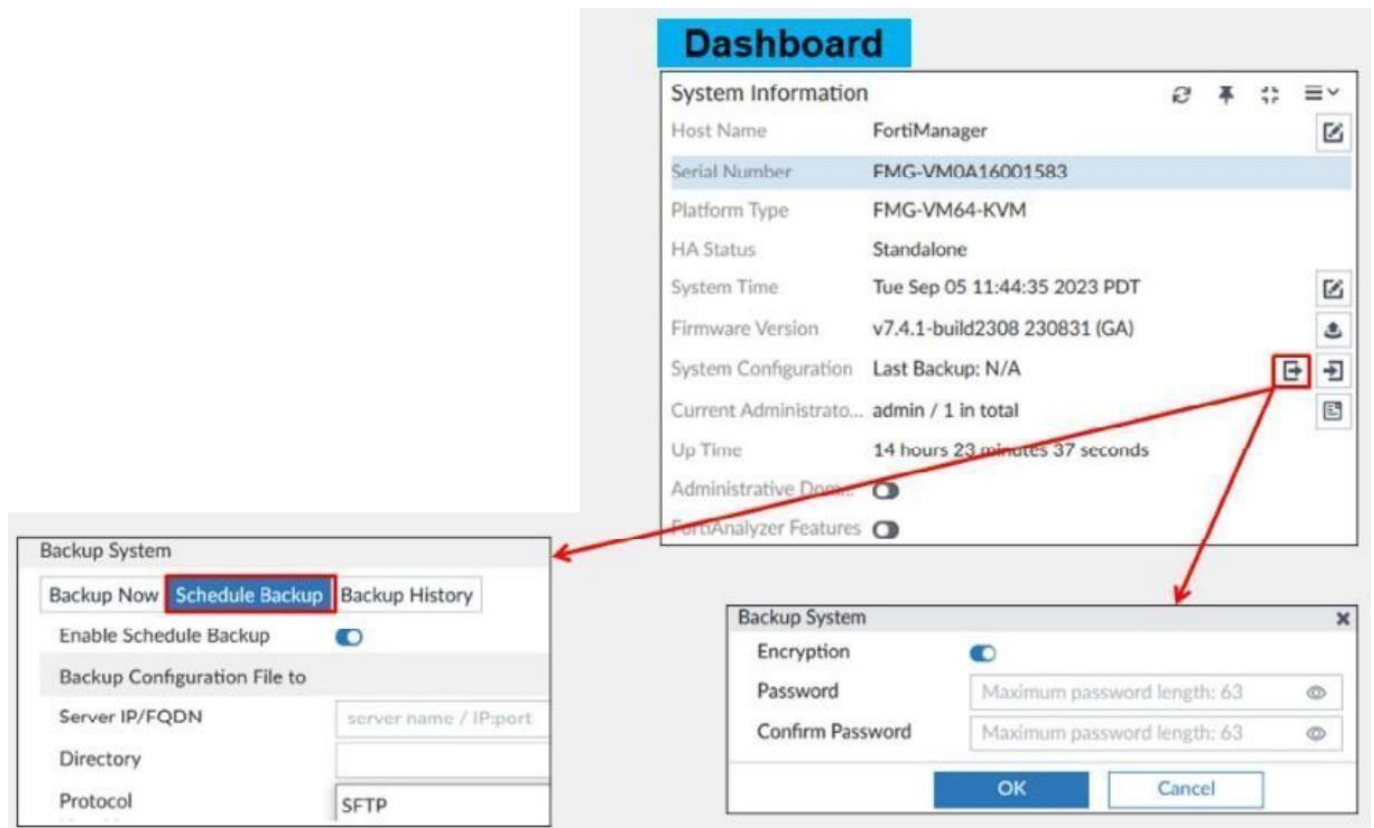
Backs up:

- All devices
- Global database
- Flash configuration

Does not back up logs, FortiGuard objects, and saved firmware images

Scheduled backups from the GUI and CLI

Supports FTP, SCP and SFTP



Question: 30

View the following exhibit:

```
#diagnose fmupdate view-serverlist fds
Fortiguard Server Comm: Enabled
Server Override Mode: Loose
FDS server list :
Index Address          Port    TimeZone  Distance  Source
-----
*0    10.0.1.50             8890    -5         0         CLI
1     96.45.33.89           443     -5         0         FDNI
2     96.45.32.81           443     -5         0         FDNI
....
38  fds1.fortinet.com     443     -5         0         DEFAULT
```

How will FortiManager try to get updates for antivirus and IPS?

- A. From public FDNI server with highest index number only
- B. From the default server fds1.fortinet.com
- C. From the list of configured override servers with ability to fall back to public FDN servers
- D. From the configured override server list only

Answer: C

Explanation:

FortiGate/FortiMail Server Override - Antivirus and IPS

Override server address configuration depends upon server override mode.

When Server Override Mode is set to Strict, FortiManager gets updates only from the server in the list. There is no fallback to other public servers, even if this configured servers is not available.

However, if you set Server Override Mode to Loose, FortiManager first tries to get updates from the configured server list and; if that server becomes unavailable, FortiManager falls back to other public FDS servers to get updates.

The image shows the FortiManager configuration interface and CLI output for the Server Override Mode. The configuration interface on the left shows the 'Strict (Access Override Server Only)' mode selected. The CLI output on the right shows the 'Strict' mode configuration with a single server listed. A blue callout points to the 'Strict' mode selection, stating 'Strict mode does not allow fallback to public servers'. Another blue callout points to the 'CLI' source in the server list, stating 'Indicates a manual override configured'. A third blue callout points to the 'Loose' mode selection, stating 'Loose mode allows fallback to public servers'.

Strict mode does not allow fallback to public servers

```
# diagnose fmupdate view-serverlist fds
FortiGuard Server Comm : Enabled
Server Override Mode   : Strict
FDS server list       :
Index Address         Port  TimeZone Distance Source
-----
*0 10.20.1.1          443   -8        0        CLI
```

Indicates a manual override configured

Loose mode allows fallback to public servers

```
# diagnose fmupdate view-serverlist fds
FortiGuard Server Comm : Enabled
Server Override Mode   : Loose
FDS server list       :
Index Address         Port  TimeZone Distance Source
-----
*0 10.2.1.1          443   -8        0        CLI
 1 fdsl.fortinet.com  443   -8        0        DEFAULT
```

Question: 31

Which two statements about Security Fabric integration with FortiManager are true? (Choose two.)

- A. The Fabric View module enables you to generate the Security Fabric ratings for Security Fabric devices.
- B. The Security Fabric settings are part of the device-level settings.
- C. The Fabric View module enables you to view the Security Fabric ratings for Security Fabric devices.
- D. The Security Fabric license, group name, and password are required for the FortiManager Security Fabric integration.

Answer: A, C

Explanation:

Two statements about Security Fabric integration with FortiManager that are true are:

A. The Fabric View module enables you to generate the Security Fabric ratings for Security Fabric devices.

The Fabric View module in FortiManager allows administrators to generate Security Fabric ratings, which assess the security posture of the entire Security Fabric environment.

C. The Fabric View module enables you to view the Security Fabric ratings for Security Fabric devices.

In addition to generating ratings, the Fabric View module provides visibility into the Security Fabric ratings for all connected devices, offering a consolidated view of security across the fabric.

Options B and D are incorrect because:

B is misleading as the Security Fabric settings are generally configured and managed separately

from other device-level settings.

D is incorrect as there is no specific requirement for a Security Fabric license, group name, and password solely for FortiManager integration.

FortiManager References:

Refer to FortiManager 7.4 Security Fabric Integration Guide: Managing Security Fabric and Generating Security Fabric Ratings.

Question: 32

An administrator, Trainer, who is assigned the Super_User profile, is trying to approve a workflow session that was submitted by another administrator, Student. However, Trainer is unable to approve the workflow session.

Session List

View Diff					
☐	ID	Name	User	Date Submitt...	Approved/To... Comments
☒	1	Firewall p..	Student	2017-06-01...	0/1 firewall policies

+ Add Comment

[Student] - 2017-06-01
13:31:35
firewall policies
[Student] - 2017-06-01
16:29:27

What can prevent an admin account that has Super_User rights over the device from approving a workflow session?

- A. Trainer is not a part of workflow approval group
- B. Student, who submitted the workflow session, must first self-approve the request
- C. Trainer does not have full rights over this ADOM
- D. Trainer must close Student's workflow session before approving the request

Answer: A

Explanation:

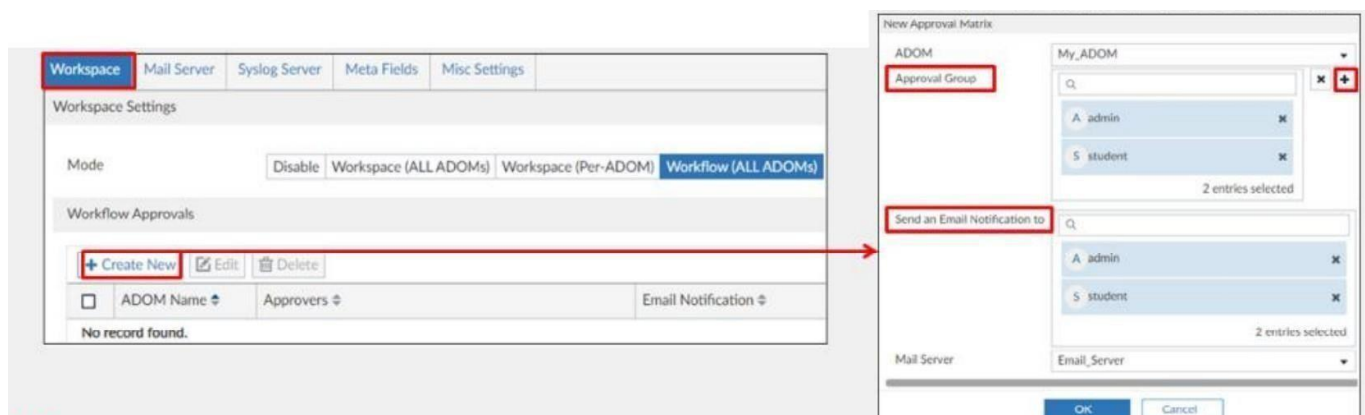
Workflow Permissions

An administrator must be part of an approval group before they can approve a session.

- Regardless of which administrator profile an administrator account is part of
- Administrator will also need to have access to the ADOM in which the session was created in order to approve it
- Being part of the Super_Admin profile is not enough to approve a session

On the GUI, the approval matrix must be configured before the workflow sessions are allowed.

- ADOM: Select the ADOM you want to apply workflow mode to.
- Approval Group #1: Add the administrators who will approve the changes in the ADOM
- Send email notification to: Send administrators email notification when another administrator makes changes and submits the changes for approval
- Mail server: Select the email server that FortiManager will use to send its notifications on the Mail Server pane.



Question: 33

View the following exhibit



Which of the following statements are true if FortiManager and Fortigate are behind the NAT devices?
(Choose two)

- A. During discovery, the FortiManager NATed IP address is not set by default on FortiGate
- B. FortiGate is discovered by FortiManager through the FortiGate NATed IP address
- C. FortiGate can announce itself to FortiManager only if the FortiManager IP address is configured on FortiGate under central management
- D. If the FCFM tunnel is torn down, FortiManager will try to re-establish the FGFM tunnel

Answer: A,B

Explanation:

What if both devices are behind a NAT device?

- Then, the FortiGate device is discovered by FortiManager through the FortiGate NATed IP address.
- Just like it was in the NATed FortiManager scenario, the FortiManager NATed IP address in this scenario is not configured under the FortiGate central management configuration.
- FortiManager doesn't attempt to reestablish the FortiGate to FortiManager (FGFM) tunnel to the FortiGate NATed IP address, if the tunnel is interrupted.
- If the FortiManager NATed IP address is configured on FortiGate under the central management configuration, FortiGate tries to reestablish the FGFM tunnel, if it is torn down.



Question: 34

An administrator wants to delete an address object that is currently referenced in a firewall policy. Which one of the following statements is true?

- A. FortiManager will not allow the administrator to delete a referenced address object
- B. FortiManager will replace the deleted address object with the none address object in the referenced firewall policy

- C. FortiManager will disable the status of the referenced firewall policy
- D. FortiManager will replace the deleted address object with all address object in the referenced firewall policy

Answer: B

Explanation:

Used Objects

On FortiManager, it is possible to delete a used object. FortiManager will display a warning message stating that the object is currently used by other firewall policies or objects. To view the references of this object, click Where Used.

However, if you delete a used object, FortiManager will replace it with a none object. The none object is equal to null, which means any traffic that meets that firewall policy will be locked, unless there is a broader policy that still meets the traffic requirement, or a policy defined to allow all traffic (catch all).



You should double-check all references to objects before deleting them, to avoid unintended firewall policy behavior.

+ Create New Edit Delete Section Policy Lookup Collapse All							
☐	#	Name	From	To	Source	Destination	Schedule
☐	1	Full_Access	port3	port1	none	all	always

Question: 35

An administrator's PC crashes before the administrator can submit a workflow session for approval. After the PC is restarted, the administrator notices that the ADOM was locked from the session before the crash.

How can the administrator unlock the ADOM?

- A. Log in using the same administrator account to unlock the ADOM
- B. Log in Super_User in order to unlock the ADOM
- C. Delete the previous admin session manually through the FortiManager or CLI
- D. Restore the configuration from a previous backup

Answer: C

Explanation:

When a FortiManager session is interrupted, such as when the management computer crashes, or when you close a browser with an open GUI session, that session remains open for the system configured timeout, or until an administrator manually deletes it. When a session associated with a locked ADOM is interrupted, the read-write access to that ADOM is not possible for other administrators until the interrupted session is deleted.

You can delete administrator session on the GUI or the CLI. After the session is deleted, the ADOM will be unlocked immediately.

The screenshot displays the FortiManager interface. On the left, the 'System Information' tab shows details for FortiManager, including Host Name (FortiManager), Serial Number (FMG-VM0A16001583), Platform Type (FMG-VM64-KVM), HA Status (Standalone), System Time (Fri Oct 06 12:14:10 2023 PDT), Firmware Version (v7.4.1-build2308 230831 (GA)), System Configuration (Last Backup: Fri Sep 15 08:19:57 2023), Current Administrator (admin / 1 in total), Up Time (3 days 3 hours 31 minutes 3 seconds), and Administrative Domain (blue icon).

On the right, the 'Admin Session List' table shows active sessions. A red box highlights the 'Delete' button. A blue callout points to the 'Delete' button with the text 'Select the session and click Delete'. The table has columns: User Name, Profile, IP Address, Current ADOM, Start Time, and Idle Time. Two sessions are listed: 'admin (Current)' and 'admin'. The 'admin' session is selected with a checkmark.

Below the table, the CLI output for the command 'diagnose sys admin-session list' is shown. A red box highlights the 'session_id: 6671' field. A blue callout points to this field with the text 'Use session_id to end the previous session'. The CLI output also shows the command 'diagnose sys admin-session kill 6671'.

A yellow box at the bottom left contains the text: 'When a session with a locked ADOM is interrupted, other administrators cannot access that ADOM until the session is closed.'

Question: 36

In the event that the primary FortiManager fails, which of the following actions must be performed to return the FortiManager HA to working state?

- A. Reboot one of the secondary devices to promote it automatically to the primary role, and reconfigure all other secondary devices to point to the new primary device
- B. Manually promote one of the secondary devices to the primary role, and reconfigure all other secondary devices to point to the new primary device
- C. Secondary device with highest priority will automatically be promoted to the primary role, and manually reconfigure all other secondary devices to point to the new primary device
- D. FortiManager HA state transition is transparent to administrators and does not require any reconfiguration

Answer: B

Explanation:

There are two HA failover modes:

Manual

- When the primary device fails, you must manually reconfigure one of the secondary devices to become the new primary device
- Must reconfigure all other secondary devices to point to the new primary device
- Don't need to reboot devices when promoting from secondary to primary
- The new primary unit will keep its IP address
- The FortiManager IP address registered on FortiGate will be automatically changed when the new primary unit is selected

**VRRP (Automatic failover)**

- Must configure VIP (virtual IP for the cluster), VRRP interface, priorities for each member, and monitored IP
- When the monitored interface for the primary is down, HA failover will occur automatically
- The secondary FortiManager with the highest priority automatically becomes the primary

**Question: 37**

An administrator has assigned a global policy package to a new ADOM called ADOM1. What will happen if the administrator tries to create a new policy package in ADOM1?

- A. When the new policy package is created, FortiManager automatically assigns the global policy package to the new policy package
- B. When creating a new policy package, the administrator can select the option to assign the global policy package to the new policy package
- C. When a new policy package is created, the administrator must assign the global policy package from the global ADOM
- D. When a new policy package is created, the administrator needs to reapply the global policy package to ADOM1

Answer: B

Explanation:

Viewing Assigned Global Policy

After you assign the global database ADOM objects, they appear on the Policy & Objects pane that ADOM. All global objects start with "g" and are edited or deleted in the global database ADOM only. A word of caution. Never create address objects or security profiles starting with letter "g" on an ADOM database or directly on the managed device because this may create conflicts with global database

ADOM objects and cause configuration failure issues.

You can assign only one global policy package to an individual ADOM policy package. Assigning an additional global policy package to the same individual ADOM policy package removes previously assigned policies. Also, you cannot edit and move the header and footer policies between the rules in an individual ADOM policy package.

You must install policy packages on the managed devices for the new rules to work. A header policy is installed at the top of the list of the firewall rules on the target device.

Header policy created in global database ADOM

All global objects start with "g" and are edited or deleted in global ADOM only

Caution: Never create address objects or profiles starting with letter "g" on the ADOMs or managed devices

Local-FortiGate now includes the header policy created in global database ADOM

#	Name	From	To	Source	Destination	Schedule	Service	Action
1	Block_Telnet	any	any	gall	gall	galways	gTELNET	Deny

After installation, the global policy shows at the top locally on FortiGate

ID	Name	From	To	Source	Destination	Schedule	Service	Action
1073741827	Block_Telnet	any	any	gall	gall	galways	gTELNET	DENY
1	Full_Access	port3	port1	all	all	always	HTTP HTTPS	ACCEPT
0	Implicit Deny	any	any	all	all	always	ALL	DENY

Question: 38

Exhibit one.

The screenshot displays the ADOM configuration interface with the following sections:

- Navigation:** Device Manager, Device & Groups, Firmware, License, Provisioning Templates, Scripts, SD-WAN. User: ADOM: My_ADOM, Unlock.
- Left Panel:** Save, Install Wizard, System Templates (2), default, Training (highlighted), Threat Weight, Certificate Templates.
- DNS:** Primary DNS Server: 192.168.1.111, Secondary DNS Server: 192.168.1.112, Local Domain Name: [empty]. [Apply]
- Alert Email:** SMTP Server: [empty], Authentication: [Enable], SMTP User: [empty], Password: [empty]. [Apply]
- SNMP:** SNMP Agent: [Enable].
 - SNMP v1/v2c:** [Create New] [Edit] [View] [Delete]. Table with columns: Community Name, Queries, Traps.
 - SNMP v3:** [Create New] [Edit] [View] [Delete]. Table with columns: User Name, Queries, Traps, Security Level.[Apply]
- NTP Server:** [Synchronize with NTP Server] [Apply]
- Admin Settings:** HTTP Port: 80, HTTPS Port: 443, SSH Port: 22, SSH v1 compatibility: OFF, Idle Timeout: 5 (1 - 480 mins), Enable SCP: OFF, Switch Controller: OFF, [Redirections to...].
- View Settings:** Language: English, Lines per page: 50 (20 - 1000), Theme: Green, IPv6 Support on GUI: OFF. [Apply]
- Replacement Message:** [Edit] [Reset]. Table with columns: Name, Description, Import.
 - Administration (0) [Import]
 - Alert Email (0) [Import]
 - Authentication (0) [Import]
 - Device Detection Portal (0) [Import]

Exhibit two.

Install Preview

Virtual Domain:

global, root

```
config system ntp
unset ntpsync
end
config system email-server
unset server
unset security
end
config log fortianalyzer setting
unset status
unset server
unset upload-option
unset reliable
unset serial
end
config system central-management
config server-list
purge
end
end
config system global
unset admintimeout
unset admin-https-redirect
end
config system dns
set primary 192.168.1.111
set secondary 192.168.1.112
end
config system snmp sysinfo
```

An administrator created a new system template named Training with two new DNS addresses on FortiManager. During the installation preview stage, the administrator notices that many unset commands need to be pushed.

What can be the main reason for these unset commands?

- A. The ADOM is locked by another administrator
- B. The Training system template has other default settings
- C. The DNS addresses in the default system settings are the same as the Training system template
- D. The Training system template does not have assigned devices

Answer: B

Explanation:

System Templates

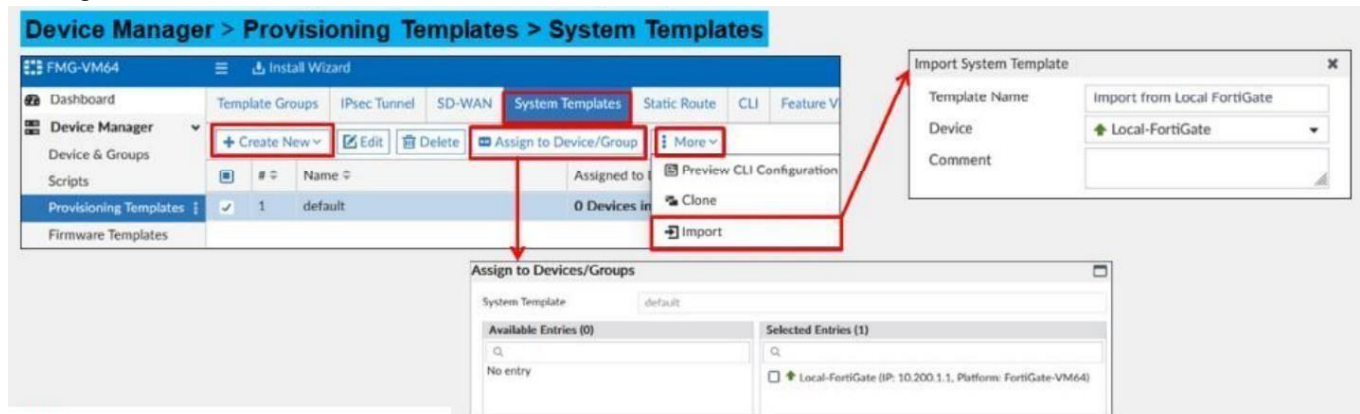
The System Templates page contains one generic template named default, which is a subset of the model device configurations and contains several widgets as DNS, Alert Email, Admin Settings, and several others.

If you do not make changes, the default settings template will be configured.

You can create a new system template and configure the settings in the included widgets, or you can import the settings from a specific managed device to inherit the system-level settings of that managed device.

You can use the Assign to Device/Group button to associate device with a template.

Applying these templates to multiple devices within the same ADOM facilitates identical device-level settings across all those devices.



Question: 39

An administrator is experiencing unusual behavior with database integrity on FortiManager.

Which of the following best practices the administrator should follow before executing database integrity commands? (Choose two)

- A. Apply a hard shutdown (power off) to recover the database
- B. Lock all the ADOMs to avoid conflicts
- C. Backup the FortiManager database
- D. Have all the administrators log off

Answer: C, D

Explanation:

If you are experiencing unusual behavior on FortiManager, check for issues with database integrity.

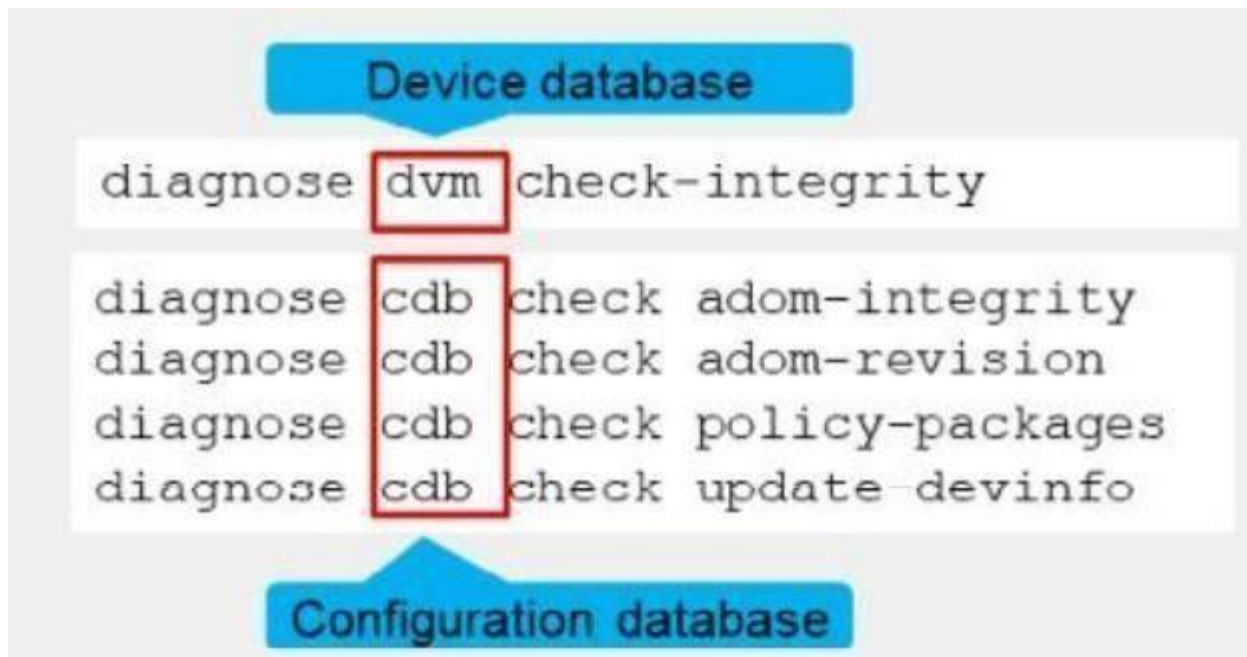
Before executing database integrity commands, you should:

Backup the FortiManager database

Have all the administrators log off

Have all the ADOMs unlocked, if workspace mode is enabled

Integrity checks modify the FortiManager database if errors are found.



Having a backup is helpful when you don't want to keep changes that were made by the integrity commands, and you need to restore the FortiManager configuration.

Question: 40

Which action can you perform in advanced ADOM mode?

- A. You can assign a FortiGate device and its VDOMs to a single ADOM only
- B. You can assign different VDOMs from FortiGate to different ADOMs
- C. You can assign the same VDOM from FortiGate to different ADOMs
- D. You can assign only VDOMs in a selected ADOM

Answer: B

Explanation:

ADOM Device Modes

An ADOM can work in two device modes: Normal, which is the default mode, and Advanced.



Normal

- You can add a FortiGate device and its VDOMs to a single ADOM only
- You cannot assign different FortiGate virtual domains (VDOMs) to different FortiManager ADOMs

Advanced

- You can assign different VDOMs from the same FortiGate from different ADOMs
- Applies globally to all ADOMs

Question: 41

Which statements about a Security Fabric Rating are true? (Choose two)

- A. It includes the same three scorecards available on FortiGate
- B. Security Fabric ratings for all FortiGate devices managed by FortiManager are shown
- C. Optimization scorecard is not available on FortiManager
- D. Fabric Coverage scorecard is not available on FortiManager

Answer: A, B

Explanation:

Security Fabric Rating

The Security Rating feature includes checks that can help you make improvements to your organization's network, such as enforce passwords security, apply recommended login attempt thresholds, encourage two-factor authentication, and so on. You can view Security Fabric ratings for all FortiGate devices managed by FortiManager.

Displays the reported Security Fabric ratings

The Security Rating page includes the same three scorecards available on FortiGate

- Security Posture
- Fabric Coverage
- Optimization

Provides executive summary of the three largest areas of security focus in the Security Fabric

Question: 42

View the following exhibit

The screenshot displays the FortiGate GUI with the following components:

- Device Manager** tab: Shows 4 Managed FortiGate devices. A list of devices is visible: Local-FortiGate, Remote-FortiGate, root [NAT] (Management), Student[NAT], and Trainer [NAT].
- Policy Packages** tab: Shows the configuration for a Shared Package (IPv4 Policy). The **Installation Targets** section lists: Remote-FortiGate, root [NAT][Management], Student[NAT], and Local-PortiGate.
- Interface Pair View** table:

Seq.#	Install On	Name	From	To
1	Remote-FortiGate(Student)	Ping_Access	port3	port1
2	Remote-FortiGate(Student)	Web	port3	port1
3	Installation Targets	Source_Device	port3	port1

Which one of the following statements is true regarding installation targets in use Install On column?

- A. Policy seq=3 will be not installed on any managed device
- B. Policy seq=3 will be installed on all managed devices and VDOMs that are listed under Installation Targets
- C. The Install On column value represents successful installation on the managed devices
- D. Policy seq=3 will be installed on the Trainer [NAT] VDOM only

Answer: B

Explanation:

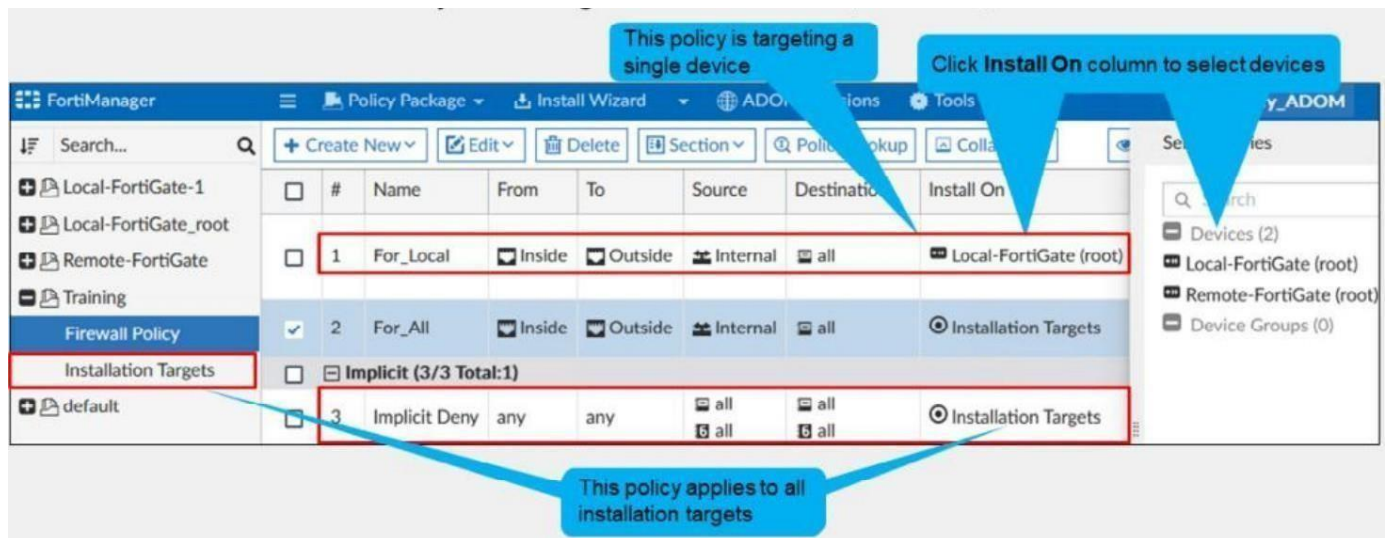
Installation Target - Per Policy

What if you need to share a policy package among many devices, except for only a few policies for specific FortiGate devices?

You can perform granular installation targets per rule in the actual policy by clicking Install On column.

This allows you to target devices to add, remove, or set to defaults.

So, by using an installation target, you can share a policy package among multiple devices, and define rules per device in the policy. Shared policy packages are helpful in environments where many devices need to share common policies (except for a few policies that you can target per device) and eliminate the need for multiple policy packages.



Question: 43

Refer to the exhibit

```
FortiManager # diagnose dvm device list
--- There are currently 1 devices/vdoms managed ---

TYPE          OID    SN      HA    IP          NAME          ADOM      IPS          FIRMWARE
fmg/faz enabled 157  FGVM01.. -    10.200.1.1   Local-FortiGate My_ADOM   14.00641 (regular) 6.0 MR2 (866)
|- STATUS: dev-db: modified; conf: in sync; cond: pending; dm: retrieved; conn: up

|- vdom:[3]root flags:0 adom:My_ADOM pkg:[imported]Local-FortiGate
```

Which two statements about the output are true?

- A. Configuration changes directly made on FortiGate have been automatically updated to device-level database
- B. The latest history for the managed FortiGate does not match with the device-level database
- C. The latest revision history for the managed FortiGate does match with the FortiGate running configuration
- D. Configuration changes have been installed to FortiGate and represents FortiGate configuration has been changed

Answer: B, C

Explanation:

Use the diagnose dvm device list command to display details of all managed and unregistered devices. Each FortiGate is assigned an object ID (OID), and its configuration is stored in its own device-level database (dev-db).

In this example, changes have been made in FortiManager to the device-level settings. That is why the CLI output is showing dev-db: modified and the cond is showing as pending. After you install the changes on FortiGate, the output displays dev-db: not modified and cond: OK.

This command also shows whether the FGFM tunnel between FortiGate and FortiManager is up or down.

Object ID is used to reference the managed device for various CLI commands

HA: Blank if in standalone mode, otherwise FGT HA mode is displayed

FortiGate connecting IP address

Name of the ADOM

Firmware version running on FortiGate

```

FortiManager # diagnose dvm device list
--- There are currently 1 devices/vdoms managed ---
--- There are currently 1 devices/vdoms count for license ---

TYPE      OID      SN      HA      IP      NAME      ADOM      IPS      FIRMWARE
fmgfaz-managed 170    FGVM010000077646 -    10.0.1.200  ISEW      ADOM2    6.00741 (regular) 7.0 MR4 (2463)
|- STATUS: dev-db: modified, conf: in sync, cond: pending, dm: retrieved, conn: up
|- vdom: [?] root flags:0 adom:ADOM2 pkg:[imported] ISEW root
  
```

Device setting status
Configuration changes made on FortiManager

Sync status
Latest revision history in sync with the FortiGate running configuration

Config status
Configuration changes need to be installed

Connection status
with managed device

Question: 44

View the following exhibit

When using Quick Install (Device DB) option to install configuration changes to manage FortiGate, which of the following statements are true? (Choose two)

- A. Installs device-level changes to FortiGate without launching the Install Wizard
- B. Will not create new revision in the revision history
- C. Once initiated, the install process cannot be canceled and changes will be installed on the managed device
- D. Provides the option to preview configuration changes prior to installing them

Answer: A, C

Explanation:

Wizards

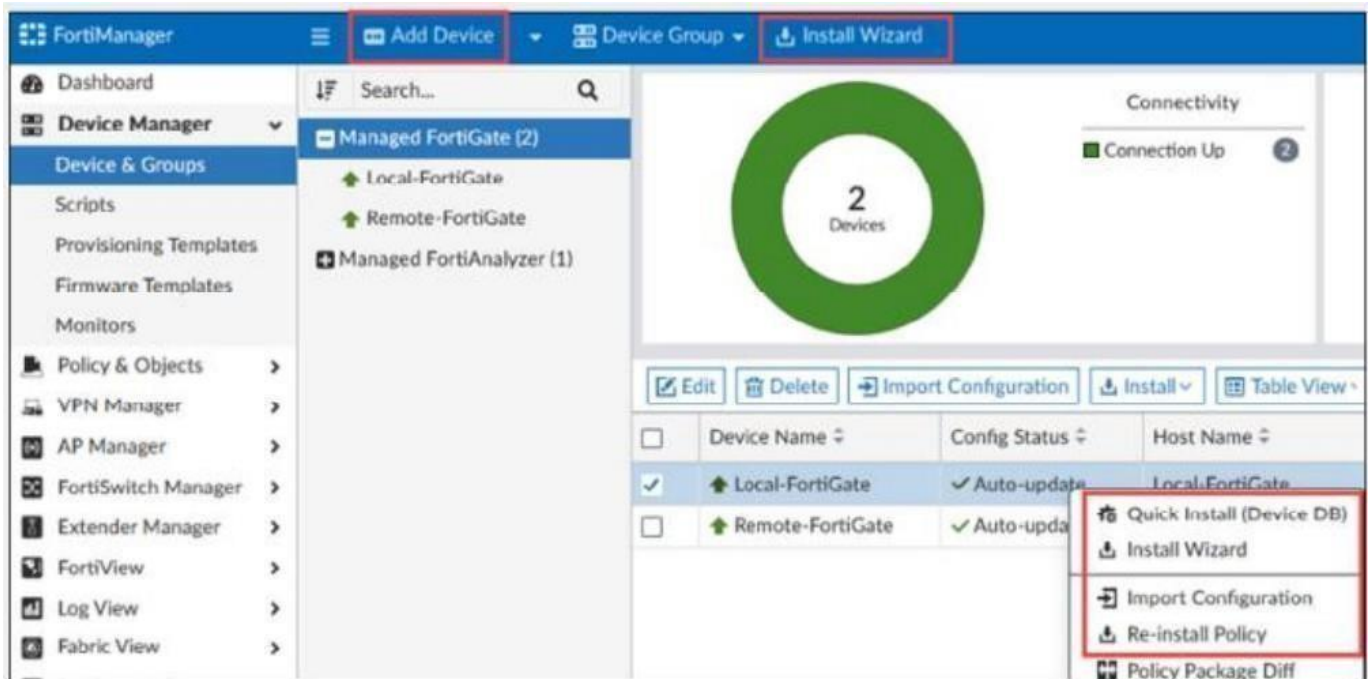
Under Device & Groups you can find several device and installation wizards to help you perform administrative and maintenance tasks. Using these tools can help you shorten the amount of time it takes to do many common tasks.

There are 4 main wizards:

- **Add Device:** Adds device to central management and imports their configurations
- **Install Wizard:** Installs changes to device settings and/or policies to the managed devices. It allows you to preview the changes and, if the administrator doesn't agree with the changes, cancel and modify them.
- **Import Configuration:** Imports interface mapping, policy database, and objects associated with the managed devices into a policy package under the Policy & Objects pane. You can run it while

using the Add Device wizard and any time from the managed device list.

- Re-install Policy: Installs the policy package already assigned to the device quickly. It also shows you a preview of the changes that will be installed on the managed device.
- Quick Install (Device DB): Pushes device configuration from the FortiManager device layer to a FortiGate device. This operation does not have an installation preview, and you cannot cancel this operation.



Question: 45

What is the purpose of the Policy Check feature on FortiManager?

- A. To find and merge duplicate policies in the policy package
- B. To find and delete disabled firewall policies in the policy package
- C. To find and provide recommendation for optimizing policies in a policy package
- D. To find and provide recommendation to combine multiple separate policy packages into one common policy package

Answer: C

Explanation:

Policy Check

Policy Check provides recommendations only on what improvements can be made - it does not perform any changes. Helps you to optimize firewall rules to potentially reduce the size of policy package database.

It uses an algorithm to evaluate policy objects based on:

- Source and destination interface policy objects
- Source and destination address policy objects

- Service and schedule policy objects

Policy Check for:

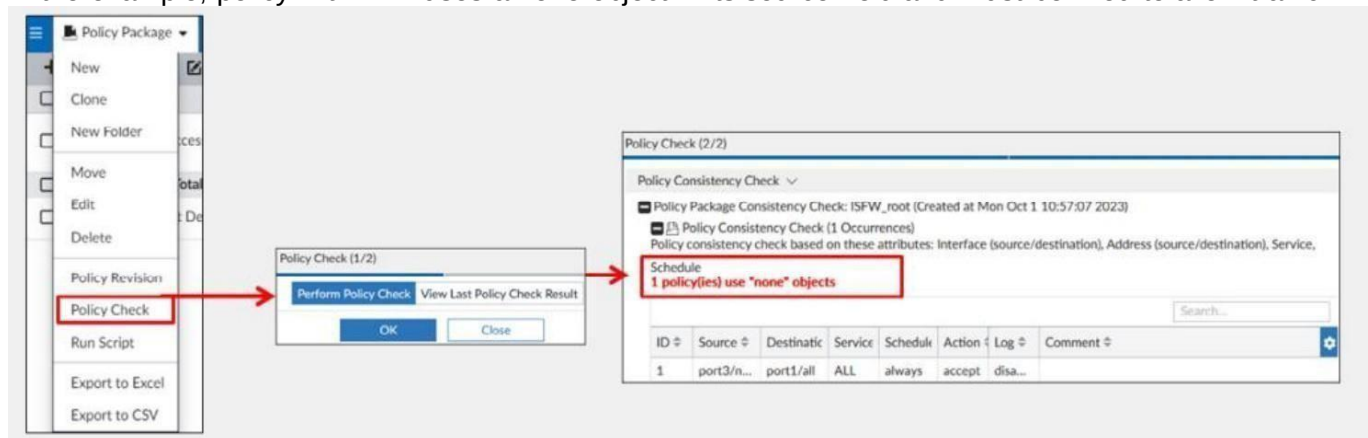
- Duplication, where two objects have identical definitions
- Shadowing, where one object completely shadows another object of the same type
- Overlap, where one object partially overlaps another object of the same type

Orphaning, where an object has been defined, but has not been used anywhere

To perform policy check, select a policy package, and then, in the Policy Package drop-down list, click Policy Check. In the Policy Check dialog box, you can select one of the following options:

Perform Policy Check: This performs a policy check for consistency and provides any conflicts that may prevent your devices from passing traffic.

View Last Policy Check Result: This allows you to view the results of the most recent consistency check. In the example, policy with ID 1 uses a none object in its source field and must be fixed to allow traffic.



Question: 46

Refer to exhibit

Start to import config from device(Local-FortiGate) vdom(root) to adom(My_ADOM), package(Local-FortiGate_root)

"firewall service category",SKIPPED,"(name=General, oid=697, DUPLICATE)"

"firewall address",SUCCESS,"(name=LOCAL_SUBNET, oid=684, new object)"

"firewall service custom",SUCCESS,"(name=ALL, oid=863, update previous object)"

"firewall policy",SUCCESS,"(name=1, oid = 1090, new object)"

Which statement about the object named ALL is true?

- A. FortiManager updated the object ALL using the FortiGate value in its database
- B. FortiManager created the object ALL as a unique entity in its database, which can be only used by this managed FortiGate
- C. FortiManager installed the object ALL with the updated value
- D. FortiManager updated the object ALL using the FortiManager value database

Answer: A

Explanation:

After the import is complete, the wizard provides the Policy Import Summary and the Download Import Report. You can download the import report, which is only available on the Import Device page. You can view the report using any text editor.

As a best practice, you should download the report.

The import report provides information about FortiGate, the ADOM name on FortiManager, and the policy package name. The report also provides additional information, such as the objects that have been added as new objects. Existing objects that have the same values locally on FortiGate and FortiManager are referred to as DUPLICATE. If the value of an existing object is changed, FortiManager updates that in its database and shows update previous object in the import report.



- You can save import report in .txt format

```
Start to import config from device(Local-FortiGate) vdom(root) to adom(root),
package(Local-FortiGate)
"application list",SUCCESS,"(name=wifi-default, oid=2459, update previous object)"
...
"authentication setting",SUCCESS,"(name=, oid=3393, new object) "
...
"firewall address",SKIPPED,"(name=all, oid=2264, DUPLICATE) "
"firewall address",SKIPPED,"(name=none, oid=2265, DUPLICATE) "
"firewall address",SUCCESS,"(name=REMOTE_ETH1, oid=2266, new object) "
"firewall address",SUCCESS,"(name=REMOTE_SUBNET, oid=2267, new object) "
```

Question: 47

Which two methods can use to link the model device to the real device?

- A. License number and IP address
- B. Serial number and pre-shared key
- C. Only serial number
- D. IP address and serial number

Answer: B

Explanation:

Add Device Wizard - Model Device

Supports zero-touch, on-site FortiGate deployment by automatically promoting a model device to a managed device.

Provisions a device in FortiManager, that is not yet online.

You can link a device by either:

- Its serial number: The serial number of the device you want to add.
- A pre-shared key: A unique pre-shared key for each device you want to add. Useful when the exact device model is not known.

Zero-touch provisioning involves the following general steps:

- Add model device and configure settings in FortiManager
- Connect FortiGate to the network and configure it for central management
- Confirm deployment is successful (it may take several minutes)

Question: 48

View the following exhibit:

Create New CLI Script

Script Name	Config	
Comments	Write a comment	0/255
Type	CLI Script	
Run Script on	Remote FortiGate Directly(via CLI)	

[\[View Sample Script\]](#)

Script Detail

```
config vpn ipsec phase1-interface
edit "H25_0"
set auto-discovery-sender enable
next
end
config system interface
edit "H25_0"
set vdom "root"
set ip 172.16.1.1 255.255.255.255
set remote-ip 172.16.1.254
next
end
config router bgp
set as 65100
set router-id 172.16.1.1
config neighbor-group
```

Advanced Device Filters

Which of the following statements are true if the scripts is executed using Remote FortiGate Directly (via

CLI) option? (Choose two)

- A. FortiGate will auto-update the FortiManager's device-level database
- B. You must install these changes using Install Wizard
- C. FortiManager will create a new revision history
- D. FortiManager provides a preview of CLI commands before executing the script on managed FortiGate

Answer: A, C

Explanation:

FortiManager creates a configuration revision when:

- FortiGate is added to FortiManager
- Device changes are installed from FortiManager
- FortiGate configurations is retrieved from FortiManager
- A local change on FortiGate causes an automatic update
- Scripts run directly on remote devices also cause automatic updates and create a revision history

Scripts can be run on:

- Device Database
- Policy Package or ADOM Database
- Remote FortiGate Directly (via CLI)

A script can be executed directly on the device, and you don't need to install the changes using the installation wizard.

As the changes are directly installed on the managed device, no option is provided to check the configuration changes through FortiManager before executing them.

Question: 49

What does a policy package status of Conflict indicate?

- A. The policy package does not have a FortiGate as the installation target
- B. The policy configuration has never been imported after a device was registered on FortiManager
- C. The policy package configuration has been changed on both FortiManager and the managed device independently
- D. The policy package reports inconsistencies and conflicts during a Policy Consistency Check

Answer: C

Explanation:

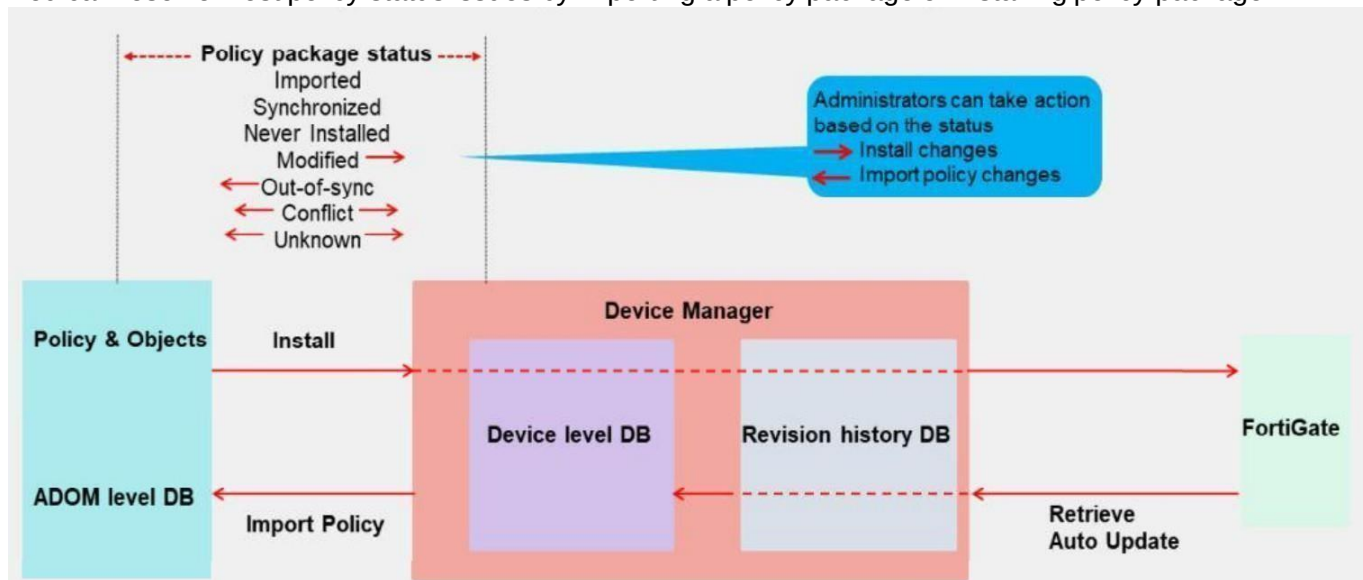
Status of Policy Package on FortiManager

After every retrieve, auto-update, and installation operation, FortiManager stores the FortiGate configuration in the revision history.

The illustration on this slide shows the status of the policy package:

- Imported: Indicates that a policy package was successfully imported for managed device.
- Synchronized: Indicates that a policies and objects are synchronized between FortiManager and the managed device.
- Never Installed: Policy package was never created, hence it was never imported for a managed device.
- Modified: Policy package configuration is changed on FortiManager and changes have not yet pushed to the managed device
- Out-of-sync: The latest policy package does not match the policies and objects configuration on the latest revision history because of configuration changes made locally on FortiGate or a previous partial installation failure. You should perform a retrieve, and then import policies from FortiManager.
- Conflict: If you make policy configuration changes locally on FortiGate and don't import the changes into the policy packages, and you also made the changes on FortiManager, the status enters conflict state. Depending on the configuration changes, you can either import a policy package or install the changes from FortiManager.
- Unknown: FortiManager is unable to determine the policy package status.

You can resolve most policy status issues by importing a policy package or installing policy package.



Question: 50

Refer to the exhibit

Edit Address

Address Name

LAN

Type

IP/Netmask

IP/Netmask

192.168.1.0/255.255.255.0

Interface

any

Static Route Configuration

OFF

Comments

Add to Groups

Click to add ...

Advanced Options >

Per-Device Mapping

ON

+ Add

Edit

Delete

	Name	VDOM	Details
☐	Remote-FortiGate	root	IP/Netmask:10.200.1.0/255.255.255.0

An administrator has created a firewall address object which is used in multiple policy package for multiple FortiGate devices in an ADOM.

When the installation operation is performed, which IP/Netmask will be installed on managed device for the firewall address object?

- A. The FortiManager administrator can choose the value for the firewall address object in the Install Wizard for Remote-FortiGate
- B. 10.200.1.0/24 on Remote-FortiGate
- C. 192.168.0.1/24 on Remote-FortiGate
- D. If no dynamic mapping is defined for other FortiGate devices, the object will not be installed

Answer: B

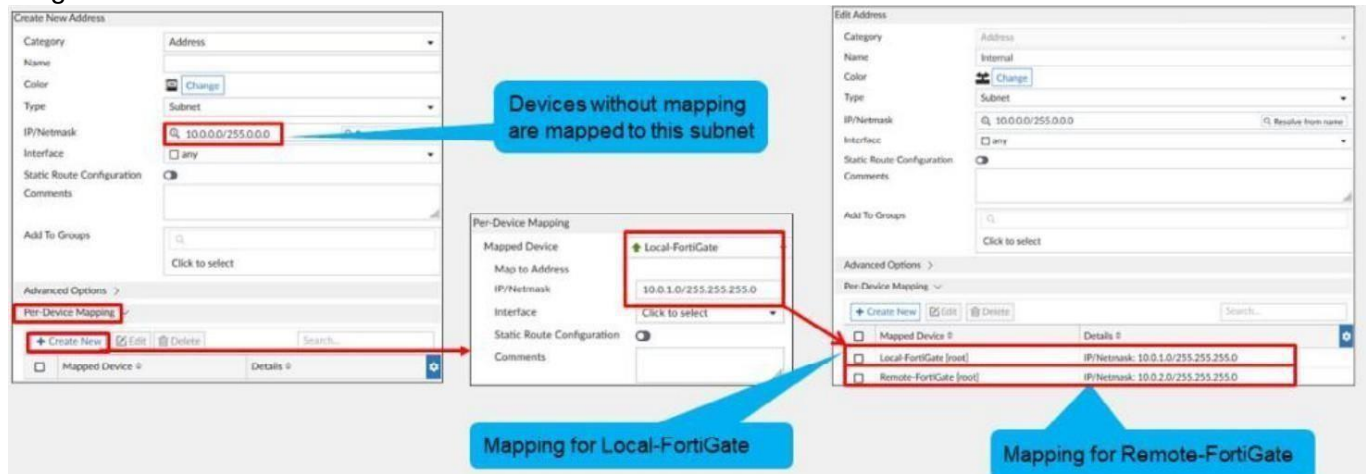
Explanation:

Dynamic Objects

All objects in an ADOM are managed by a single database unique to the ADOM. Many objects now include the option to enable dynamic mapping. You can use dynamic objects to map a single logical object to a unique definition per device. You can dynamically map common features such as addresses, interfaces, virtual IPs, and IP pools. A common example is a firewall address. You may have a common name for an address object, but have different value depending on the device it is installed on.

In the example shown on this slide, the dynamic address object Internal refers to the internal network address of the managed firewalls. The object has a default value of 10.0.0.0/8. The mapping rules are defined per device. For Remote-FortiGate, the address object Internal refers to 10.0.2.0/24, whereas for Local-FortiGate the same object refers to 10.0.1.0/24. The device in the ADOM that do not have dynamic mapping for Internal have a default value of 10.0.0.0/8

To add devices for dynamic mapping, select Per-Device Mapping arrow, and then, in the Per-Device Mapping section, click Create New. In the pop-up window that opens, select the device and set the IP ranges/subnet.



Question: 51

You are moving managed FortiGate devices from one ADOM to a new ADOM. Which statement correctly describes the expected result?

- A. Policy packages will be imported into the new ADOM automatically
- B. Any pending device settings will be installed automatically
- C. The shared policy package will not be moved to the new ADOM
- D. Any unused objects from a previous ADOM are moved to the new ADOM automatically

Answer: C

Explanation:

Moving FortiGate From One ADOM to Another

Considerations before moving devices:

When you move a device from one ADOM to another, policies and objects (used and unused) don't move to the new ADOM.

If using a shared policy package, it is not moved to the new ADOM

When FortiGate device is upgraded, it is best to keep them in the same ADOM and use ADOM upgrade

Moving FortiGate devices to a new ADOM introduces additional work and certain complications.

After moving the devices:

If you need to move a device from one ADOM to another, run Import Configuration wizard to import the policy package into the new ADOM.

Can use CLI to import unused objects if needed

execute fmpolicy copy-adom-object

Configuration of manager panes like VPN manager does not move

FortiManager administrator must reconfigure all the VPN configuration

Question: 52

Which of the following statements are true regarding ADOM revisions? (Choose two)

- A. ADOM revisions can save the current state of all policy packages and objects for an ADOM
- B. ADOM revisions can create System Checkpoints for the FortiManager configuration
- C. ADOM revisions can save the current size of the whole ADOM
- D. ADOM revisions can significantly increase the size of the configuration backups

Answer: A,D

Explanation:

ADOM Revisions

ADOM revisions allow you to save locally in FortiManager snapshots of the policy packages, objects, and VPN console settings contained in an ADOM.

Warning: Keep in mind that ADOM revisions can significantly increase the size of configuration backup.

Question: 53

When installation is performed from the FortiManager, what is the recovery logic used between FortiManager and FortiGate for an FGFM tunnel?

- A. FortiManager will not push the CLI commands as a part of the installation that will cause the tunnel to go down.
- B. FortiManager will revert and install a previous configuration revision on the managed FortiGate
- C. FortiGate will reject the CLI commands that will cause the tunnel to go down.
- D. After 15 minutes, FortiGate will unset all CLI commands that were part of the installation that caused the tunnel to go down

Answer: D

Explanation:

Recovery Logic

When an installation is performed from FortiManager to FortiGate, FortiManager always tries to ensure connectivity with the managed FortiGate device. If the connection fails FortiManager tries to recover the FGFM tunnel by unsetting the command that caused the tunnel to go down.

To make configuration changes to FortiGate, FortiManager sends set and unset CLI commands to the managed FortiGate device:

Set commands, needed to apply the configuration changes

Unset commands, to recover the configuration changes

When applying changes, FortiGate:

Apply the set commands, using memory only, nothing written to a configuration file

Test the FGFM connection to FortiManager

If the connection fails to reestablish, FortiGate applies the unset command after 15 minutes (not configurable and not based on stock timeout values).

If the connection fails remains down, FortiGate reboots and recovers the previous configuration from its configuration file (optional)

Can be enabled from the FortiManager CLI

```
config system dm
set rollback-allow-reboot enable
end
```

Note: Default disabled

Question: 54

View the following exhibit

Edit Address

Address Name

Training

Type

IP/Netmask

IP/Network

192.168.1.0/255.255.255.0

Interface

any

Static Route Configuration

OFF

Comments

0/255

Add to Groups

Click to add

Advanced Options >

Per-Device Mapping

ON

+ Add  Edit  Delete

☐

Name

VDOM Details

☐

Local-FortiGate root

IP/Netmask 10.0.10/255.255.255.0

An administrator has created a firewall address object, Training, which is used in the Local-FortiGate policy package. If the Training firewall address object is used in different managed device other than Local-FortiGate, which IP Netmask will be installed?

- A. It will create firewall address group on Local-FortiGate with 192.168.0.1/24 and 10.0.1.0/24 object values
- B. 10.0.1.0/24
- C. Local-FortiGate will automatically choose an IP Network based on its network interface settings
- D. 192.168.1.0/24

Answer: D

Explanation:

All objects in an ADOM are managed by a single database unique to the ADOM. Many objects now include the option to enable dynamic mapping. You can use dynamic objects to map a single logical object to a unique definition per device. You can dynamically map common features such as addresses, interfaces, virtual IPs, and IP pools. A common example is a firewall address. You may have a common name for an address object, but have different value depending on the device it is installed on.

In the example shown on the next slide, the dynamic address object Internal refers to the internal network address of the managed firewalls. The object has a default value of 10.0.0.0/8. The mapping rules are defined per device. For Remote-FortiGate, the address object Internal refers to 10.0.2.0/24, whereas for Local-FortiGate the same object refers to 10.0.1.0/24. The device in the ADOM that do not have dynamic mapping for Internal have a default value of 10.0.0.0/8

To add devices for dynamic mapping, select Per-Device Mapping arrow, and then, in the Per-Device Mapping section, click Create New. In the pop-up window that opens, select the device and set the IP ranges/subnet.

The screenshot displays the FortiGate configuration interface for the 'Internal' address object. It is divided into three main sections:

- Create New Address:** Shows the 'Internal' address object with a default IP/Netmask of 10.0.0.0/255.0.0.0. A red box highlights the 'Per-Device Mapping' section, and a blue callout points to it with the text 'Devices without mapping are mapped to this subnet'.
- Per-Device Mapping:** A table showing mappings for different devices. A red box highlights the 'Local-FortiGate' entry, and a blue callout points to it with the text 'Mapping for Local-FortiGate'.
- Edit Address:** Shows the 'Internal' address object with a default IP/Netmask of 10.0.0.0/255.0.0.0. A red box highlights the 'Per-Device Mapping' section, and a blue callout points to it with the text 'Mapping for Remote-FortiGate'.

Mapped Device	Details
Local-FortiGate (root)	IP/Netmask: 10.0.1.0/255.255.255.0
Remote-FortiGate (root)	IP/Netmask: 10.0.2.0/255.255.255.0

Question: 55

An administrator would like to create an SD-WAN using central management in the Training ADOM. To create SD-WAN using central management, which two steps must be completed? (Choose two)

- A. Enable SD-WAN central management in the Training ADOM
- B. Configure and install the SD-WAN firewall policy and SD-WAN static route before installing the SD-WAN template settings
- C. Specify a gateway address when you create a default SD-WAN static route
- D. Remove all the interface references such as routes or policies that will be a part of SD-WAN member interfaces

Answer: A,B

Explanation:

Enabling central SD-WAN management

Central SD-WAN management can be enabled per ADOM. When enabled, the SD-WAN tab shows the following items on the left pane:

- Assigned Devices
- SD-WAN Templates
- Interface Members
- Health-Check Servers
- Monitor

When you create a SD-WAN configuration using central management, it's necessary to remove all the interface references such as routes or policies that will be a part of SD-WAN member interfaces

Question: 56

Refer to the exhibit.

Import Device - Local-FortiGate [root]

Create a new policy package for import.

Policy Package Name	Local-FortiGate
Folder	root
Policy Selection	☒ Import All (3) ☐ Select Policies and Profile Groups to Import
Object Selection	☐ Import only policy dependent objects ☒ Import all objects

An administrator is importing a new device to FortiManager and has selected the options shown in the exhibit.

What will happen if the administrator makes changes and installs the modified policy package on this managed FortiGate?

- A. The unused objects that are not tied to the firewall policies will be installed on FortiGate
- B. The unused objects that are not tied to the firewall policies will remain as read-only locally on

FortiGate

- C. The unused objects that are not tied to the firewall policies locally on FortiGate will be deleted
- D. The unused objects that are not tied to the firewall policies in policy package will be deleted from the FortiManager database

Answer: C

Explanation:

Import Device - Local-FortiGate [root]

Create a new policy package for import.

Policy Package Name: Local-FortiGate

Folder: root

Policy Selection:

- ☒ Import All (3)
- ☐ Select Policies and Profile Groups to Import

Object Selection:

- ☐ Import only policy dependent objects
- ☒ Import all objects

You can choose whether to import all configured objects, or only the objects referenced by the current firewall policies. Regardless of whether you choose to import only policy-dependent object or all objects, the system will delete orphan (unused) objects that are not tied to policies locally on FortiGate in the next installation.

But if you choose to import all objects, then the system imports all used and unused objects in the FortiManager ADOM object database and can use them later by referencing the policies on FortiManager and installing them on the managed devices.

When occurs the first installation in any device, you may see many changes, because objects may be renamed during the import process and unused objects removed from the device configuration.

Question: 57

Which statements about a FortiManager HA cluster are true? (Choose two)

- A. Cluster members can be on different locations
- B. A FortiManager HA cluster supports an active-active configuration
- C. FortiGuard database is not synchronized between FortiManager HA cluster members
- D. Global database is not synchronized between FortiManager HA cluster members

Answer: A, C

Explanation:

The HA heartbeat packets use TCP port 5199. FortiManager HA supports geographic redundancy so the primary unit and secondary units can be in different locations attached to different networks as long as

communication is possible between them.

What is synchronized?

All device configuration (including global databases)

All configuration revision

There are a few configuration settings, FortiGuard databases, and logs that are not synchronized between the primary and secondary devices. The FortiGuard databases and packages are downloaded separately, and each device can provide FortiGuard services to managed devices.

Question: 58

An administrator with the Super_User profile is unable to log in to FortiManager because of an authentication failure message. Which troubleshooting step should you take to resolve the issue?

- A. Make sure the administrator IP address is part of the trusted hosts
- B. Make sure Offline Mode is disabled
- C. Make sure FortiManager Access is enabled in the administrator profile
- D. Make sure ADOMs are enabled and the administrator has access to the Global ADOM

Answer: B

Explanation:

Depending on your deployment, you may want to divide FortiManager administrative tasks among multiple employees by creating additional administrator accounts.

You can control and restrict administrator access using several methods, but the following are the must used:

- ADOMs: Administrators have access to only those ADOMs to which they are assigned.
- Administrative profiles: The level of access is determined by the administrative profile selected. When the administrators are restricted to specific ADOMs, you can assign different profiles for each ADOM, or the same profile for all ADOMs.
- Trusted hosts: This option restricts administrators to log in only from specific IP addresses or subnets. The trusted host you define apply to both the GUI and the CLI when accessed through SSH. This feature is available for IPv4 and IPv6 addresses.

• Most used access control methods are:

- ADOMs
- Administrative profiles
- Trusted hosts

Select one or more ADOMs

Select administrative profile type

Trusted hosts restrict login access to specific IP addresses or subnets

System Settings > Administrators

Create New Administrator

User Name: Training

Avatar: T + Add Photo - Remove Photo

Description:

Admin Type: LOCAL

New Password:

Confirm Password:

FortiToken Cloud: Disable FortiToken Mobile Email SMS

Administrative Domain: All ADOMs All ADOMs except specified ones Specify

Admin Profile: Single Per-ADOM Restricted_User

Specify Device Group to Access: All Packages Specify

Policy Package: None

JSON API Access:

Theme Mode: Use Global Theme Use Own Theme

Trusted Hosts: 0.0.0.0/0.0.0.0 255.255.255.255/255.255.255.255

OK Cancel

Question: 59

Refer to exhibit

Create New ADOM

Name: Training

Type: FortiGate 6.2 6.0 5.6 5.4

Comments:

Devices: + Select Device

Name	IP Address	Platform
Click to select devices for this ADOM.		

Mode: ☒ Normal ☐ Backup

Which two statements about ADOM set in Normal mode on FortiManager are true? (Choose two)

- A. It allows making configuration changes managed devices on FortiManager panes
- B. You cannot assign the same ADOM to multiple administrators
- C. FortiManager automatically installs the configuration difference in revision on the managed FortiManager
- D. It supports the FortiManager script feature

Answer: A,D

Explanation:

Normal mode ADOMs

When creating an ADOM in Normal Mode, the ADOM is considered Read/Write, where you are able to make changes to the ADOM and managed devices from the FortiManager. FortiGate units in the ADOM will query their own configuration every 5 seconds. If there has been a configuration change, the FortiGate unit will send a diff revision on the change to the FortiManager using the FGFM protocol.

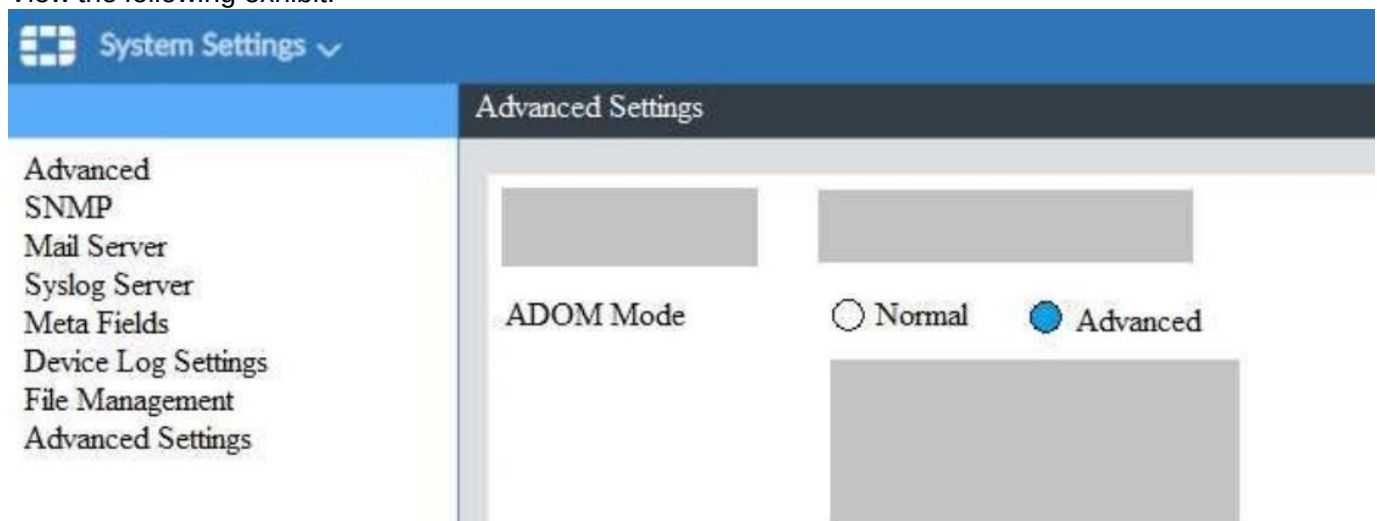
Backup mode ADOMs

When creating an ADOM in Backup Mode, the ADOM is considered Read Only, where you cannot make changes to the ADOM and managed devices from FortiManager. Changes are made via scripts, which are run on the managed device, or through the device's GUI or CLI directly.

The mode can be set to Backup mode enables you to configure an ADOM where all the devices that are added to the ADOM will only have their configuration backed up. Configuration changes cannot be made to the devices in backup ADOM. You can push any existing revisions to managed devices.

Question: 60

View the following exhibit.



Based on the configuration setting, which one of the following statements is true?

- A. The setting disables concurrent ADOM access and adds ADOM locking
- B. The setting allows automatic updates to the policy package configuration for a managed device
- C. The setting enables the ADOMs feature on FortiManager
- D. This setting allows you to assign different VDOMs from the same FortiGate to different ADOMs.

Answer: D

Explanation:

Administrative Domains (ADOMs)

ADOMs allow you to group devices logically for administrators to manage.

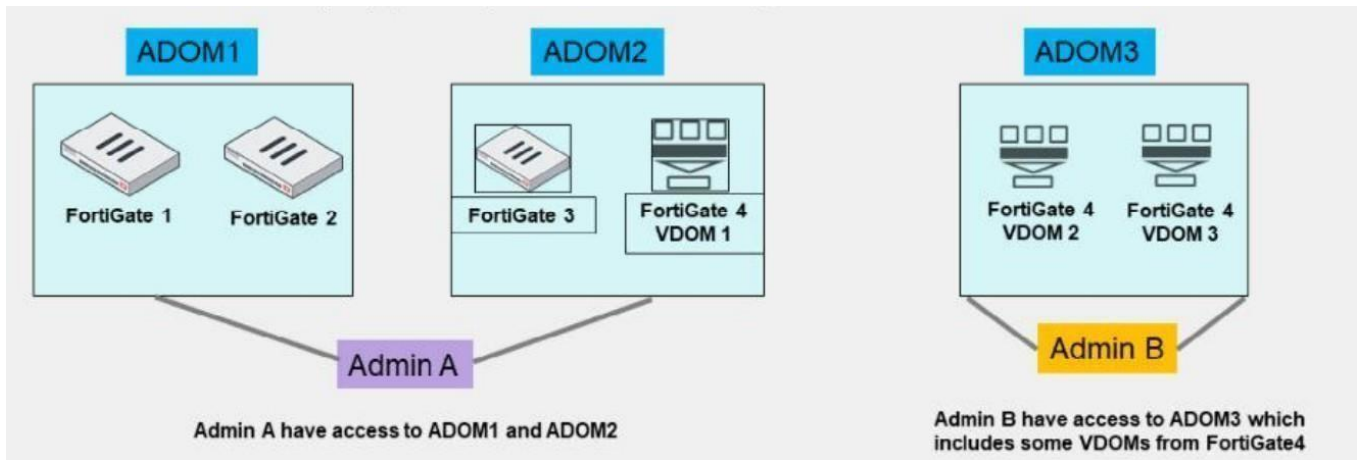
A device is assigned to an ADOM

An administrators are assigned to one or more ADOMs

ADOMs can work in two device modes: normal or advanced

In advanced mode, each FortiGate virtual domain (VDOM) can be assigned to a different ADOM

Accounts with the Super_User profile can manage devices in all ADOMs



Question: 61

Refer to the exhibit

```
config system global
set workspace-mode normal
end
```

Given the configuration shown in the exhibit, which two statements are true? (Choose two)

- A. It disables concurrent read-write access to an ADOM
- B. It allows two or more administrators to make configuration changes at the same time, in the same ADOM
- C. It allows the same administrator to lock more than one ADOM at the same time
- D. It is used to validate administrator login attempts through external servers

Answer: A, C

Explanation:

Enabling workspace mode allows you to lock different items in FortiManager, effectively preventing concurrent read/write access to any locked item. Only the administrator that initiated the lock has read/write access to that item, while all other administrators have read-only access.

You can lock entire ADOMs, as well as specific devices, policy packages, objects, and individual policies.

After you make all required configuration changes, you can unlock the item manually. Another way that you can release the locks is to log out of FortiManager.

- ADOM

🔒
ADOM: My_ADOM
- Specific devices

Device Name
⌵

🔒
Local-FortiGate
- Policy packages

Local-FortiGate_root
⌵

🔒
Firewall Policy

🔒
Installation Targets

Locking an ADOM automatically removes locks on devices and policy packages that you have locked within that ADOM

If another administrator locked devices or policy packages, you cannot lock the ADOM containing those devices or policy packages

Question: 62

Refer to the exhibit.

Starting Log (Run the device)**Start installing**

Local-FortiGate \$ config user device

Local-FortiGate (device) \$ edit "mydevice"

new entry 'mydevice' added

Local-FortiGate (mydevice) \$ next

MAC address can not be 0

Node_check_object fail!for mac 00:00:00:00:00:00

Attribute 'mac' value '00:00:00:00:00:00' checkingfail -33

Command fail. Return code 1

Local-FortiGate (device) \$ end

...

Local-FortiGate \$ config firewall policy

Local-FortiGate (policy) \$ edit 2

New entry '2' added

Local-FortiGate (2) \$ set name "Device_policy"

Local-FortiGate (2) \$ set uuid 64...

Local-FortiGate (2) \$ set srcintf "port3"

Local-FortiGate (2) \$ set dstintf "port1"

Local-FortiGate (2) \$ set srcaddr "all"

Local-FortiGate (2) \$ set dstaddr "all"

Local-FortiGate (2) \$ set action accept

Local-FortiGate (2) \$ set schedule "always"

Local-FortiGate (2) \$ set service "ALL"

Local-FortiGate (2) \$ set devices "mydevice"

Entry not found in datasource

Value parse error before 'mydevice'

Command fail. Return code -3

Local-FortiGate (2) \$ set nat enable

Local-FortiGate (2) \$ next

Local-FortiGate (policy) \$ end

...

An administrator has configured the command shown in the exhibit on FortiManager. A configuration change has been installed from FortiManager to the managed FortiGate that causes the FGFM tunnel to go down for more than 15 minutes. What is the purpose of this command?

- A. It allows FortiGate to reboot and restore a previously working firmware image
- B. It allows FortiGate to reboot and recover the previous configuration from its configuration file
- C. It allows the FortiManager to revert and install a previous configuration revision on the managed FortiGate
- D. It allows FortiGate to unset central management settings

Answer: B

Explanation:

Recovery Logic

When an installation is performed from FortiManager to FortiGate, FortiManager always tries to ensure connectivity with the managed FortiGate device. If the connection fails FortiManager tries to recover the FGFM tunnel by unsetting the command that caused the tunnel to go down.

To make configuration changes to FortiGate, FortiManager sends set and unset CLI commands to the managed FortiGate device:

Set commands, needed to apply the configuration changes

Unset commands, to recover the configuration changes

When applying changes, FortiGate:

Apply the set commands, using memory only, nothing written to a configuration file

Test the FGFM connection to FortiManager

If the connection fails to reestablish, FortiGate applies the unset command after 15 minutes (not configurable and not based on stock timeout values).

If the connection fails remains down, FortiGate reboots and recovers the previous configuration from its configuration file (optional)

Can be enabled from the FortiManager CLI

```
config system dm
set rollback-allow-reboot enable
end
```

Note: Default disabled

Question: 63

What happens to a device Config Status if you execute a script using the Remote FortiGate Directly (via CLI) option?

- A. It will be tagged as Modified
- B. It will be tagged as Auto-Updated
- C. It will be tagged as Processing
- D. It will be tagged as Pending

Answer: B

Explanation:

Remote FortiGate Directly (via CLI)

A script can be executed directly on the device, and you don't need to install the changes using the installation wizard.

As the changes are directly installed on the managed device, no option is provided to check the

configuration changes through FortiManager before executing them.
Managed device will be tagged as Auto-Updated

Question: 64

Which of the following statements are true regarding SD-WAN Central Management? (Choose three)

- A. SD-WAN settings can be installed on multiple FortiGate devices at the same time
- B. The first step in creating an SD-WAN using FortiManager is to create two SD-WAN firewall policies.
- C. You can create multiple SD-WAN interfaces per VDOM
- D. When you configure an SD-WAN, you must specify at least two member interfaces
- E. SD-WAN must be enabled on per-ADOM basis

Answer: A, D, E

Explanation:

Central SD-WAN management can be enabled per ADOM. When enabled, the SD-WAN tab shows the following items on the left pane:

Assigned Devices

SD-WAN Templates

Interface Members

Health-Check Servers

Monitor

SD-WAN templates help you do the following:

Deploy a single SD-WAN template from FortiManager across multiple FortiGate devices.

Perform a zero-touch deployment without manual configuration locally at the FortiGate devices.

Roll out a uniform SD-WAN configuration across your network.

Eliminate errors in SD-WAN configuration across multiple FortiGate devices since the SD-WAN template is applied centrally from FortiManager.

Monitor network Performance SLA across multiple FortiGate devices centrally from FortiManager.

Monitor the performance of your SD-WAN with multiple views.

Question: 65

View the following exhibit.

Starting Log (Run the device)**Start installing****Local-FortiGate \$ config user device****Local-FortiGate (device) \$ edit "mydevice"****new entry 'mydevice' added****Local-FortiGate (mydevice) \$ next****MAC address can not be 0****Node_check_object fail!for mac 00:00:00:00:00:00****Attribute 'mac' value '00:00:00:00:00:00' checkingfail -33****Command fail. Return code 1****Local-FortiGate (device) \$ end****...****Local-FortiGate \$ config firewall policy****Local-FortiGate (policy) \$ edit 2****New entry '2' added****Local-FortiGate (2) \$ set name "Device_policy"****Local-FortiGate (2) \$ set uuid 64...****Local-FortiGate (2) \$ set srcintf "port3"****Local-FortiGate (2) \$ set dstintf "port1"****Local-FortiGate (2) \$ set srcaddr "all"****Local-FortiGate (2) \$ set dstaddr "all"****Local-FortiGate (2) \$ set action accept****Local-FortiGate (2) \$ set schedule "always"****Local-FortiGate (2) \$ set service "ALL"****Local-FortiGate (2) \$ set devices "mydevice"****Entry not found in datasource****Value parse error before 'mydevice'****Command fail. Return code -3****Local-FortiGate (2) \$ set nat enable****Local-FortiGate (2) \$ next****Local-FortiGate (policy) \$ end****...**

Which statement is true regarding this failed installation log?

- A. Policy ID 2 is installed without a source device
- B. Policy ID 2 will not be installed

- C. Policy ID 2 is installed without a source address
- D. Policy ID 2 is installed in disabled state

Answer: A

Explanation:

In this scenario, the administrator tried to create a new object device entry called "mydevice" but not set a valid mac value in its information, hence the object was not created. After, when the policy ID 2 was created it try to set the object called "mydevice" in its source device, but this object doesn't exist, hence the policy was installed without a source device.

Question: 66

Which API is available on FortiManager?

- A. SOAP API
- B. JSON API
- C. UML API
- D. RPC API

Answer: B

Explanation:

FortiManager - Application Program Interface (API)

The FortiManager JSON API allows you to perform configuration and monitoring operations on a FortiManager appliance or VM. The JSON API is based on JSON-RPC, a remote procedure call protocol encoded in JSON. The API allows you to do many of the same tasks as the FortiManager GUI using FortiPortal or other third-party applications.

Question: 67

An administrator run the reload failure command: diagnose test deploymanager reload config on FortiManager. What does this command do?

- A. It downloads the latest configuration from the specified FortiGate and performs a reload operation on the device database
- B. It installs the provisioning template configuration on the specified FortiGate
- C. It compares and provides differences in configuration on FortiManager with the current running configuration of the specified FortiGate
- D. It installs the latest configuration on the specified FortiGate and update the revision history database

Answer: A

Explanation:

Failed Reload

In this example, the configuration is correctly retrieved and saved in te revision history; however, the problem occurs when updating the device database.

An operation that fails to update the device-level database from the revision history database.

You can troubleshoot reload failures to see at which stage the configuration is failing to load into the device-level database.

Caused by inconsistent or corrupt FortiGate configuration

Can be due to not following proper upgrade path

When you execute the reload failure command, FortiManager connects to FortiGate and downloads its configuration file. Then, FortiManager performs a reload operation on the device database.

diagnose test deployment reloadconf

```
# diagnose test deploymanager reloadconf 183
Retrieving configuration file from FGT...
Configuration file import succeeded.
Reloading configuration file...
Configuration reload succeeded.
```

There are two possible outcomes:

If there are no errors in the FortiGate configuration, the reload is successful, and the device-level database is updated with the FortiGate configuration. However, note that a new revision history is not created.

If there are errors in the FortiGate configuration, the output of the reload command indicates the point in the configuration at which the device-level database failed to update.

Question: 68

What are the factory default settings on FortiManager? (Choose three)

- A. Reports and Event Monitor panes are enabled
- B. Username is admin
- C. Port1 Interface IP address is 192.168.1.99/24
- D. FortiAnalyzer features are disabled
- E. Password is fortinet

Answer: B, C, D

Explanation:

Factory Default Settings

Use factory default settings to initially log in to your FortiManager and bring your own network configuration.

Default settings:

Port 1 is default management port

Port1 interface IP: 192.168.1.99/24

HTTPS and SSH protocols are enabled for management access

admin/blank

You will be prompted to set a new password

FortiAnalyzer features are disabled by default. If FortiAnalyzer features are enabled, you cannot add FortiAnalyzer to FortiManager. You will also not be able to configure FortiManager high availability (HA).

Question: 69

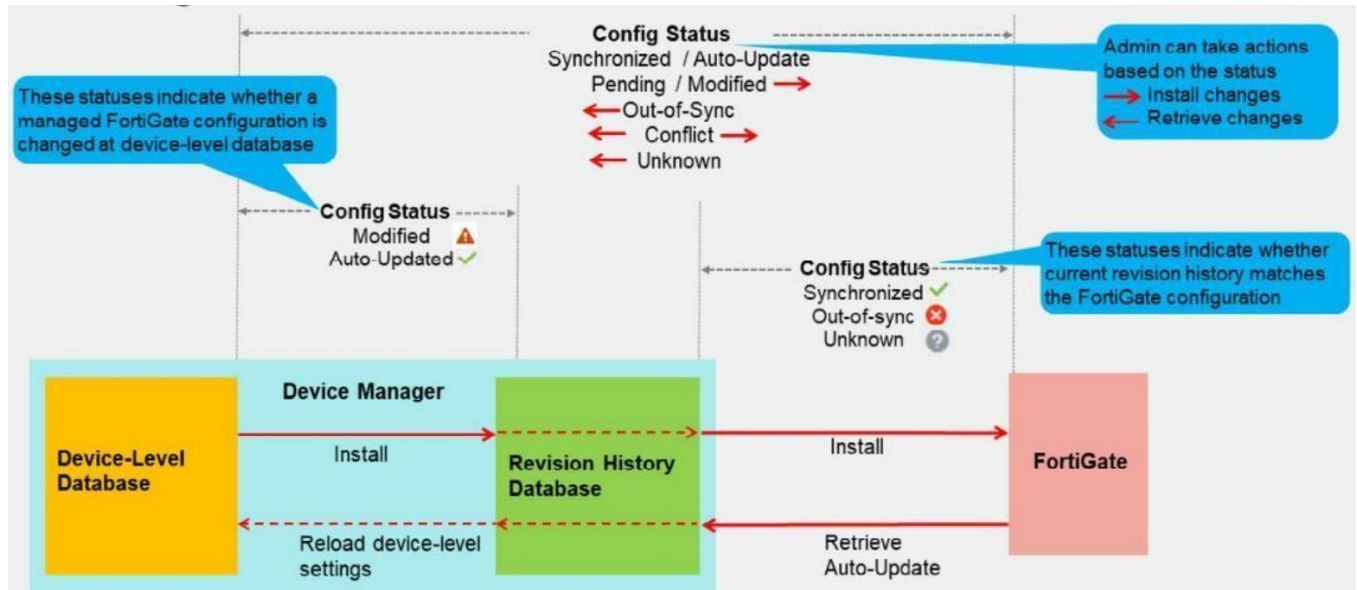
How does FortiManager determine if the configuration status of a managed device is Out-of-Sync?

- A. It compares the last ADOM revision with the FortiGate configuration
- B. It compares the device-level database with the FortiGate configuration
- C. The FortiGate is not reachable
- D. It compares the current revision history with the FortiGate configuration

Answer: D

Explanation:**Out-of-Sync**

The latest revision history configuration entry does not match the configuration on the FortiGate due to configuration changes made locally on FortiGate or a previous partial install failure. It is recommended that you perform a retrieve from the FortiManager.

**Question: 70**

An administrator would like to create an SD-WAN default static route for a newly created SD-WAN using the FortiManager GUI. Both port1 and port2 are part of the SD-WAN member interfaces. Which interface must the administrator select in the static route device drop-down list?

- A. virtual-wan-link
- B. auto-discovery
- C. Port1
- D. Port2

Answer: B**Explanation:**

Understanding the Failed Installation Log

CLI commands were sent to FortiGate, however FortiGate rejected specific commands because of missing virtual-wan-link interface.

Failed installation:

```

Starting log (Run on device)Start installing
Local-FortiGate $ config firewall policy
Local-FortiGate (policy) $ edit 4
Local-FortiGate (4) $ set name "sd-wan"
Local-FortiGate (4) $ set uuid 9f4cf75a-1de9-51eb-dcdc-48868fd5ca82
Local-FortiGate (4) $ set srcintf "port3"
Local-FortiGate (4) $ set dstintf "virtual-wan-link"
node_check_object fail! for name virtual-wan-link

value parse error before 'virtual-wan-link'
Command fail. Return code -651
Local-FortiGate (4) $ set srcaddr "all"
Local-FortiGate (4) $ set dstaddr "all"
Local-FortiGate (4) $ set action accept
Local-FortiGate (4) $ set schedule "always"
Local-FortiGate (4) $ set service "ALL"
Local-FortiGate (4) $ set nat enable
Local-FortiGate (4) $ next
Attribute 'dstintf' MUST be set.
Command fail. Return code 1
Local-FortiGate (policy) $ end

```

New firewall policy named sd-wan failed to install due to unavailable interface

In this case, a new firewall policy named sd-wan was not added (failed) because of an unavailable virtual-wan-link interface on FortiGate.

In this scenario, the administrator did not follow the correct steps to enable the SD-WAN interface before pushing the SD-WAN policy, and FortiGate rejected the addition of the SD-WAN firewall policy.

Question: 71

Refer to the exhibit:

```

# execute top
%Cpu(s):  0.0 us,   6.2 sy,   0.0 ni,   0.0 id, 93.8 wa  ...
PID USER      PR  NI   VIRT  RES  %CPU  %MEM   TIME+ S  COMMAND
 1163 root        20   0   17.6m   2.1m   7.1   0.1   0:00.05 R   top
    1 root        20   0  602.2m  14.9m   0.0   0.7   0:11.67 S   /bin/initXXXXXXXXXXXX
    2 root        20   0    0.0m   0.0m   0.0   0.0   0:00.00 S   [kthreadd]
28993 root        20   0  676.8m  27.8m  87.0   1.4  686:18.68 S   ../fgdlinkd..
20510 root        20   0  239.0m  92.7m  11.6   4.6  48:02.83 S   fgdupd

```

How much RAM is the web filter and email filter database consuming for download?

- A. 0.1
- B. 0.7
- C. 4.6
- D. 1.4

Answer: D

Explanation:

When you enable web and anti-spam services for the first time, it may take several hours to download and merge the databases. During this time, you will notice higher I/O wait times and a spike in CPU usage related to web and email processes on FortiManager.

fgdlinkd

Responsible for downloading web filter and email filter databases

fgdupd

Responsible for database merging and consolidating smaller delta files into larger files

fgdsvr

Serves FortiGate and FortiClient for web filter and email filter requests

Question: 72

What prerequisite should be considered when using FortiManager to manage FortiAnalyzer devices?

- A. The highest log rate FortiManager supports is limited
- B. Download the licence to enable this features
- C. Check if FMG is in advanced mode
- D. Ensure free performance for processing logs

Answer: C

Explanation:

Note that you cannot enable advanced ADOM mode when FortiManager is managing FortiAnalyzer. You can add and then manage most of the FortiAnalyzer features from FortiManager. This is the recommended solution for environments with a high volume of logs.

Logs are stored on FortiAnalyzer, not FortiManager

After you add a FortiAnalyzer device, the following panes are enabled automatically on FortiManager:

FortiView and LogView: Provide visibility of FortiAnalyzer logs

Incidents & Events: Allows you to manage FortiAnalyzer incidents, events handlers, and work with the Threat Hunting tool and Outbreak Alerts, when available

Reports: Allow you to manage FortiAnalyzer reports

FortiManager can manage multiple FortiAnalyzer devices, but each FortiAnalyzer must be in its own ADOM.

You can add the same FortiAnalyzer device to more than one ADOM. When you do this, the FortiAnalyzer features and visibility in the ADOM are limited to the logging devices included in that ADOM.

Question: 73

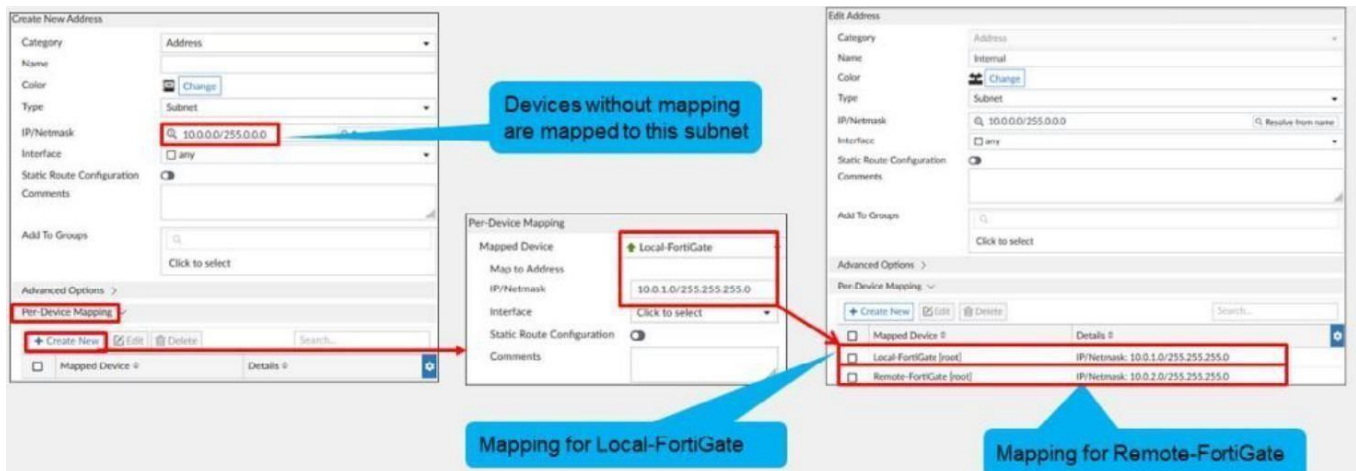
What is the purpose of dynamic objects?

- A. To merge duplicate objects automatically
- B. To map a single logical object to a unique definition per device
- C. To avoid conflicts if two or more administrators try to change the same setting at same time
- D. To use the same object in different ADOMs

Answer: B

Explanation:

All objects in an ADOM are managed by a single database unique to the ADOM. Many objects now include the option to enable dynamic mapping. You can use dynamic objects to map a single logical object to a unique definition per device. You can dynamically map common features such as addresses, interfaces, virtual IPs, and IP pools. A common example is a firewall address. You may have a common name for an address object, but have different value depending on the device it is installed on.



Question: 74

Refer to the exhibit:



What statement is true according to the ADOM?

- A. ADOM in workflow mode
- B. ADOM locked by another administrator
- C. ADOM locked by you
- D. ADOM unlocked

Answer: C

Explanation:



Question: 75

Which statement about FortiManager used as a local FDS is true?

- A. It provides devices with a faster connection to updates
- B. It provides updates for registered devices only
- C. It reduce the management cost with FortiGuard services
- D. It provides an independent database

Answer: A

Explanation:

Configuring FortiManager to act as a local FortiGuard server allows you to:

Synchronize the FortiGuard updates and packages with the public FortiGuard Distribution Network (FDN)

Download all antivirus and IPS packages, and web filter and email filter databases on FortiManager

Reduce internet connection load

Provide device a faster connection to updates

Redistribute the packages to many devices

In some high-security environments, internet access for internal FortiGate is restricted

Question: 76

What does the fgfm-sock-timeout command do?

- A. It sets the idle timeout setting for unlocking an ADOM
- B. It sets the idle time out setting for communication between FortiManager and the public FortiGuard server
- C. It sets the idle time out setting for configuration change
- D. It sets the idle time out setting for communication between FortiManager and FortiGate

Answer: D

Explanation:

You can configure the timeout and interval by setting the following parameters:

fgfm-sock-timeout: the maximum FortiManager or FortiGate communication socket idle time, in seconds

fgfm_keepalive_itvl: the interval at which the FortiGate sends a keepalive signal to a FortiManager device to keep the FortiManager or FortiGate communication protocol active

Question: 77

Which criteria must you take into consideration when naming objects in the ADOM database?

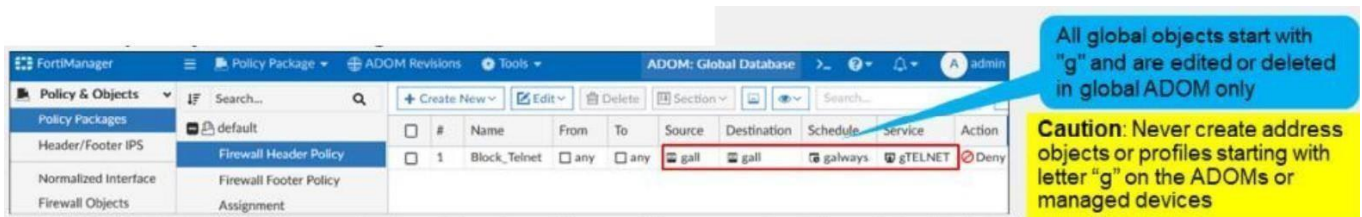
- A. Object names must not start with the letter "g" in the ADOM database
- B. All object names must be based on the name of the ADOM they belong to
- C. Objects must have unique names across all ADOMs
- D. Objects must include the underscore "_" character

Answer: A

Explanation:

After you assign the global database ADOM objects, they appear on the Policy & Objects pane that ADOM. All global objects start with "g" and are edited or deleted in the global database ADOM only.

A word of caution. Never create address objects or security profiles starting with letter "g" on an ADOM database or directly on the managed device because this may create conflicts with global database ADOM objects and cause configuration failure issues.



Question: 78

An administrator has moved a FortiGate from one ADOM to another. Which statements are true about this change?

- A. VPN manager settings move to new ADOM
- B. Policy package status is set to "Never installed"
- C. Policies and objects don't move to the new ADOM
- D. SNMP configuration don't move to the new ADOM

Answer: B, C

Explanation:

Considerations before moving devices:

When you move a device from one ADOM to another, policies and objects (used and unused) don't move to the new ADOM.

If using a shared policy package, it is not moved to the new ADOM

When FortiGate device are upgraded, it is best to keep them in the same ADOM and use ADOM upgrade

Moving FortiGate devices to a new ADOM introduces additional work and certain complications.

After moving the devices:

If you need to move a device from one ADOM to another, run Import Configuration wizard to import the policy package into the new ADOM.

Can use CLI to import unused objects if needed

execute fmpolicy copy-adom-object

Configuration of manager panes like VPN manager does not move

FortiManager administrator must reconfigure all the VPN configuration

Question: 79

Which command is useful when troubleshooting ADOM-level issues?

- A. execute fmpolicy print-adom-database
- B. execute fmpolicy print-device-database
- C. execute fmpolicy print-adom-object
- D. execute fmpolicy print-device-object

Answer: A

Explanation:

Command	Information
<code>execute fmpolicy print-adom-database <adom> <output></code>	Exports the entire ADOM database to a file. You can display the information on screen but the output is extremely verbose.
<code>execute fmpolicy print-adom-package <adom> <policy package/template name> <package name> <category name> <object name></code>	Firewall policies contained in a specific policy package in the ADOM
<code>execute fmpolicy print-adom-object <adom> <category name></code>	Individual objects in an ADOM

After exporting the entire ADOM database, you can use the `execute fmpolicy upload-print-log` command to upload the resulting file to an FTP, SFTP, or SCP server.

Question: 80

Which statements is true about Meta Fields?

- A. You can make meta fields required or optional
- B. Meta fields can be used in any ADOM
- C. It can be used in provisioning templates
- D. It can be used as variables in scripts

Answer: A

Explanation:

Meta Fields allow administrators to add additional attributes to several types of objects and administrator accounts. You can make meta fields required or optional. When meta fields are required, administrators must supply additional information when they create an associated object. For example, if you create a required meta field for a device object, administrators must define a value for the meta field for all devices.

Meta fields cannot be used as variables in scripts or provisioning templates. Instead, you can use ADOM-level metadata variables for that purpose.

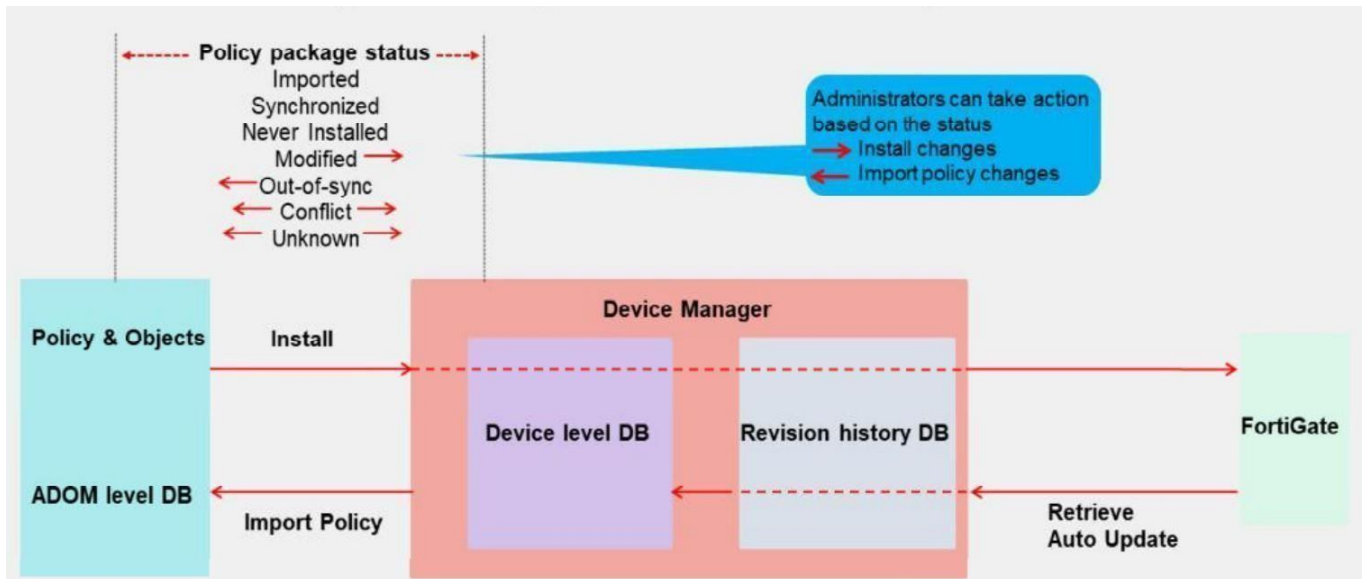
Question: 81

On FortiManager, where do you install policy packages on managed devices?

- A. At the Root ADOM layer
- B. At the device manager layer
- C. At the ADOM layer
- D. At the Global ADOM layer

Answer: C

Explanation:



Question: 82

An administrator is configuring a new restricted admin profile.
Which two permission types can the administrator configure? (Choose two.)

- A. Device registration
- B. Web Filter profile
- C. FortiGuard configuration
- D. IPS sensor

Answer: B, D

Explanation:

Profiles with the Restricted Admin type allows you to delegate administrators the management of web filtering profile, IPS sensors, and application sensor associated with their ADOM. When a restricted administrator logs in to the FortiManager, they enter the Restricted Admin Mode.

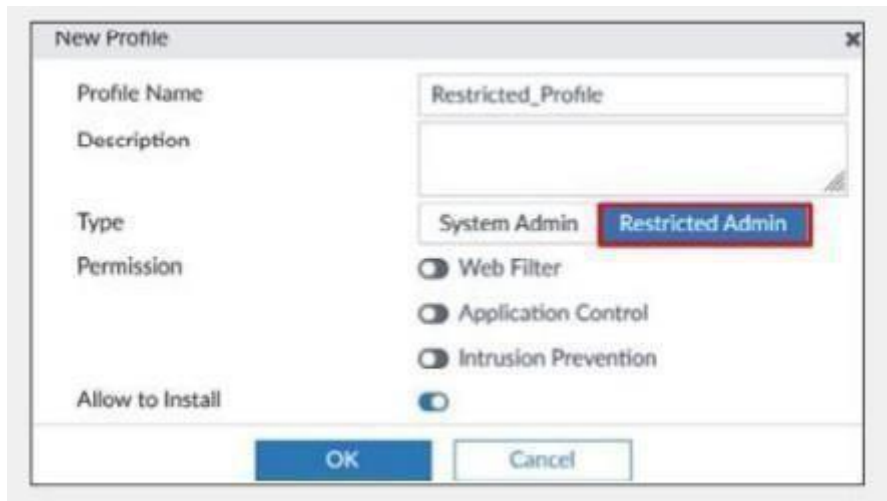
Delegated management of specific security profiles for each ADOM

Scope is only three security profiles

Intrusion protection system (IPS) sensors

Web filter profiles

Application sensors



Question: 83

If FortiGate is behind a NAT device, which statements are true?

- A. Configure the NAT device IP address on FortiManager
- B. Only FortiManager tries to reestablish the FGFM tunnel automatically, if it is torn down
- C. Only FortiGate tries to reestablish the FGFM tunnel automatically, if it is torn down
- D. FortiManager can discover FortiGate through the FortiGate NATed IP address

Answer: C, D

Explanation:

FortiManager can discover FortiGate through the FortiGate NATed IP address. FortiGate can also announce itself to FortiManager.

FortiManager doesn't automatically attempt to reestablish the FGFM tunnel, if it is torn down. Clicking the Refresh Device icon in the Managed FortiGate forces a one-time connection attempt. If the FGFM tunnel is interrupted, only FortiGate attempts to re-establish the connection.

Question: 84

Which information is included when you perform a FortiManager backup?

- A. Firmware images
- B. FortiGuard objects
- C. Logs and firmware images
- D. Flash configuration

Answer: D

Explanation:

Backs up:

All devices

Global database

Flash configuration

Does not back up logs, FortiGuard objects, and saved firmware images

Scheduled backups from the GUI and CLI

Supports FTP, SCP and SFTP

Question: 85

A remote device managed has problems to connect with FortiManager, which options should be consider to make a troubleshooting? (Choose two)

- A. Ensure that managed device is not locked by workmode normal
- B. Enable port UDP 541 through the communication
- C. Ensure that FortiManager is not in offline mode
- D. Ensure that management ports and device IP are defined correctly in its Virtual IP

Answer: C, D

Explanation:

Enabling offline mode stops FGFM management protocol

By default, offline mode is enabled when a backup is restored

You cannot manage devices when offline mode is enabled.

Question: 86

Where can you upload the resulting file after you export the entire ADOM database?

- A. SFTP server
- B. AD server
- C. MAIL server
- D. HTTP server

Answer: C

Explanation:

Command	Information
<code>execute fmpolicy print-adom-database <adom> <output></code>	Exports the entire ADOM database to a file. You can display the information on screen but the output is extremely verbose.
<code>execute fmpolicy print-adom-package <adom> <policy package/template name> <package name> <category name> <object name></code>	Firewall policies contained in a specific policy package in the ADOM
<code>execute fmpolicy print-adom-object <adom> <category name></code>	Individual objects in an ADOM

After exporting the entire ADOM database, you can use the `execute fmpolicy upload-print-log` command to upload the resulting file to an FTP, SFTP, or SCP server.

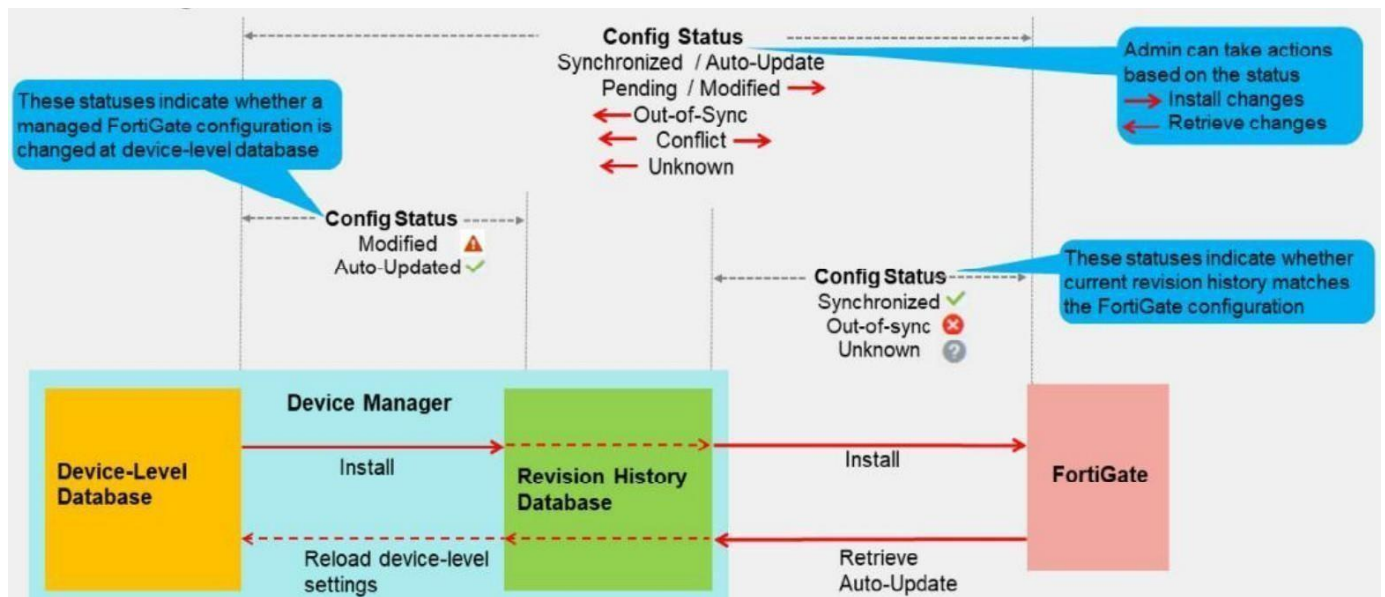
Question: 87

How does FortiManager determine if the configuration status of a managed device is set to Out-of-Sync?

- A. It compares the ADOM-level database with the FortiGate configuration.
- B. It compares the device-level database with the FortiGate configuration.
- C. It compares the current revision history with the FortiGate configuration.
- D. It compares the provisioning template with the FortiGate configuration.

Answer: C

Explanation:



Question: 88

An administrator needs to install a policy on managed devices in a backup ADOM. Which of the following options should be considered?

- A. You can assign the same policy to different FortiGate from different ADOMs
- B. Use a script to make changes from FortiManager
- C. Managed devices share a common policy package in an ADOM
- D. Script must be run on Policy Package or ADOM Database

Answer: B

Explanation:

Backup Mode

ADOM is read-only

Not all management panes available

Must make changes directly on managed devices

Only scripts can be used to make changes from FortiManager to the managed devices

Question: 89

What does a status of Copy Failed indicate?

- A. An operation that failed to set a Revision history
- B. An operation that failed to copy the ADOM-level policy or object to a device database
- C. An operation that failed to set changes to a modified device
- D. An operation that failed to copy the device database from the revision history

Answer: B

Explanation:

When you perform a policy package installation, the copy operation is the first operation that FortiManager performs, before you the actual installation. Is the operation in which FortiManager tries to copy the ADOM-level object or policy to the device database. It is the opposite of the import operation.

Copy failure issues are usually caused by having incorrect or missing object dependencies, or invalid parameters being used when copying from the ADOM database to the device database due to corruption or inconsistencies.

Question: 90

Why should the ADOM version match the FortiGate firmware version?

- A. To minimize CLI syntax issues between FortiGate and FortiManager
- B. To enable geographic regions
- C. To reduce policy management
- D. To keep the FortiGate licenses up to date

Answer: A

Explanation:

When you organize managed FortiGate devices, it is highly recommended that you group them based on their FortiOS firmware version. This is because valid command syntax varies by firmware version, which affects scrip compatibility and other features.

Question: 91

Refer to the following exhibit:

```
nr-retry           : 1
retry              : enable
retry-intvl        : 15
rollback-allow-reboot: enable
```

Which of the following statements are true based on this configuration? (Choose three)

- A. FortiManager will not push the CLI commands as a part of the installation that will cause the tunnel to go down.
- B. If connection remains down before unset CLI commands, the FortiGate unit reboots to recover the previous configuration from its config file
- C. After 15 minutes, FortiManager retry to set CLI commands that were part of the installation that caused the tunnel to go down
- D. FortiGate will reject the CLI commands that will cause the tunnel to go down
- E. After 15 minutes, FortiGate will unset all CLI commands that were part of the installation that caused the tunnel to go down

Answer: B, C, E

Explanation:

For each install:

The FortiManager sends the following to the FortiGate:

- a listing of the set commands needed to apply the configurations changes
- a listing of the unset commands that would revert the configuration changes

The FortiGate uses the following logic when applying changes:

- apply the set commands, using memory only, nothing written to a configuration file
- test the fgfm connection to the FortiManager
- if the connection goes down, it applies the unset commands
- retest the fgfm connection

if connection remains down, the FortiGate unit reboots to recover the previous configuration from its config file

The final step above is optional and can be enabled and disabled via the CLI using the following command:

```
config system dm
set rollback-allow-reboot {enable |disable}
end
```

Question: 92

What is synchronized in cluster HA? (Choose two)

- A. FortiGuard databases
- B. Global databases
- C. SNMP
- D. All configuration revision

Answer: B, D

Explanation:

What is synchronized?

All device configuration (including global databases)

All configuration revision

What is not synchronized?

Configuration settings for:

Interfaces, routes, HA, SNMP, log, FortiAnalyzer feature configuration

FortiGuard

Antivirus/IPS packages, web filter, antispam databases

Local logs and Alert Message Console logs (data)

FortiGate logs sent to FortiManager

Question: 93

If FortiManager is behind a NAT device, which statements are true? (Choose three)

- A. Only FortiManager can discover the device
- B. It's recommended to configure a NATed IP address of FortiManager with the set mgmt-addr command under the config system admin setting
- C. Only FortiManager tries to reestablish the FGFM tunnel, if it is torn down
- D. Configure the NAT device IP address on FortiGate
- E. It's recommended to configure a policy route on FortiManager to know the IP original of managed device

Answer: A, B, C

Explanation:

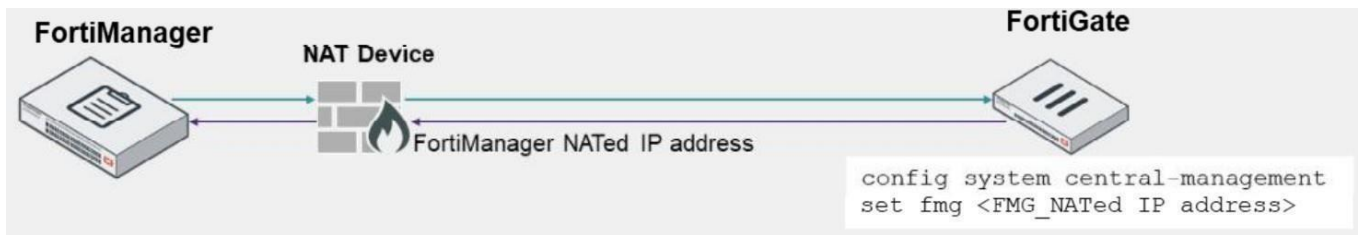
FortiManager Behind a NAT Device

Only FortiManager can discover the new device

During discovery, the FortiManager NATed IP address is not set on FortiGate

Only FortiManager tries to reestablish the FGFM tunnel, if it is torn down

This is because, by default, the NATed FortiManager IP address is not configured on FortiGate central management



How can FortiGate announce itself to the NATed FortiManager, or try to re-establish the FGFM tunnel if it is torn down?

You can configure the FortiManager NATed IP address on FortiGate under the central management configuration. This allows FortiGate to announce itself to FortiManager and try to re-establish the FGFM tunnel, if it is torn down. Configuring the FortiManager NATed IP address on FortiGate allows both FortiManager and FortiGate to re-establish the FGFM tunnel. Also, if you configure the FortiManager NATed IP address under the FortiManager system administrator settings, FortiManager sets this address on FortiGate during the discovery process.

```
config system central-management
set fmg
```

Additionally, if your FortiManager has an FQDN associated, you can also configure it so FortiGate devices can use it to announce themselves and reconnect to FortiManager if needed.

If FortiManager is behind a NAT device, it is recommended that you configure the FortiManager NATed IP address using system admin settings on FortiManager. You can also configure the FortiManager FQDN if it has one associated.

```
config system admin setting
set mgmt-addr <FMG_NATed IP address>
set mgmt-fqdn <FMG fqdn>
```

Question: 94

What command can you use to check and repair the filesystem in FortiManager?

- A. diagnose dvm check-integrity
- B. diagnose dvm task repair
- C. diagnose sys fsck harddisk
- D. diagnose system print df

Answer: C

Explanation:

```
# diagnose system fsck harddisk
```

This operation will check and repair the file system, then reboot the system.

Question: 95

Which command can be used to sniff the FGFM protocol communication between FortiManager and FortiGate?

- A. diagnose sniffer packet any "port 80" 4
- B. diagnose sniffer packet any "port 541" 4
- C. diagnose sniffer packet any "port 443" 4
- D. diagnose sniffer packet any "port 8113" 4

Answer: B

Explanation:

Functionality	Port(s)
Destination Ports	
Remote management of a FortiGate device	TCP 541 (IPv4), TCP 542 (IPv6)
Firmware image download	TCP 443
Antivirus, IPS, Web Filtering, Antispam updates from FortiManager to FortiGuard Distribution Servers	TCP 443
DNS lookup	UDP 53
Syslog	UDP 514
Listening Ports	
FortiGuard Antivirus and IPS update push	UDP 9443
FortiGuard Antivirus or IPS update request from a FortiGate device	TCP 8890
FortiGuard Antispam or Web Filtering rating lookup from a FortiClient or FortiGate	UDP 53 or 8888
Web Services (a.k.a XML API)	TCP 8080
Web GUI and JSON API	TCP 443
Antivirus and IPS updates for FortiClient	TCP 80
HA heartbeat or synchronization (FortiManager HA cluster)	TCP 5199
Two FortiManagers configured in cascade mode	
FortiManagers in cascade mode both listening and destination ports	TCP 8891, 8900 and 8901

Question: 96

An administrator is having trouble when trying to add FortiGate using the Add Device wizard on FortiManager.

Which two options can be used to troubleshoot this issue? (Choose two.)

- A. Make sure Offline Mode is disabled
- B. Make sure UDP port 541 between FortiManager and FortiGate is allowed.
- C. A configuration sync failure
- D. Make sure FMG-Access on the FortiGate interface facing FortiManager is enabled.

Answer: A, D

Explanation:

FGFM Management Protocol

The management protocol FGFM runs on both FortiGate (fgfmd) and FortiManager (fgfmsd). FortiManager and FortiGate create a secure tunnel using port TCP 541. Being TCP-based, the connection works with port-based NAT, which allows both FortiGate and FortiManager to be behind a NAT device. On the FortiGate side, you must enable the FMG-Access setting for each interface facing FortiManager.

Question: 97

Which statement about locking an ADOM is true?

- A. Other administrators have read/write access
- B. The changes made are installed without wizard
- C. It automatically removes lock on device and policy packages
- D. Another FortiGate devices in the same ADOM can be locked

Answer: C

Explanation:

Once enabled, you can create locks at different levels:

- ADOM



- Specific devices



- Policy packages



- Objects



Locking an ADOM automatically removes locks on devices and policy packages that you have locked within that ADOM

If another administrator locked devices or policy packages, you cannot lock the ADOM containing those devices or policy packages

Question: 98

Which of the following statements about Database Integrity are true? (Choose two)

- A. If you cannot resolve a data integrity issue, you can perform a factory reset on FortiManager, and then restore the configuration using a good backup
- B. Apply a hard shutdown (power off) can repair the database
- C. If you need to fix database errors, back up before making any changes
- D. An unexpected shut-down cannot damage the file system and database

Answer: A, C

Explanation:

Before upgrading, it is recommended that you check the integrity of FortiManager databases using the following CLI commands. If you find any errors, you can fix the errors before the upgrade.

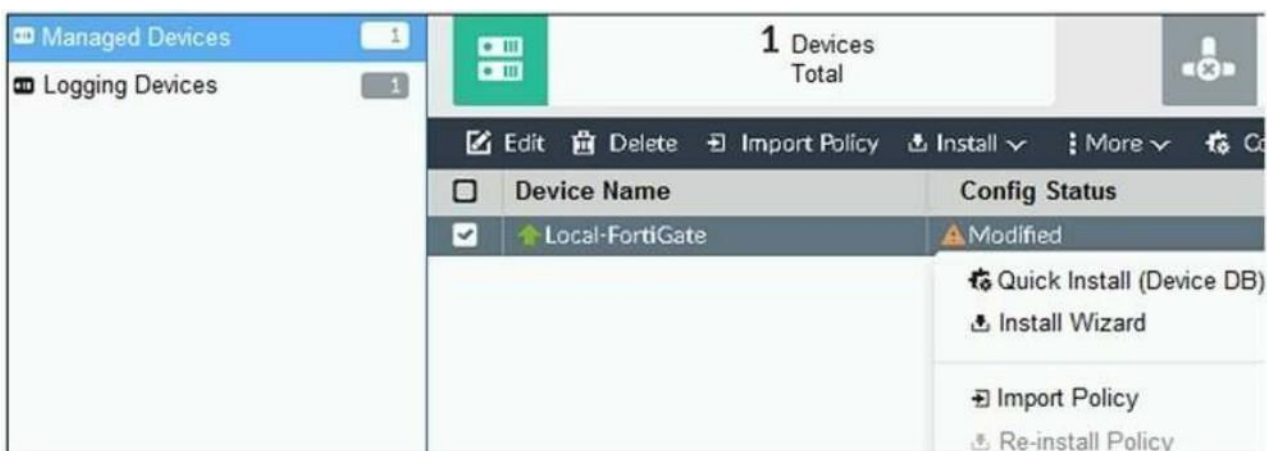
If you need to fix database errors, back up before making any changes.

Before running integrity check commands, ensure only one admin is logged in and no objects are locked.

If workspace mode is enabled, you must unlock all ADOMs before running any integrity commands.

Question: 99

Refer to exhibit.



You are using the Quick Install option to install configuration changes on the managed FortiGate. Which two statements correctly describe the result? (Choose two)

- A. It cannot be canceled once initiated and changes will be installed on the managed device
- B. It will not create a new version in the revision history
- C. It installs device-level changes to FortiGate without launching the Install Wizard
- D. It provides the option to preview configuration changes prior to installing them

Answer: A, C

Explanation:

The Quick Install (Device DB) option allows you to perform a quick installation of device-level settings without launching the Install Wizard.

When you use this option, you cannot preview the changes prior to installing them. Administrators should be certain of the changes made before using this option because the install can't be cancelled after the process is initiated.

If unsure about the changes, administrators are encouraged to use the Install Wizard, so that they can preview the changes before committing them.

Question: 100

An administrator needs to add one or more options to an existing list without modifying the other objects. Which API should be used?

- A. Add
- B. Select
- C. Append
- D. Set

Answer: C

Explanation:

For multi-option values, four different keywords can be used:

- 1) append: add one or more options to an existing list
- 2) select: clear all options except for those specified (override the existing entries values). It is the same as set
- 3) unselect: remove one or more options from an existing list
- 4) clear: remove all the options from an existing list. It's the same as unset (reset the list to default value)

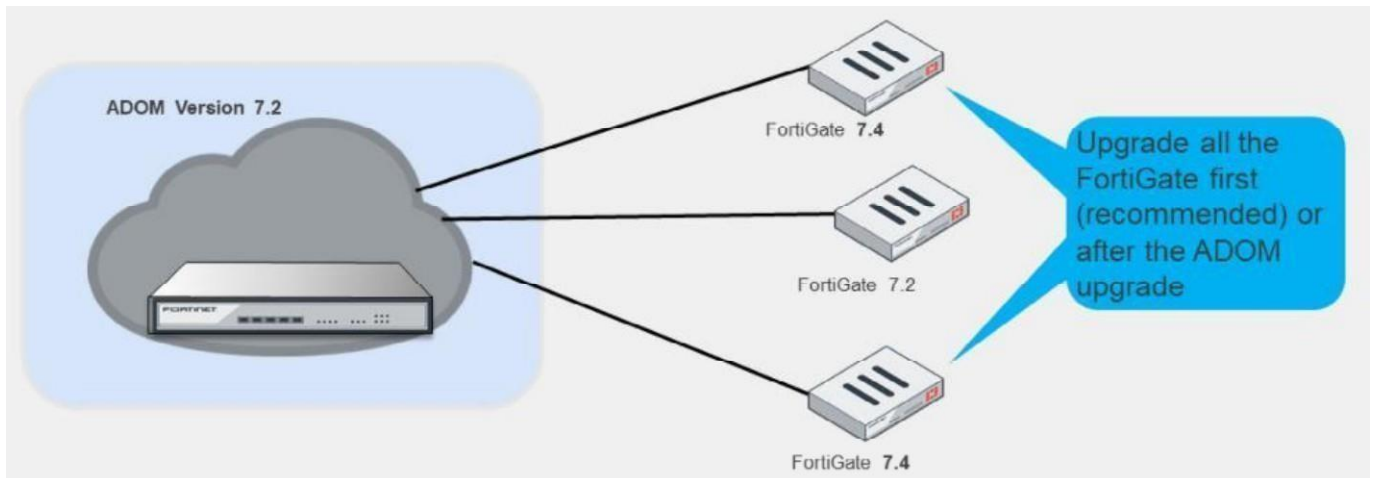
Question: 101

What is the most recommended sequence of upgrading an existing ADOM?

- A. Upgrade the ADOM first, and then upgrade all the devices in the ADOM
- B. Upgrade the ADOM first, and then upgrade all the devices in the ADOM
- C. Upgrade FortiManager replicate the same version in the ADOM
- D. Upgrade all the devices in the ADOM first, and then upgrade the ADOM

Answer: D

Explanation:



Question: 102

Which statement about meta fields is true?

- A. Meta fields can be used as variables in templates
- B. Allow administrator to add additional attributes to several FortiManager objects
- C. Meta fields can be used as variables in scripts
- D. If you create an optional metafield for a device object, administrators must define a value for the meta field for all devices.

Answer: B

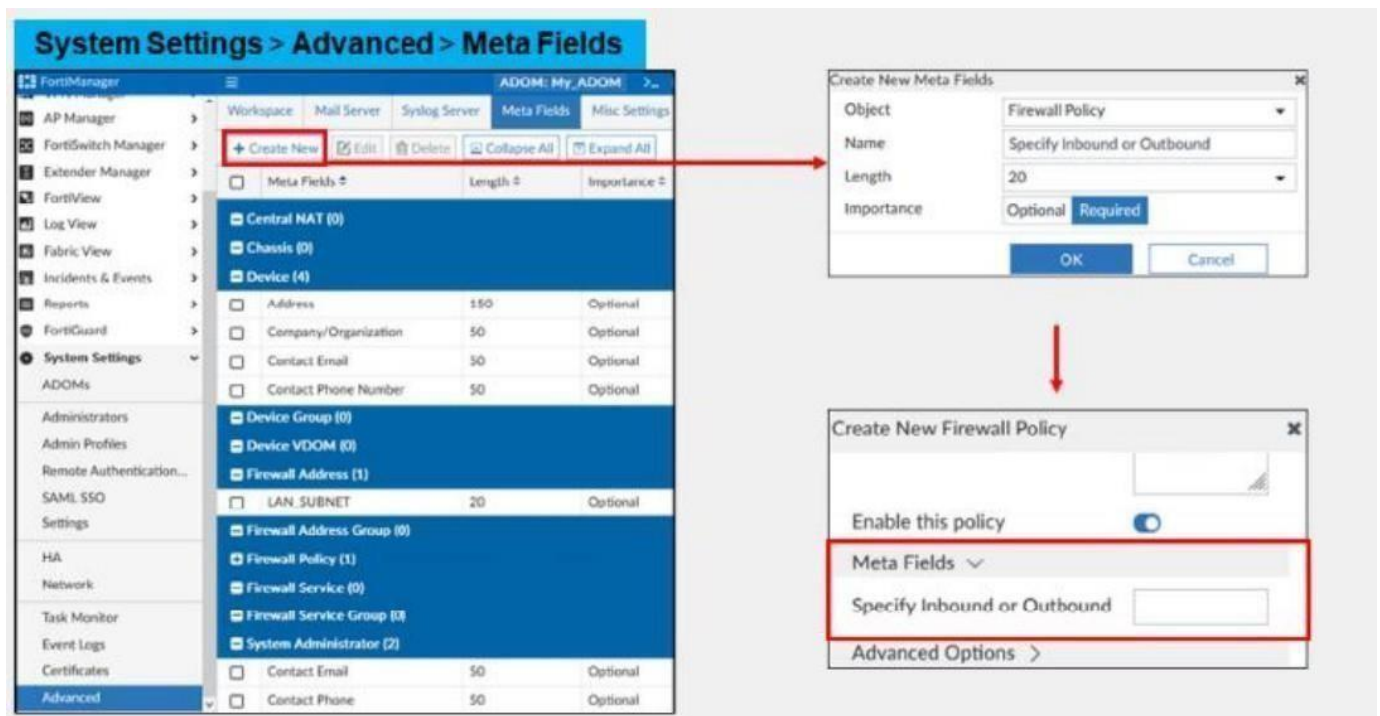
Explanation:

Meta Fields allow administrator to add additional attributes to several types of objects and administrator accounts.

You can make meta fields required or optional.

Required: Administrators must supply additional information when they create an associated object. If you create a required metafield for a device object, administrators must define a value for the meta field for all devices.

Meta Fields cannot be used as variables in scripts or provisioning templates. Instead, you can use ADOM-level metadata variables for that purpose.



Question: 103

Which statement is true of the global database ADOM on FortiManager?

- A. Global objects are identified to start with "o"
- B. You can assign only one global policy package to an individual ADOM policy package
- C. Assigning an additional global policy package to the same individual ADOM policy package is merged with previously assigned policies
- D. You can edit and move the header and footer policies between the rules in an individual ADOM policy package.

Answer: B

Explanation:

Viewing Assigned Global Policy

After you assign the global database ADOM objects, they appear on the Policy & Objects pane that ADOM. All global objects start with "g" and are edited or deleted in the global database ADOM only.

A word of caution. Never create address objects or security profiles starting with letter "g" on an ADOM database or directly on the managed device because this may create conflicts with global database ADOM objects and cause configuration failure issues.

You can assign only one global policy package to an individual ADOM policy package. Assigning an additional global policy package to the same individual ADOM policy package removes previously assigned policies. Also, you cannot edit and move the header and footer policies between the rules in an individual ADOM policy package.

You must install policy packages on the managed devices for the new rules to work. A header policy is installed at the top of the list of the firewall rules on the target device.

Question: 104

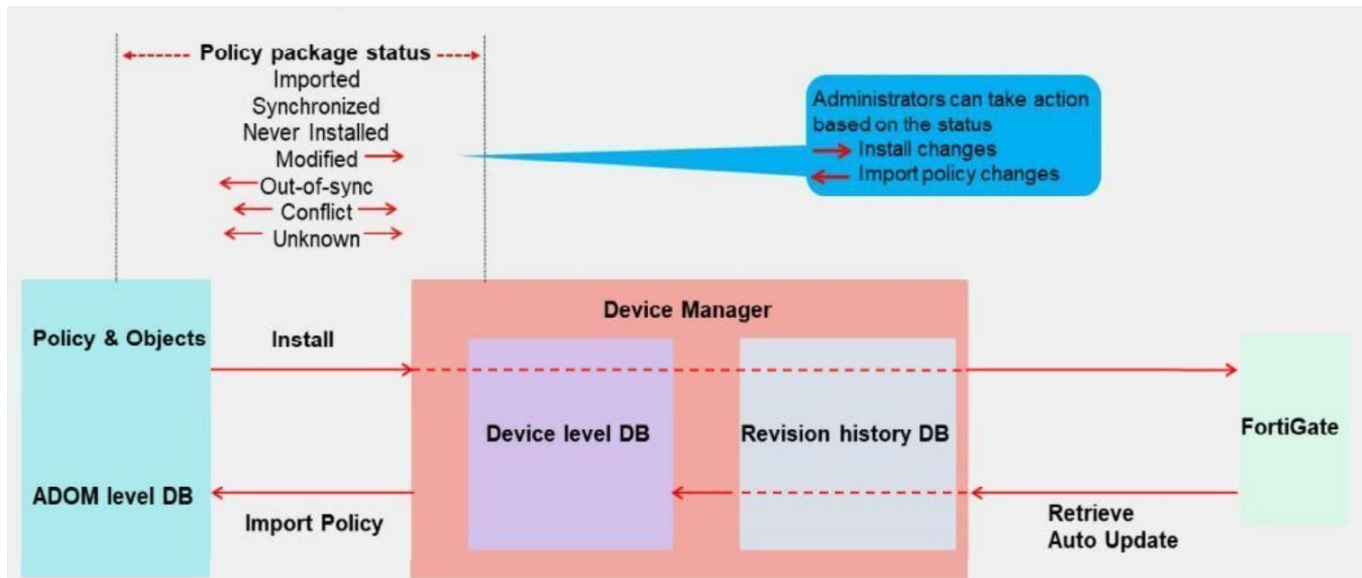
An administrator configured a new firewall policy on FortiManager and has not yet pushed the changes

to the managed FortiGate. In which database will the configuration be saved?

- A. Revision history database
- B. Policy check database
- C. Device-level database
- D. ADOM-level database

Answer: D

Explanation:



Question: 105

Refer to the next keep alive message:

```
# FGfMs: Timeout[360] for sock.  
FGfMs (FGVM..-10.0.1.254): Cleanup  
devid=307 tunnel_ip from DVM  
(ret=0)
```

Which of the following statements are true? (Choose two)

- A. The managed device didn't receive a keepalive and the tunnel is torn down
- B. There were no responses to keepalive messages during the sending interval
- C. FortiManager didn't receive a keepalive and the tunnel is torn down
- D. There was no responses to the keepalive messages for duration of the sock timeout value

Answer: C, D

Explanation:

Keepalive messages are sent from FortiGate at configured intervals. If there are no responses to the keepalive messages for duration of the sock timeout value, the tunnel is torn down and both ends attempt to reestablish it.

You can configure the timeout and interval by setting the following parameters:

fgfm-sock-timeout: the maximum FortiManager or FortiGate communication socket idle time, in seconds

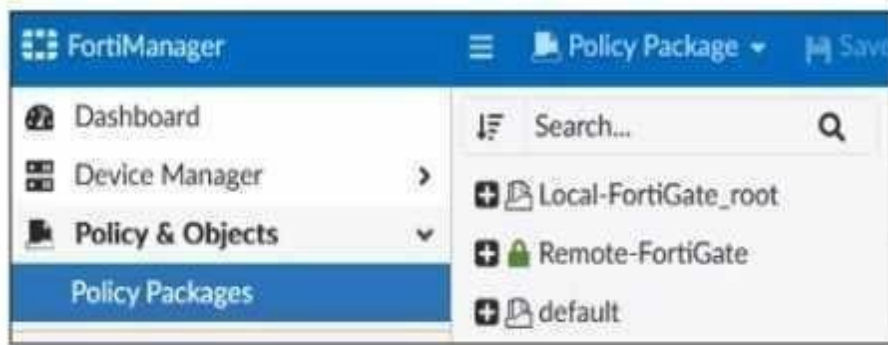
fgfm_keepalive_itvl: the interval at which the FortiGate sends a keepalive signal to a FortiManager device to keep the FortiManager or FortiGate communication protocol active

```
# FGfMs: Timeout[360] for sock.  
FGfMs (FGVM..-10.0.1.254): Cleanup  
devid=307 tunnel_ip from DVM  
(ret=0)
```

FortiManager didn't receive a
keepalive and the tunnel is torn down

Question: 106

Exhibit.



Given the configuration shown in the exhibit, which two statements are true? (Choose two.)

- A. An administrator can also lock the Local-FortiGate_root policy package.
- B. FortiManager is in workflow mode.
- C. The FortiManager ADOM is locked by the administrator.
- D. The FortiManager ADOM workspace mode is set to Normal

Answer: A, B

Explanation:

When considering the FortiManager configuration and the implications of workflow mode and ADOM locking, it's essential to understand how these features function:

Option A - "An administrator can also lock the Local-FortiGate_root policy package":

In FortiManager, administrators have the ability to lock policy packages, including the Local-FortiGate_root policy package. Locking a policy package prevents any changes to it until the lock is removed, ensuring that the configuration remains stable during modifications or when multiple administrators are working on the same configurations. This feature is particularly useful in environments where changes need to be controlled and reviewed before application.

FortiManager Reference: "Locking policy packages allows for controlled management of configurations and prevents unintentional changes." (FortiManager 7.4 Administration Guide, Policy Management)

Option B - "FortiManager is in workflow mode":

When FortiManager is set to workflow mode, it introduces a process where all changes to policies and objects must go through an approval process before they are implemented. This mode enhances oversight and prevents unauthorized changes, making it a critical feature for organizations requiring strict change management protocols. Workflow mode ensures that all modifications are reviewed, thereby improving overall security and compliance.

FortiManager Reference: "In workflow mode, changes to the configuration must be approved before being applied to the device, enhancing change management." (FortiManager 7.4 Administration Guide, Workflow Management)

Option C - "The FortiManager ADOM is locked by the administrator":

This statement is false unless explicitly indicated in the exhibit. An ADOM being locked means that no changes can be made to the objects or policies within that ADOM until it is unlocked. The locking status

must be actively managed by the administrator to ensure that changes are made only when necessary.

Option D - "The FortiManager ADOM workspace mode is set to Normal":

This statement is false in the context of workflow mode. If FortiManager is in workflow mode, it cannot be in Normal workspace mode simultaneously. Workflow mode signifies a different operational state where review and approval processes are in place, indicating that the ADOM's workspace mode is likely set to Workflow rather than Normal.

Question: 107

An administrator enabled workspace mode and now wants to delete an address object that is currently referenced in a firewall policy. Which two results can the administrator expect? (Choose two.)

- A. FortiManager will temporarily change the status of the referenced firewall policy to disabled.
- B. FortiManager will disable the status of the address object until the changes are installed.
- C. FortiManager will not allow the administrator to delete a referenced address object until they lock the ADOM.
- D. FortiManager will replace the deleted address object with the none address object in the referenced firewall policy.

Answer: C, D

Explanation:

When operating in workspace mode on FortiManager 7.4, the administrator must understand how object references and deletions work:

Option C- "FortiManager will not allow the administrator to delete a referenced address object until they lock the ADOM": In workspace mode, all changes are managed within an Administrative Domain (ADOM) scope. When an object (like an address object) is referenced in a policy, FortiManager prevents its deletion to maintain configuration integrity. The ADOM must be locked by the administrator to make changes to any referenced objects. This locking mechanism ensures that no unintended deletions or changes occur that could disrupt the policies or configuration.

FortiManager Reference: "In workspace mode, changes to objects or policies require the ADOM to be locked. If an object is referenced, you must lock the ADOM before deleting or modifying the object."
(FortiManager 7.4 Administration Guide, Section on Workspace Mode and ADOM Management)

Option D- "FortiManager will replace the deleted address object with the none address object in the referenced firewall policy": If the administrator attempts to delete an address object that is currently referenced by a firewall policy, FortiManager will replace the deleted object with the 'none' address object. This is done to maintain the policy structure and avoid policy corruption due to a missing reference. This behavior ensures that the firewall policy remains syntactically correct, even though the specific address object is no longer in use.

FortiManager Reference: "When a referenced object is deleted, FortiManager will replace it with a 'none' object in the policy. This behavior is to ensure the integrity and continuity of the policy configurations."
(FortiManager 7.4 Administration Guide, Object Management and Policy Handling in Workspace Mode)

Question: 108

Refer to the exhibit.

Edit Address

Category: Address

Name: LOCAL_SUBNET

Color: Change

Type: Subnet

IP/Netmask: 192.168.1.0/255.255.255.0 Resolve from name

Interface: any

Static Route Configuration: ☐

Comments:

Add To Groups:

Advanced Options >

Per-Device Mapping

☐	Mapped Device	Details
☐	Local-FortiGate [root]	IP/Netmask: 192.168.1.0/255.255.255.240

An administrator has created a firewall address object that is used in multiple policy packages for multiple FortiGate devices in an ADOM.

After the installation operation is performed, which IP/netmask is shown on FortiManager for this firewall address object for devices without a Per-Device Mapping set?

- A. FortiManager generates an error for each FortiGate without a per-device mapping defined for that object.
- B. 192.168.1.0/24
- C. 192.168.1.0/28
- D. FortiManager replaces the address object to none.

Answer: B

Explanation:

Option B: 192.168.1.0/24 is the correct answer. In FortiManager, when a firewall address object is defined and used across multiple policy packages without any Per-Device Mapping, the default value configured in the object definition (192.168.1.0/255.255.255.0) is applied to all devices. The exhibit shows that the address object LOCAL_SUBNET has a default IP/netmask of 192.168.1.0/24. Therefore, FortiManager will use this default value for any FortiGate device that does not have a specific Per-Device Mapping configured.

Explanation of Incorrect Options:

Option A: FortiManager generates an error for each FortiGate without a per-device mapping defined for that object is incorrect because FortiManager does not generate an error when a Per-Device Mapping is not set. Instead, it uses the default value provided in the object definition.

Option C: 192.168.1.0/28 is incorrect because the default value is 192.168.1.0/24, as seen in the exhibit, not /28.

Option D: FortiManager replaces the address object to none is incorrect because FortiManager does not replace address objects to "none" when a Per-Device Mapping is missing; it uses the default value instead.

FortiManager References:

Refer to the FortiManager 7.4 Administration Guide, specifically in sections related to "Address Object Management" and "Per-Device Mapping," which detail the behavior of address objects without specific device mappings.

Question: 109

Which statement about the policy lock feature on FortiManager is true?

- A. Policy locking is available in workspace normal mode.
- B. Locking a policy takes precedence over a locked ADOM.
- C. When a policy is locked, the ADOM that contains it is also locked.
- D. Administrators in the approval group can work concurrently on a locked policy.

Answer: A

Explanation:

The statement that is true about the policy lock feature on FortiManager is:

- A. Policy locking is available in workspace normal mode.

In FortiManager, when working in "workspace-mode normal," policies can be locked by administrators to prevent other administrators from editing them simultaneously. This ensures that only one administrator makes changes at any given time, reducing conflicts or mistakes due to concurrent modifications.

Statements B, C, and D are incorrect because:

B. is incorrect since locking a policy does not override a locked ADOM. The ADOM lock takes precedence.

C. is incorrect because when a policy is locked, it does not necessarily mean the ADOM is locked.

D. is incorrect because administrators in the approval group cannot work concurrently on a locked policy; the policy lock prevents concurrent modifications.

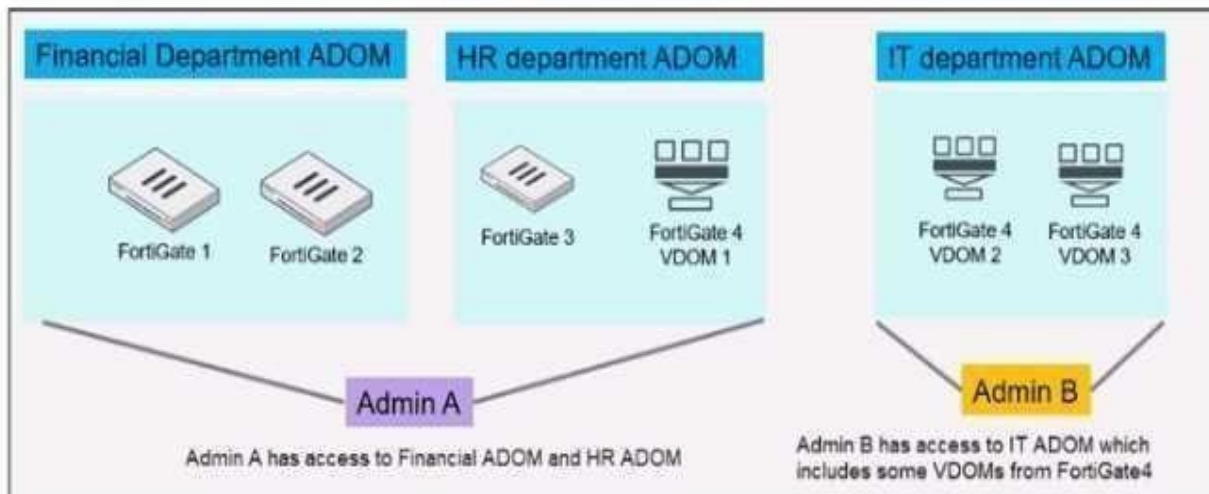
FortiManager References:

Refer to FortiManager 7.4 Administrator Guide: Policy and Objects > Policy Locking to understand how

the policy lock feature functions in different workspace modes.

Question: 110

Exhibit.



An administrator would like to create three ADOMs on FortiManager with different access levels based on departments. What two conclusions can you draw from the design shown in the exhibit? (Choose two.)

- A. The FortiManager administrator must set the ADOM device mode to Advanced
- B. Policies and objects databases can be shared between the Financial and HR ADOMs.
- C. An administrator with the super user profile can access all the VDOMs.
- D. The administrator must configure FortiManager in workspace normal mode.

Answer: A, C

Explanation:

Based on the exhibit, the FortiManager administrator is setting up three ADOMs (Administrative Domains) that correspond to different departments (Financial, HR, and IT). Each ADOM has specific FortiGate devices or VDOMs (Virtual Domains) assigned to it, with different administrators managing the ADOMs.

Explanation of Options:

- A. The FortiManager administrator must set the ADOM device mode to Advanced.

This is true. In FortiManager, when there are VDOMs (Virtual Domains) involved, you must set the ADOM to Advanced mode to manage VDOMs properly. The IT department ADOM includes different VDOMs from FortiGate 4 (VDOM 2 and VDOM 3), which means the ADOM mode must be in Advanced to support managing VDOMs separately from other ADOMs.

- B. Policies and objects databases can be shared between the Financial and HR ADOMs.

This is false. By default, ADOMs are separate, and policies and objects cannot be shared between them unless they are specifically designed to do so. The exhibit shows distinct ADOMs for each department, implying no direct sharing of policies and objects between Financial and HR ADOMs.

C. An administrator with the super user profile can access all the VDOMs.

This is true. A FortiManager administrator with the super user profile has full access to all ADOMs and VDOMs, regardless of how access is restricted for individual administrators. In this case, an admin with the super user profile could access Financial, HR, and IT ADOMs, including all the VDOMs from FortiGate 4.

D. The administrator must configure FortiManager in workspace normal mode.

This is false. There is no requirement mentioned in the exhibit or scenario that mandates using workspace normal mode. Workspace mode is more related to how configuration changes are managed (locking, editing, etc.), but it doesn't affect the creation or access control of ADOMs.

Conclusion:

A is correct because Advanced mode is necessary for managing VDOMs within ADOMs.

C is correct because a super user can access all VDOMs and ADOMs without restrictions.

Question: 111

Refer to the exhibit.

Managed FortiGate devices

Managed FortiGate (4)

- ISFW (3)
 - root
 - Student
 - Trainer
- Local-FortiGate
- Managed FortiAnalyzer (1)
 - FAZVM64-KVM

2 Devices

☐	Device Name
☐	Training
☐	ISFW
☐	root [NAT] (Management)
☐	Student [NAT]
☐	Trainer [NAT]
☐	Local-FortiGate*

FortiManager policy package

Policy Package

- Local-FortiGate_root
- Remote-FortiGate
- Shared_Package
 - Firewall Header Policy
 - Firewall Policy
 - Installation Targets
- default

☐	Installation Target
☐	Local-FortiGate
☐	ISFW
☐	root [NAT] (Management)
☐	Trainer [NAT]
☐	Student [NAT]

FortiManager policy package

☐	#	Name	Install On	From	To
☐	1	Ping_Access	ISFW (root) ISFW (Student)	port3	port1
☐	2	Web	Local-FortiGate (root) ISFW (Student)	port3	port1
☐	3	Source_Device	Installation Targets	port3	port1
☐	Implicit (4/4 Total:1)				
☐	4	Implicit Deny	Installation Targets	any	any

Given the configuration shown in the exhibit, which two conclusions can you draw from the installation targets in the Install On column? (Choose two.)

- A. Policy seq.S will be installed on all managed devices and VDOMs that are listed under Installation Targets
- B. Policy seq.# 3 will be skipped because no installation targets are specified.
- C. Policy seq.# 2 will not be installed on the Local-FortiGate root VDOM because there is no root VDOM in the Installation Target
- D. Policy seq.# 1 will be installed on the ISFW device root[NAT] and Student[NAT] VDOMs only.

Answer: A, D

Explanation:

Option A: Policy seq.S will be installed on all managed devices and VDOMs that are listed under Installation Targets.This is correct. The "Install On" column indicates that the policy is targeted for installation on all listed managed devices and VDOMs under Installation Targets.

Option D: Policy seq.# 1 will be installed on the ISFW device root[NAT] and Student[NAT] VDOMs only.This is correct. Policy sequence #1 specifies that it will be installed only on the ISFW device and the VDOMs 'root[NAT]' and 'Student[NAT]' as indicated by the "Install On" column.

Explanation of Incorrect Options:

Option B: Policy seq.# 3 will be skipped because no installation targets are specifiedis incorrect because it is clearly listed under "Installation Targets," which means it will be installed according to the specified configuration.

Option C: Policy seq.# 2 will not be installed on the Local-FortiGate root VDOM because there is no root VDOM in the Installation Target is incorrect as the exhibit does not show any specific exclusion for seq.# 2 on the Local-FortiGate root VDOM.

FortiManager References:

Refer to the FortiManager Administration Guide sections on "Policy Packages" and "Policy Installation Targets" for more details.

Question: 112

An administrator wants to create a policy on an ADOM that is in backup mode and install it on a FortiGate device in the same ADOM. How can the administrator perform this task?

- A. The administrator must use the Policy & Objects section to create a policy first.
- B. The administrator must use a FortiManager script.
- C. The administrator must disable the FortiManager offline mode first.
- D. The administrator must change the ADOM mode to Advanced to bring the FortiManager online.

Answer: B

Explanation:

To create and install a policy on a FortiGate device in an ADOM (Administrative Domain) that is in backup mode, the administrator must use a FortiManager script. This is because backup mode restricts direct configuration changes, and scripts can be used to push specific configuration changes without altering the ADOM mode.

Options A, C, and D are incorrect because:

A. requires the ADOM to be in normal or advanced mode to create policies directly in the Policy & Objects section.

C. suggests disabling offline mode, which is irrelevant to the backup mode configuration.

D. implies changing the ADOM mode, which is unnecessary if using a script to perform the task.

FortiManager References:

Refer to FortiManager 7.4 Administrator Guide: Working with ADOMs and Using Scripts for managing policies in backup mode.

Question: 113

Refer to the exhibit.

FortiManager log

```
-----Executing time: -----  
  
Starting log (Run on device)  
  
Local-FortiGate $ config user local  
Local-FortiGate (local) $ edit student  
Local-FortiGate (student) $ set type ldap  
Local-FortiGate (student) $ set status enable  
Local-FortiGate (student) $ next  
Attribute 'ldap-server' MUST be set.  
Command fail. Return code 1  
Local-FortiGate (local) $ end  
Local-FortiGate $ config firewall policy  
Local-FortiGate (policy) $ edit 2  
Local-FortiGate (2) $ set srcintf port3  
Local-FortiGate (2) $ set dstintf port1  
Local-FortiGate (2) $ set srcaddr all  
Local-FortiGate (2) $ set dstaddr all  
Local-FortiGate (2) $ set action accept  
Local-FortiGate (2) $ set schedule always  
Local-FortiGate (2) $ set service ALL  
Local-FortiGate (2) $ set users student  
entry not found in datasource  
  
value parse error before 'student'  
Command fail. Return code -3  
Local-FortiGate (2) $ set nat enable  
Local-FortiGate (2) $ next  
Local-FortiGate (policy) $ end  
Local-FortiGate $  
  
-----End of Log-----
```

What can you conclude from the failed installation log shown in the exhibit?

- A. Policy ID 2 is installed in the disabled state.
- B. Policy ID 2 is installed without the remote user student.
- C. Policy ID 2 will not be installed.
- D. Policy ID 2 is installed without a source address.

Answer: B

Explanation:

From the log provided in the exhibit, several conclusions can be drawn regarding the installation of Policy ID 2:

The installation process fails when attempting to set the LDAP user "student". The log shows:

"Attribute 'ldap-server' MUST be set.": This error indicates that when setting up the LDAP user configuration, a mandatory field (ldap-server) is missing. As a result, the configuration could not be completed for the user.

"entry not found in datasource" and "value parse error before 'student'": These errors confirm that the user "student" could not be properly added due to a configuration issue (such as the missing LDAP server).

Because of these errors, while other configuration elements (such as source and destination interfaces, actions, and services) are properly set, the user configuration for "student" is not applied.

Evaluation of the answer options:

A. Policy ID 2 is installed in the disabled state.

This is false. There is no indication in the log that Policy ID 2 is disabled.

B. Policy ID 2 is installed without the remote user student.

This is true. Due to the failure in setting the "student" user (as indicated by the errors), the policy will be installed without that user being properly configured.

C. Policy ID 2 will not be installed.

This is false. The policy is installed, but the user configuration failed. The rest of the policy configuration appears to have proceeded without critical errors that would prevent the installation.

D. Policy ID 2 is installed without a source address.

This is false. The log shows that the source address is properly set to "all" (set srcaddr all), so this is not the cause of the issue.

From the log exhibit, we see errors related to the "ldap-server" attribute not being set and an error with the entry "student" not being found in the datasource. This indicates that Policy ID 2 will not be installed due to missing or incorrect data required for successful installation. The "Command fail. Return code -3" confirms the installation failure, so the correct answer is C.

Options A, B, and D are incorrect because:

A suggests the policy is installed in a disabled state, which isn't supported by the log.

B and D suggest partial installation, but the error messages indicate a complete failure to install Policy ID 2.

FortiManager References:

Refer to FortiManager 7.4 Troubleshooting Guide: Common Errors and Log Interpretation.

Question: 114

In the event that one of the secondary FortiManager devices fails, which action must be performed to return the FortiManager HA manual mode to a working state?

- A. The FortiManager HA state transition is transparent to administrators and does not require any reconfiguration.
- B. Reboot the failed device to remove its IP from the primary device.
- C. Manually promote one of the working secondary devices to the primary role, and reboot the old primary device to remove the peer IP of the failed device.
- D. Reconfigure the primary device to remove the peer IP of the failed device.

Answer: C

Explanation:

When a secondary FortiManager device fails in HA manual mode, an administrator must manually promote one of the working secondary devices to the primary role and reboot the old primary device to remove the peer IP of the failed device. This ensures the HA configuration is updated correctly, and the network remains resilient.

Options A, B, and D are incorrect because:

- A. suggests the transition is transparent, which is true only in automatic mode, not in manual mode.
- B. and D. imply simpler steps that do not fully address the HA reconfiguration process in manual mode.

FortiManager References:

Refer to FortiManager 7.4 High Availability (HA) Configuration Guide: Manual Mode Configuration and Failover Procedures.

Question: 115

An administrator has assigned a global policy package to custom ADOM1. Then the administrator creates a new policy package. Fortinet. in the custom ADOM1. What happens to the Fortinet policy package when it is created?

- A. You must assign the global policy package from the global ADOM.
- B. The global policy package is automatically assigned.
- C. You must reapply the global policy package to ADOM1.
- D. You can select the option to assign the global policies.

Answer: B

Explanation:

When a new policy package is created in a custom ADOM that already has a global policy package assigned, the global policy package is automatically assigned to the new policy package. This behavior ensures consistent policy enforcement across different ADOMs.

Options A, C, and D are incorrect because:

A and C incorrectly suggest that manual reassignment or reapplication is needed.

D implies optional assignment, whereas it is automatically done.

FortiManager References:

Refer to FortiManager 7.4 Administrator Guide: Working with Global and Custom ADOM Policy Packages

Question: 116

Which output is displayed right after moving the ISFW device from one ADOM to another?

A)

```
FortiManager # diagnose dvm device list ISFW
--- There are currently 4 devices/vdoms managed ---
--- There are currently 4 devices/vdoms count for license ---

TYPE          OID    SN              HA    IP          NAME      ADOM    IPS          FIRMWARE
fmgfaz-managed 325    FGVM010000077646 -    10.0.1.200  ISFW      ADOM74  6.00741 (regular) 7.0 MR4 (2463)
|- STATUS: dev-db: not modified; conf: in sync; cond: OK; dm: retrieved; conn: up
|- vdom:[3]root flags:0 adom:ADOM74 pkg:[unknown]ISFW
```

B)

```
FortiManager # diagnose dvm device list ISFW
--- There are currently 4 devices/vdoms managed ---
--- There are currently 4 devices/vdoms count for license ---

TYPE          OID    SN              HA    IP          NAME      ADOM    IPS          FIRMWARE
fmgfaz-managed 325    FGVM010000077646 -    10.0.1.200  ISFW      ADOM74  6.00741 (regular) 7.0 MR4 (2463)
|- STATUS: dev-db: not modified; conf: in sync; cond: OK; dm: autoupdated; conn: up
|- vdom:[3]root flags:1 adom:ADOM74 pkg:[out-of-sync]ISFW
```

C)

```
FortiManager # FortiManager # diagnose dvm device list ISFW
--- There are currently 4 devices/vdoms managed ---
--- There are currently 4 devices/vdoms count for license ---

TYPE          OID    SN              HA    IP          NAME      ADOM    IPS          FIRMWARE
fmgfaz-managed 325    FGVM010000077646 -    10.0.1.200  ISFW      ADOM74  6.00741 (regular) 7.0 MR4 (2463)
|- STATUS: dev-db: not modified; conf: in sync; cond: OK; dm: installed; conn: up
|- vdom:[3]root flags:0 adom:ADOM74 pkg:[never-installed]
```

D)

```
FortiManager # diagnose dvm device list ISFW
--- There are currently 4 devices/vdoms managed ---
--- There are currently 4 devices/vdoms count for license ---

TYPE          OID    SN              HA    IP          NAME      ADOM    IPS          FIRMWARE
fmgfaz-managed 325    FGVM010000077646 -    10.0.1.200  ISFW      ADOM74  6.00741 (regular) 7.0 MR4 (2463)
|- STATUS: dev-db: not modified; conf: in sync; cond: OK; dm: installed; conn: up
|- vdom:[3]root flags:0 adom:ADOM74 pkg:[imported]ISFW
```

- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer: A

Explanation:

When a FortiGate device, like the ISFW (Internal Segmentation Firewall), is moved from one ADOM to another in FortiManager, the status of the device in the new ADOM will temporarily show some level of inconsistency or unknown state until the ADOM fully syncs and integrates the device.

In the provided options, we are analyzing the FortiManager diagnose dvm device list output for the ISFW device.

Explanation of the Outputs:

Option A:

The output shows that the device has the following status:

dev-db: not modified

conf: in sync

cond: OK

dm: retrieved

The key part here is the pkg: [unknown]. This suggests that the configuration package for the ADOM in the new environment is still in an unknown state, which happens right after moving the device to a new ADOM. FortiManager needs time to process the device's configuration before syncing it properly.

Option B:

This output shows the pkg: [out-of-sync]. This occurs after some configuration mismatch is identified, but it is not the immediate output after moving a device to a new ADOM.

Option C:

This output shows pkg: [never-installed], which indicates that no package was ever installed on the device. This status typically appears when a device is newly added to FortiManager but not immediately after moving it between ADOMs.

Option D:

This output shows pkg: [imported], which indicates that the device configuration has been successfully imported into the new ADOM. This would occur after the device is fully synced, but not immediately after moving the device to a new ADOM.

Conclusion:

The output that is displayed immediately after moving the ISFW device from one ADOM to another is Option A, where the package status is still unknown (pkg: [unknown]) because FortiManager has not yet fully synchronized the device's configuration in the new ADOM.

Question: 117

Refer to the exhibit.



A junior administrator is troubleshooting a FortiManager connectivity issue that is occurring with a managed FortiGate device.

Given the FortiManager device manager settings shown in the exhibit, what can you conclude from this scenario?

- A. The administrator must refresh the device to restore connectivity.
- B. FortiManager lost internet connectivity, therefore, the device appears to be down.
- C. The administrator can reclaim the FortiGate to FortiManager protocol (FGFM) tunnel to get the device online.
- D. The administrator recently restored a FortiManager configuration file.

Answer: C

Explanation:

Option C: The administrator can reclaim the FortiGate to FortiManager protocol (FGFM) tunnel to get the device online. This is the correct answer. The exhibit shows a device in "Unknown" status, which indicates that the FortiManager cannot currently communicate with the device. Reclaiming the FGFM tunnel will help to restore connectivity by re-establishing the management tunnel between the FortiManager and the FortiGate.

Explanation of Incorrect Options:

Option A: The administrator must refresh the device to restore connectivity is incorrect because refreshing the device is unlikely to solve the connection issue when the status is "Unknown."

Option B: FortiManager lost internet connectivity, therefore, the device appears to be down is incorrect because FortiManager does not require internet connectivity to manage a FortiGate; it needs a direct connection to the device.

Option D: The administrator recently restored a FortiManager configuration file is incorrect because the exhibit does not indicate a recent restoration of configuration.

FortiManager References:

Refer to "FortiManager Administration Guide" and the section on "Device Management and Connectivity" for more information about reclaiming FGFM tunnels.

Question: 118

Refer to the exhibit.



You are using the Quick Install option to install configuration changes on the managed FortiGate. Which two statements correctly describe the result? (Choose two.)

- A. It installs provisioning template changes on the FortiGate device.
- B. It provides the option to preview only the policy package changes before installing them.
- C. It installs all the changes in the device database first and the administrator must reinstall the changes on the FortiGate device.
- D. It installs device-level changes on the FortiGate device without launching the Install Wizard

Answer: B, D

Explanation:

Option B: It provides the option to preview only the policy package changes before installing them. This is correct. The Quick Install option in FortiManager provides a preview of policy changes before they are applied, allowing administrators to review and confirm the changes.

Option D: It installs device-level changes on the FortiGate device without launching the Install Wizard. This is correct. Quick Install allows for the immediate installation of device-level changes, such as interface or routing configurations, directly onto the FortiGate without going through the full Install Wizard.

Explanation of Incorrect Options:

Option A: It installs provisioning template changes on the FortiGate device is incorrect because Quick Install does not specifically deal with provisioning templates.

Option C: It installs all the changes in the device database first and the administrator must reinstall the changes on the FortiGate device is incorrect because Quick Install directly applies changes to the FortiGate device, not requiring a separate reinstall step.

FortiManager References:

Refer to "FortiManager Administration Guide" for details on "Quick Install" functionality under "Device Management."

Question: 119

Refer to the exhibit.

FortiManager CLI output

```
FortiManager # execute top
top - 13:08:23 up 1 day,  1:01,  0 users,  load average: 2.40, 3.19, 3.34

Tasks: 188 total,   2 running, 186 sleeping,   0 stopped,   0 zombie

%Cpu(s): 15.4 us,  7.7 sy,  0.0 ni, 76.9 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st

MiB Mem : 7955.5 total,   2235.6 free,  2895.6 used,  2824.1 buff/cache

MiB Swap: 2048.0 total,   2048.0 free,    0.0 used.  4011.0 avail Mem

  PID USER      PR  NI   VIRT   RES %CPU  %MEM     TIME+ S COMMAND
 1163 root        20   0   17.6m   2.1m  7.1   0.1   0:00.05 R top
    1 root        20   0 602.2m  14.9m  0.0   0.7   0:11.67 S /bin/initXXXXXXXXXX
    2 root        20   0    0.0m   0.0m  0.0   0.0   0:00.00 S [kthreadd]
 1462 root        20   0 303.2m 248.0m  0.0   3.1   0:14.72 S fwmsvrd
 1463 root        20   0 288.2m 232.3m  0.0   2.9   0:16.47 S fgdlinkd
 1465 root        20   0 383.7m 328.0m  0.0   4.1   0:15.26 S fgdsvr
 1467 root        20   0  84.0m  23.6m  0.0   0.3   0:00.06 S /bin/fgdhttpd
 1468 root        20   0  63.9m  13.1m  0.0   0.2   0:13.00 S fgdupd
 1469 root        20   0  63.5m  12.6m  0.0   0.2   0:00.07 S fmtr_svr
 1470 root        20   0   6.3m   3.5m  0.0   0.0   0:00.09 S /bin/webconsole
 1471 root        20   0 996.4m 850.6m  0.0  10.7   0:00.01 S srchd
 1475 root        20   0 996.4m 120.6m  0.0   1.5   0:00.00 S fclinkd
```

What percent of the available RAM is being used by the process in charge of downloading the web and email filter databases from the public FortiGuard servers?

- A. 2.9
- B. 3.1
- C. 1.5
- D. 4.1

Answer: A

Explanation:

In the exhibit, the FortiManager CLI output displays the results of the `top` command, which shows system processes, CPU usage, and memory (RAM) usage. We are specifically looking for the process responsible for downloading the web and email filter databases from the public FortiGuard servers. This process is typically handled by the `fgdlinkd` process.

Key information from the output:

The `fgdlinkd` process is listed with a PID of 1463.

The `%MEM` column shows that this process is using 2.9% of the available RAM.

Evaluation of Options:

A. 2.9: This is correct. The `fgdlinkd` process, which handles the web and email filter database downloads, is using 2.9% of the available memory, as indicated in the `%MEM` column.

B. 3.1: This is incorrect. The 3.1% memory usage belongs to the fwmsvr process, not the fgdlkd process.

C. 1.5: This is incorrect. The 1.5% memory usage belongs to the fclinkd process, not the fgdlkd process.

D. 4.1: This is incorrect. The 4.1% memory usage belongs to the fgdsr process, not the fgdlkd process.

Question: 120

Refer to the exhibit.

FortiManager script

Create New Script

Script Name

Routing

Comments

Type

CLI Script

Run script on

Device Database

Script details

Search...

1 config router prefix-list

2 edit public

3 config rule

4 edit 1

5 set prefix 0.0.0.0/0

6 set action permit

7 next

8 edit 2

9 set prefix 8.8.8.8/32

10 set action deny

11 end

Revert All Changes

Advanced Device Filters >

Which two results occur if the script is run using the Device Database option? (Choose two.)

- A. You must install these changes on a managed device using the Install Wizard.
- B. The successful execution of a script on the Device Database creates a new revision history.
- C. The script history shows successful installation of the script on the remote FortiGate device.
- D. The device Config Status is tagged as Modified.

Answer: A, D

Explanation:

If the script is run using the "Device Database" option on FortiManager, the following occurs:

- A. You must install these changes on a managed device using the Install Wizard.

Running the script on the Device Database updates only the configuration in the FortiManager's database, not on the actual FortiGate device. To apply the changes, you need to use the Install Wizard to push these configurations to the managed device.

- D. The device Config Status is tagged as Modified.

After running the script on the Device Database, FortiManager tags the device's configuration status as "Modified," indicating that there are pending changes that have not yet been installed on the device.

Options B and C are incorrect because:

B suggests a new revision history is created, but this only happens when changes are actually installed on the managed device.

C implies the script is directly executed on the FortiGate, which is not the case when using the Device Database option.

FortiManager References:

Refer to FortiManager 7.4 Administrator Guide: Scripting and Configuration Management.

Question: 121

Push updates are failing on a FortiGate device that is located behind a NAT device. Which two settings should the administrator check? (Choose two.)

- A. That the override server IP address is set on FortiManager and the NAT device
- B. That the external IP address on the NAT device is set to DHCP and configured with the virtual IP
- C. That the NAT device IP address and correct ports are configured on FortiManager
- D. That the virtual IP address and correct ports are set on the NAT device

Answer: A, D

Explanation:

When push updates are failing on a FortiGate device behind a NAT device, the administrator should check:

- A. That the override server IP address is set on FortiManager and the NAT device.

The override server IP should be configured to ensure that FortiManager uses the correct IP address that can traverse the NAT to reach the FortiGate device.

D. That the virtual IP address and correct ports are set on the NAT device.

The NAT device must have the correct virtual IP (VIP) configured to map the FortiGate's internal IP to an external address, along with the correct ports needed for communication.

Options B and C are incorrect because:

B. suggests setting the external IP on the NAT device to DHCP, which is not relevant to solving the push update issue.

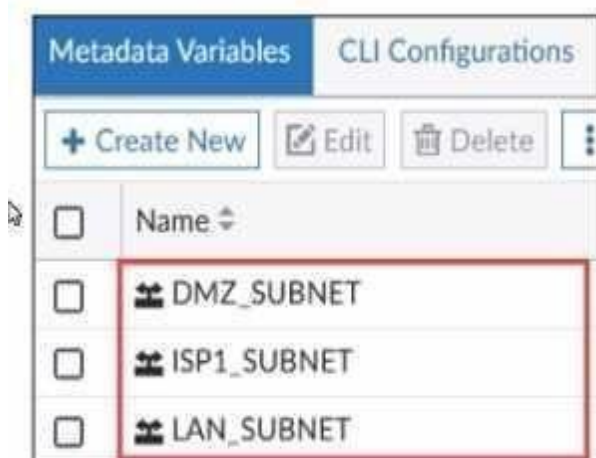
C. implies configuring NAT device IP and ports on FortiManager, which is less likely needed compared to configuring the correct VIP and ports.

FortiManager References:

Refer to FortiManager 7.4 Administrator Guide: Device Management and NAT Configuration.

Question: 122

Exhibit.



What is true about the objects highlighted in the image?

- A. They can be set to optional or required.
- B. They are available across all ADOMs by default.
- C. They can be used as variables in scripts.
- D. They cannot be created in the global database ADOM.

Answer: C

Explanation:

The objects highlighted in the image (DMZ_SUBNET, ISP1_SUBNET, LAN_SUBNET) are metadata variables.

C. They can be used as variables in scripts.

These metadata variables are placeholders that can be used in FortiManager scripts to dynamically insert specific values, enabling script flexibility and scalability across multiple devices or ADOMs.

Options A, B, and D are incorrect because:

A. suggests optional or required settings, which do not apply to metadata variables.

B. implies they are available across all ADOMs by default, which is not always the case.

D. states they cannot be created in the global database ADOM, but metadata variables are typically managed within ADOMs and can be utilized globally based on specific configurations.

FortiManager References:

Refer to FortiManager 7.4 Administrator Guide: Using Metadata Variables and Script Management.

Question: 123

Which statement about the upgrade of ADOMs on FortiManager is true?

- A. To ensure database consistency, you must upgrade an ADOM before you upgrade the devices in it.
- B. Upgrading the FortiManager version upgrades all existing ADOMs automatically.
- C. You cannot import policies from a device until its FortiOS version matches the ADOM version.
- D. ADOMs using global objects can be upgraded before or after upgrading the global database ADOM.

Answer: A

Explanation:

Option A: To ensure database consistency, you must upgrade an ADOM before you upgrade the devices in it. This is the correct answer. When upgrading ADOMs on FortiManager, the ADOM must be upgraded first to match the FortiOS version of the devices it manages. This is necessary to ensure compatibility and consistency between the ADOM's database schema and the FortiGate's configuration.

Explanation of Incorrect Options:

Option B: Upgrading the FortiManager version upgrades all existing ADOMs automatically is incorrect because the ADOMs must be upgraded manually or individually after upgrading the FortiManager.

Option C: You cannot import policies from a device until its FortiOS version matches the ADOM version is incorrect because while version matching is important, it is not strictly necessary for policy import.

Option D: ADOMs using global objects can be upgraded before or after upgrading the global database ADOM is incorrect as the order of upgrade matters to maintain compatibility.

FortiManager References:

Refer to "FortiManager Upgrade Guide" for detailed procedures on upgrading ADOMs and devices.

Question: 124

An administrator created a new global policy package that includes header and footer policies and then assigned it to an ADOM. What are two outcomes of this action? (Choose two.)

- A. To assign another global policy package later to the same ADOM, you must unassign this policy first.
- B. After you assign the global policy package to an ADOM, the impacted policy packages become hidden in that ADOM.
- C. You can edit or delete all the global objects in the global ADOM.
- D. You must manually move the header and footer policies after the policy assignment.

Answer: A, C

Explanation:

Option A: To assign another global policy package later to the same ADOM, you must unassign this policy first. This is correct. FortiManager does not allow multiple global policy packages to be assigned to a single ADOM simultaneously. If you want to assign a different global policy package, the existing one must be unassigned first.

Option C: You can edit or delete all the global objects in the global ADOM. This is correct. Once a global policy package is assigned, you have the flexibility to edit or delete global objects in the global ADOM, affecting all ADOMs to which this package is assigned.

Explanation of Incorrect Options:

Option B: After you assign the global policy package to an ADOM, the impacted policy packages become hidden in that ADOM. This is incorrect because the policy packages do not become hidden; they are modified according to the global policies.

Option D: You must manually move the header and footer policies after the policy assignment. This is incorrect because header and footer policies are automatically applied when assigned.

FortiManager References:

See the "Global Policy and ADOM Management" section in the FortiManager Administration Guide.

Question: 125

What must you consider before deciding to use FortiManager to manage a FortiAnalyzer device?

- A. Confirm that FortiManager has enough storage capacity for the expected logs.
- B. Ensure that FortiAnalyzer features are installed in advance.
- C. Check whether FortiManager is part of a high availability (HA) cluster.
- D. Determine whether the VDOMs of the same FortiGate will be assigned to different ADOMs.

Answer: B

Explanation:

When deciding to use FortiManager to manage a FortiAnalyzer device, you must ensure certain conditions are met so that the integration works seamlessly. One key aspect to consider is whether the necessary FortiAnalyzer features are enabled on FortiManager.

Explanation of Options:

A. Confirm that FortiManager has enough storage capacity for the expected logs.

This is false. FortiManager is not primarily responsible for storing logs. Logs are stored on the FortiAnalyzer device, and FortiManager's role is more focused on managing configuration, policies, and pushing updates, not on handling large volumes of logs.

B. Ensure that FortiAnalyzer features are installed in advance.

This is true. Before using FortiManager to manage a FortiAnalyzer device, you must ensure that the necessary FortiAnalyzer features are properly installed and enabled on FortiManager. FortiAnalyzer's reporting and logging functions must be correctly integrated for FortiManager to manage it effectively.

C. Check whether FortiManager is part of a high availability (HA) cluster.

This is false. While HA is important for redundancy, it is not a prerequisite for managing FortiAnalyzer with FortiManager. The HA status of FortiManager does not directly affect its ability to manage a FortiAnalyzer device.

D. Determine whether the VDOMs of the same FortiGate will be assigned to different ADOMs.

This is false. VDOMs (Virtual Domains) and ADOMs (Administrative Domains) relate to the management of FortiGate devices and the segregation of administrative access within FortiManager. This is unrelated to the management of a FortiAnalyzer device.

Question: 126

Refer to the exhibit.



An administrator is about to add the FortiGate device to FortiManager using the discovery process. FortiManager is operating behind a NAT device, and the administrator configured the FortiManager NATed IP address under the FortiManager system administration settings. What is the expected result?

- A. During discovery, FortiManager uses only the FortiGate serial number to establish the connection.
- B. During discovery, FortiManager sets both the FortiManager NATed IP address and NAT device IP address on FortiGate.
- C. During discovery, FortiManager sets the NATed device IP address on FortiGate.
- D. During discovery, FortiManager sets the FortiManager NATed IP address on FortiGate.

Answer: D

Explanation:

When adding a FortiGate device to FortiManager that is operating behind a NAT device, and the FortiManager NATed IP address is configured under the system administration settings, FortiManager will set the FortiManager NATed IP address on the FortiGate device during the discovery process. This ensures that the FortiGate knows how to reach the FortiManager through the NAT device.

Options A, B, and C are incorrect because:

A. is incorrect because the discovery process also requires knowing the NATed IP to establish a connection, not just the serial number.

B. is incorrect because FortiManager does not set the NAT device's IP address on the FortiGate.

C. is incorrect because it implies that the NAT device IP is set on FortiGate, which is not the expected outcome.

FortiManager References:

Refer to FortiManager 7.4 Administrator Guide: Device Discovery and Management with NAT.

Question: 127

An administrator is in the process of copying a system template profile between ADOMs by running the following command: `execute fmprofile import-profile ADOM2 3547 /tmp/myfile` Where does this command import the system template profile from?

- A. FortiManager file system
- B. ADOM2 object database
- C. ADOM2 device database
- D. Source ADOM policy database

Answer: A

Explanation:

The command `execute fmprofile import-profile ADOM2 3547 /tmp/myfile` is used to import a system template profile from the FortiManager file system. The path `/tmp/myfile` indicates a location in the FortiManager's local file system, from which the profile will be imported into the specified ADOM.

Options B, C, and D are incorrect because:

B, C, and D suggest importing from different databases, which is not accurate since the command explicitly refers to the file system location.

FortiManager References:

Refer to FortiManager 7.4 CLI Reference Guide: Commands for Profile Management.

Question: 128

Which API method is used to create objects or overwrite existing ones?

- A. Set
- B. Add
- C. Exec
- D. Update

Answer: A

Explanation:

In the context of the FortiManager JSON API, the `set` method is used to create new objects or overwrite existing ones. The API allows administrators to manage FortiManager and its associated devices by automating tasks like configuration changes, policy updates, and object creation.

Explanation of Options:

A. Set:

This is true. The `set` method is used to create a new object if it does not exist or overwrite an existing object if it already exists. This method is frequently used in API requests to configure settings and apply changes on FortiManager.

B. Add:

This is false. The `add` method is used to add new objects without overwriting any existing ones. It is used when you want to create a new entry and ensure it doesn't conflict with or replace an existing object.

C. Exec:

This is false. The `exec` method is used to execute specific actions or commands, rather than creating or modifying objects. This is typically used for actions like running scripts or executing operational commands on FortiManager or FortiGate.

D. Update:

This is false. While "update" might seem relevant, FortiManager's API does not specifically use an "update" method for modifying or creating objects. The `set` method serves that function by both creating new objects and overwriting existing ones.

Question: 129

What is a characteristic of the FortiManager high availability (HA) feature?

- A. When a secondary unit is removed, FortiManager updates the managed devices using TCP port 5199.
- B. The primary unit synchronizes all configuration revision with the secondary units.
- C. All secondary units must be in the same network as the primary unit.
- D. Each cluster member must be upgraded manually, starting with the primary unit.

Answer: B

Explanation:

The characteristic of the FortiManager high availability (HA) feature is that the primary unit synchronizes all configuration revisions with the secondary units. This ensures that all devices in the HA cluster are up-to-date with the same configurations, providing redundancy and failover capabilities.

Options A, C, and D are incorrect because:

A. refers to a specific port number (5199), but FortiManager does not specifically use TCP port 5199 to update managed devices when a secondary unit is removed.

C. is incorrect as secondary units do not necessarily have to be in the same network as the primary unit; they just need to be able to communicate with each other.

D. is incorrect because HA upgrades can be automated and do not require manual upgrading, starting with the primary unit.

FortiManager References:

Refer to FortiManager 7.4 High Availability (HA) Guide: HA Synchronization and Configuration.

Question: 130

Which two items are included in the FortiManager backup? (Choose two.)

- A. All devices
- B. Firmware images
- C. FortiGuard database
- D. Flash configuration

Answer: A, D

Explanation:

FortiManager backups include:

A. All devices— This includes all device configurations managed by FortiManager, such as firewall policies, objects, and other settings.

D. Flash configuration— This consists of local FortiManager configurations stored in flash memory, such as system settings, scripts, and other locally-stored configurations.

Options B and C are incorrect because:

B. (Firmware images) are not typically included in a FortiManager backup. Firmware images are usually stored separately and managed through a different process.

C. (FortiGuard database) is incorrect as the FortiGuard database, which contains threat intelligence and security signatures, is not part of the standard FortiManager backup.

FortiManager References:

Refer to FortiManager 7.4 Administrator Guide: Backup and Restore Processes.

Question: 131

Refer to the exhibit which shows the Download Import Report.

```
Start to import config from device(Remote-FortiGate) vdom(root) to
adom(root), package(Remote-FortiGate_root)

"firewall address",SKIPPED,"(name=all, oid=2309, DUPLICATE)"

"firewall address",FAIL,"(name=REMOTE_SUBNET, oid=2311,
reason=interface((firewall address:REMOTE_SUBNET) any<-port6) binding
fail)"

"firewall policy",FAIL,"(name=1, oid=3070, reason=interface(interface binding
contradiction. detail: (firewall address:REMOTE_SUBNET) any<-port6) binding
fail)"
```

Why is FortiManager failing to import firewall policy ID 1?

- A. Policy ID 1 is configured from the interface any to port6. FortiManager rejects the request to import this policy because the any interface does not exist on FortiManager
- B. Policy ID 1 for this managed FortiGate already exists on FortiManager in the policy package named Remote-FortiGate.
- C. Policy ID 1 has an address object that already exists in the ADOM database with any as the interface association, and conflicts with the address object interface association locally on FortiGate.
- D. Policy ID 1 does not have the ADOM Interface mapping configured on FortiManager.

Answer: A

Explanation:

Option A: Policy ID 1 is configured from the interface any to port6. FortiManager rejects the request to import this policy because the any interface does not exist on FortiManager. This is the correct answer. FortiManager fails to import firewall policy ID 1 because it cannot map the "any" interface to a valid interface in its ADOM database. The error indicates that there is a binding failure due to an interface mismatch.

Explanation of Incorrect Options:

Option B: Policy ID 1 for this managed FortiGate already exists on FortiManager in the policy package named Remote-FortiGate is incorrect because the error is related to interface mapping, not a duplicate policy ID.

Option C: Policy ID 1 has an address object that already exists in the ADOM database with any as the interface association and conflicts with the address object interface association locally on FortiGate is incorrect because the error specifies an interface issue, not an address object conflict.

Option D: Policy ID 1 does not have the ADOM Interface mapping configured on FortiManager is incorrect because the error directly mentions a binding failure due to the "any" interface.

FortiManager References:

For more information, refer to the "Device Manager" section and "Configuration Import and Mapping" in the FortiManager Administration Guide.

Question: 132

Exhibit.

```
FortiManager # config system global
(global)# set workspace-mode normal
(global)# end
FortiManager #
```

Given the configuration shown in the exhibit, what are two results from this configuration? {Choose two.}

- A. You can validate administrator login attempts through external servers.
- B. The same administrator can lock more than one ADOM at the same time.
- C. Two or more administrators can make configuration changes at the same time, in the same ADOM.
- D. Concurrent read-write access to an ADOM is disabled.

Answer: B, D

Explanation:

The configuration shown in the exhibit sets the workspace-mode to normal. The workspace mode in FortiManager defines how configuration changes and administrative tasks are handled, specifically regarding locking and collaboration in ADOMs (Administrative Domains).

Understanding the workspace modes:

Normal Mode: In this mode, only one administrator at a time can lock and edit an ADOM. The changes made by one administrator must be completed and saved before another administrator can make changes. It prevents concurrent read-write access within the same ADOM.

Workflow Mode: This mode allows multiple administrators to work on different tasks within the same ADOM, but changes still need to be approved before being committed.

Explanation of Options:

- A. You can validate administrator login attempts through external servers.

This option is unrelated to the workspace mode. External authentication servers can be used for administrator logins, but that is a different configuration setting (not related to workspace-mode).

- B. The same administrator can lock more than one ADOM at the same time.

This is true. In Normal mode, an administrator can lock multiple ADOMs, meaning they can work on more

than one ADOM simultaneously, but each ADOM can only be accessed by one administrator at a time for read-write purposes.

C. Two or more administrators can make configuration changes at the same time, in the same ADOM.

This is false. In Normal mode, only one administrator can have read-write access to an ADOM at a time. If another administrator attempts to make changes, they must wait until the ADOM is unlocked by the first administrator.

D. Concurrent read-write access to an ADOM is disabled.

This is true. In Normal mode, concurrent read-write access is disabled. This means only one administrator at a time can make changes to an ADOM. Other administrators can view the ADOM in read-only mode but cannot make changes until the ADOM is unlocked.

Question: 133

Exhibit.

```
FortiManager # diagnose dvm device list
--- There are currently 1 devices/vdoms managed ---
--- There are currently 1 devices/vdoms count for license ---

TYPE          OID    SN          HA    IP          NAME          ADOM    IPS          FIRMWARE
fmgfaz-managed 325    FGVN010000077646 -    10.0.1.200    ISFW          ADOM2    6.00741 (regular) 7.0 MR4 (2463)
|- STATUS: dev-db: modified; conf: in sync; cond: pending; dm: retrieved; conn: up
|- vdom:[3]root-flags:1 adom:ADOM2 pkg: [imported]ISFW
```

Which two statements about the output are true? (Choose two.)

- A. The latest revision history for the managed FortiGate does not match the device-level database.
- B. Configuration changes have been installed on FortiGate, which means the FortiGate configuration has been changed.
- C. Configuration changes directly made on FortiGate have been automatically updated to the device-level database.
- D. The latest revision history for the managed FortiGate does match the FortiGate running configuration.

Answer: A, B

Explanation:

The output indicates that:

The device's status is shown as "dev-db: modified" and "conf: in sync," which means that there is a difference between the device-level database on FortiManager and the actual running configuration of the managed FortiGate. Therefore, the latest revision history for the managed FortiGate does not match the device-level database, which confirms statement A as true.

"dm: retrieved" status indicates that configuration changes have been installed on the FortiGate, confirming statement B as true. It also means that the configuration has been modified, and those changes have been pulled from the FortiGate to the FortiManager.

Statements C and D are incorrect because:

C. is incorrect as it implies an automatic update, whereas "dev-db: modified" indicates changes have been made on the FortiGate device that are not yet reflected in the FortiManager's database.

D. is incorrect because "dev-db: modified" shows that the device-level database and running configuration are not in sync.

FortiManager References:

Refer to the FortiManager 7.4 Administrator Guide: Device Manager > Device Status to understand the "dev-db" and "conf" status meanings.

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